Universität Basel

Stochastic Analysis

Lecture by Prof. Jiří Černý, HS 2025

Notes by David Flückiger

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Vorwort

Diese Notizen begleiten die Mastervorlesung “Stochastic Analysis” von Professor Jiří Černý an der Universität Basel. Sie werden gewiss nicht fehlerfrei sein, daher bitte mit Vorsicht verwenden. Die Vorlesung basiert hauptsächlich auf dem Buch von Le Gall, “Brownian Motion, Martingales, and Stochastic Calculus” (siehe [2]). Weitere Resultate stammen aus den folgenden Büchern, welche auch für vertieftes Verständnis interessant sind:

[1] Richard Durrett. *Stochastic Calculus - A Practical Introduction.* Boca Raton, Fla: CRC Press, 1996. ISBN: 978-0-849-38071-6.

[2] Jean-François Le Gall. *Measure Theory, Probability, and Stochastic Processes -*. Singapore: Springer Nature, 2022. ISBN: 978-3-031-14205-5.

[3] Ioannis Karatzas and Steven Shreve. *Brownian Motion and Stochastic Calculus* -. Berlin, Heidelberg: Springer, 2014. ISBN: 978-1-461-20949-2.

[4] Daniel Revuz and Marc Yor. *Continuous Martingales and Brownian Motion -*. Berlin Heidelberg: Springer Science & Business Media, 2013. ISBN: 978-3-662-06400-9.

[5] L. C. G. Rogers and David Williams. *Diffusions, Markov Processes and Martingales: Volume 2, Itô Calculus -*. Cambridge: Cambridge University Press, 2000. ISBN: 978-0-521-77593-9.

[6] L. C. G. Rogers and David Williams. *Diffusions, Markov Processes, and Martingales: Volume 1, Foundations -*. Cambridge: Cambridge University Press, 2000. ISBN: 978-1-107-71749-7.

[7] Daniel W. Stroock and S.R.S. Varadhan. *Multidimensional Diffusion Processes -*. Berlin, Heidelberg: Springer, 2007. ISBN: 978-3-540-28999-9.

Contents

1. Introduction

The objection of this course is to construct Brownian motion, present its most important properties and develop the infinitesimal calculus attached to it. We will also see that Brownian motion is an element of various important classes of processes, e.g. it is

- continuous martingale
- continuous Markov process
- Gaussian process
- Levy process.

We will thus use Brownian motion as a potential to give an introduction into the mathematical theory of those processes.

1.1. History

Brownian Motion is named after the botanist *Robert Brown*, who in 1827 observed the irregular notion of pollen particles suspended in water. First mathematical studies of Brownian motion appear on the turn of the 20th century. In particular Bachelier in his thesis from 1900 use Brownian motion as a model for a stock market, and, in 1905, Einstein consider Brownian motion in his fundamental work on the movement of particles in a fluid, giving one of the first (individual) confirmation of the existence of atoms and molecules. The mathematical theory of Brownian motion was then put on a firm basis with Wiener (1923).

1.2. Informal Construction

There are many ways to construct Brownian motion. From previous lectures you probably know the, probably, most natural one, as the scaling limit of a (polygonal interpolation of) random walk trajectory, scaling the line by n^{-2} and space by n^{-1}: Let $X_1, X_2, \ldots$ be independent identically distributed (i.i.d.) Bernoulli random

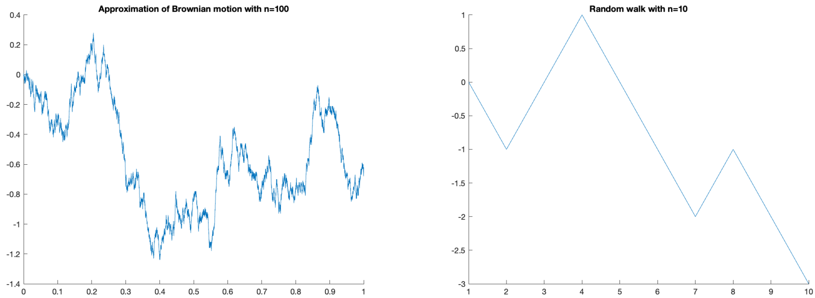

Figure 1.1.: Brownian Motion and Random Walk

variables, i.e.,

$$\mathbb{P}(X_i = -1) = \mathbb{P}(X_i = 1) = \frac{1}{2}.$$

Set $S_n := \sum_{k=1}^{n} X_k$ for $n \in \mathbb{N}$ and define S_t with $t \in \mathbb{R}_+$ by linear interpolation (see Figure). Finally, define the rescaled trajectory

$$B_t^{(n)} := \frac{S_{n^2 t}}{n}, \qquad t \geq 0. \tag{1.1}$$

From the *central limit theorem*, $B_1^{(n)} \xrightarrow{\text{law}} \mathcal{N}(0,1)$ as $n \to +\infty$, a standard normal random variable, i.e.,

$$\lim_{n \to +\infty} \mathbb{P}(B_1^{(n)} \leq a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{a} \mathrm{e}^{-x^2/2}\,\mathrm{d}x, \qquad a \in \mathbb{R}.$$

Donsker's theorem (1951) then shows that $t \mapsto B_t^{(n)}$ viewed as a random continuous function on $\mathbb{R}_+$ converges (usually in the uniform topology) to the law of Brownian motion. We will use other constructions in this lecture.

An important advantage of continuous models (like Brownian motion) vs. discrete models (like random walk) is the apparatus of "infinitesimal calculus". The development of this apparatus is rather non-trivial because the typical realization of Brownian motion trajectory is rather rough: the function $t \mapsto B_t(\omega)$ is continuous but *nowhere* differentiable and of *infinite variation.*

As an example, consider the basic formula of calculus:

$$\frac{\mathrm{d}}{\mathrm{d}t} f\big(b(t)\big) = f'\big(b(t)\big) b'(t) \quad f, b \in C^1. \tag{1.2}$$

It can be extended to $f \in C^1$ and b continuous of finite variation by writing

$$f\big(b(t)\big) = f\big(b(0)\big) + \int_0^t f'\big(b(s)\big)\,\mathrm{d}b(s) \tag{1.3}$$

where $\mathrm{d}b$ stands for the Stieltjes measure on $\mathbb{R}_+$ given by

$$\int_{(x,y]} \mathrm{d}b(s) = b(y) - b(x), \quad 0 \leq x < y < +\infty. \tag{1.4}$$

As Brownian motion is not of finite variation, this does not help when b is a Brownian motion trajectory.

Nontheless, we will develop an infinitesimal calculus from where (1.3) gets replaced by the so-called Itô's formula:

$$f(B_t) = f(B_0) + \int_0^t f'(B_s)\,\mathrm{d}B_s + \frac{1}{2}\int_0^t f''(s)\,\mathrm{d}s, \quad f \in C^2, t \geq 0 \tag{1.5}$$

or in differential notation of (1.2)

$$\mathrm{d}f(B_t) = f'(B_t)\,\mathrm{d}B_t + f''(B_t)\,\mathrm{d}t. \tag{1.6}$$

The main work will be to give a sense to the term "$\int_1^t f'(B_s)\,\mathrm{d}B_s$" appearing as (1.5), since it cannot be defined as Stieltjes integral, as explained.

With this calculus we will be able to solve certain differential equations with random perturbations, the *stochastic differential equations* (SDE's):

$$\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \underbrace{\sigma(X_t)\,\mathrm{d}B_t}_{\text{random pertubation}}. \tag{1.7}$$

We will also see the deep connection between SDE's and certain PDE's. As an example, consider $D \subseteq \mathbb{R}^d$ to be a smooth and bounded domain (i.e., ∂D has a C^∞-parameterization), and the following two PDE problems:

- *Dirichlet problem*: Given $f \in \partial D$, find u such that

$$\frac{1}{2}\Delta u = 0 \quad \text{in } D; \quad u|_{\partial D} \equiv f. \tag{1.8}$$

- *Poisson problem*: For $g \in C(D)$, find u such that

$$\begin{cases} \Delta u = g & \text{in } D, \\ u = 0 & \text{on } \partial D. \end{cases} \tag{1.9}$$

Surprisingly, both these problems can "be solved" with the help of Brownian motion. To do this let $B_t = (B_t^1, \dots, B_t^d)$ be the d-dimensional Brownian motion (i.e., $B_\cdot^1, \dots, B_\cdot^d$ are independent copies of the real-valued Brownian motion). For $x \in D$ define

$$\tau_x := \inf\{s \geq 0 : x + B_s \in \partial D\}.$$

Then the solutions to (1.8), (1.9) are given by

$$\begin{aligned} u_{\mathrm{Dirichlet}}(x) &= \mathbb{E}[f(x + B_{\tau_x})], \\ u_{\mathrm{Poisson}}(x) &= -\mathbb{E}\left[\int_0^{\tau_x} g(x + B_s)\,\mathrm{d}s\right] \end{aligned}$$

We will also be able to replace $\frac{1}{2}\Delta$ in (1.8), (1.9) by more complicated differential operators.

2. Gaussian Vectors and Processes

In the next chapter we will define Brownian motion as a particular "Gaussian process." We will give an introduction to the theory of those processes in this chapter.

2.1. One-dimensional Gaussian distribution

We recall that a random variable X is called *standard normal* or *standard Gaussian* if it has a density on $\mathbb{R}$ given by

$$p_X(x) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-x^2/2}.$$

Exercise 2.1. (1) Show that the *Laplace transform* of X is given by

$$\mathbb{E}[\mathrm{e}^{zX}] = \mathrm{e}^{z^2/2}, \quad \text{for any } z \in \mathbb{C}.$$

Hint. Consider first $z \in \mathbb{R}$ and prove and use the analyticity. In particular, show that the *characteristic function* of X is

$$\mathbb{E}[\mathrm{e}^{\mathrm{i}\xi X}] = \mathrm{e}^{-\xi^2/2}, \quad \xi \in \mathbb{R}.$$

(2) Using the properties of characteristic functions show that $X \in L^p$ (i.e., $\mathbb{E}[|X|^p] < +\infty$) and that

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[X^2] = 1, \quad \mathbb{E}[X^{2n}] = \frac{(2n)!}{2^n!}.$$

For $m \in \mathbb{R}, \sigma > 0$ we say that a random variable Y is $\mathcal{N}(m, \sigma^2)$ distributed (we write it as $Y \sim \mathcal{N}(m, \sigma^2)$) if Y can be written as

$$Y = m + \sigma X, \tag{2.1}$$

where X is standard normal random variable.

Exercise 2.2. (1) Show that (2.1) is equivalent with

(i) Y has a density

$$p_Y(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-m)^2}{2\sigma^2}\right\},$$

(ii) the characteristic function of Y is

$$\mathbb{E}[e^{i\xi Y}] = e^{im\xi - \frac{\sigma^2}{2}\xi^2}.$$

(2) Prove $\mathbb{E}[X] = m$, $\mathrm{Var}(Y) = \sigma^2$.

(3) Let Y be $\mathcal{N}(m_Y, \sigma_Y^2)$ distributed, Z be $\mathcal{N}(m_Z, \sigma_Z^2)$ distributed and Y be independent of Z. Show that $Y + Z$ is $\mathcal{N}(m_Y + m_Z, \sigma_Y^2 + \sigma_Z^2)$ distributed.

Lemma 2.3. *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables where $X_n \sim \mathcal{N}(m_n, \sigma_n^2)$. Assume that X_n converge in distribution to a random variable X. Then*

(i) X is also a Gaussian random variable, $X \sim \mathcal{N}(m, \sigma^2)$, where

$$m = \lim_{n\to+\infty} m_n, \qquad \sigma = \lim_{n\to+\infty} \sigma_n.$$

(ii) If X_n converges in probability, then it converges in L^p for any $p \in [1, +\infty)$.

Proof. *(i)*: Convergence in distribution is equivalent with

$$\mathbb{E}[e^{i\xi X_n}] = \exp\left\{im_n\xi - \frac{\sigma_n^2}{2}\xi^2\right\} \xrightarrow{n\to+\infty} \mathbb{E}[e^{i\xi X}] \quad \text{for any } x \in \mathbb{R}.$$

Taking absolute values,

$$\exp\left\{-\frac{\sigma_n^2}{2}\xi^2\right\} \xrightarrow{n\to+\infty} |\mathbb{E}[e^{i\xi X}]|,$$

which holds only if $\sigma_n \to \sigma \geq 0$ ($\sigma = +\infty$ can be excluded, since $\mathbb{E}[e^{i\xi X}]$ must be continuous at 0). Hence, for all $\xi \in \mathbb{R}$

$$e^{im_n\xi} \xrightarrow{n\to+\infty} e^{\frac{1}{2}\sigma^2\xi^2}\, \mathbb{E}[e^{i\xi X}] \tag{2.2}$$

To prove that (2.2) implies $\lim_{n\to+\infty} m_n = m$ observe first that m_n must be a bounded sequence. Indeed, assume by contradiction that for some $n_k \nearrow +\infty$ we have $m_{n_k} \nearrow$

$+\infty$. Then for any $A > 0$,

$$\mathbb{P}(X \geq A) \geq \limsup_{k\to+\infty} \mathbb{P}(X_{n_k} \geq A) \geq \frac{1}{2}.$$

Letting $A \nearrow +\infty$ we obtain a contradiction. Assume now that m_n is bounded and does not converge. Then it has at least two accumulation points $m \neq m'$ and by (2.2) $\mathrm{e}^{\mathrm{i}\xi m} = \mathrm{e}^{\mathrm{i}\xi m'}$ for any $\xi \in \mathbb{R}$. But this is possible only if $m = m'$.

(ii): X_n can be written as $X_n = m_n + \sigma_n X$ with $X \sim \mathcal{N}(0,1)$. By the first point *(i)*, m_n and σ_n are bounded sequences which implies that $\sup_{n\in\mathbb{N}} \mathbb{E}[|X_n|^q] < +\infty$ for any $q \in (0,+\infty)$. By Fatou's lemma then also $\mathbb{E}[|X|^q] < +\infty$.

Let now $p \in [1,+\infty)$. By assumption, $Y_n := |X_n - X|^p$ converges to 0 in probability. Using $q = 2p$, we see from the previous reasoning that Y_n is bounded in L^2 and so uniformly integrable, hence $Y_n \to 0$ in L^1, proving $X_n \to X$ in L^p. □

2.2. Gaussian Vectors

Let E be a d-dimensional Euclidean space with *scalar product* $\langle \cdot, \cdot \rangle_E$.[1]

Definition 2.4. An E-valued random variable X is called *Gaussian vector* if $\langle n, X\rangle_E$ is a Gaussian random variable for any $n \in E$.

Example 2.5. Let $E = \mathbb{R}^d$, $X_1, \dots, X_d$ independent Gaussian random variables. Then $X = (X_i)_{i=1}^d$ is a Gaussian vector. ♦

Exercise 2.6. Use Exercise 2.2 to show this claim.

If X is a Gaussian vector in E, then there exists $m_x \in E$ and a positive quadratic form $q_X : E \to \mathbb{R}$ such that

$$\mathbb{E}[\langle n, X\rangle_E] = \langle n, m_X\rangle_E \quad \text{and} \quad \mathrm{Var}[\langle n, X\rangle_E] = q_X(n)$$

Exercise 2.7. Prove the above claim. In order to do this consider an orthonormal basis $(e_1, \dots, e_d)$ of E and show that if $X = \sum_{k=1}^d X_k e_k$ with $X_i = \langle X, e_i\rangle_E$ then $m_X =$

[1] A Euclidean vector space is a finite-dimensional inner product space over the real numbers, see e.g., Wikipdia.

$\sum_{i=1}^d \mathbb{E}[X_i]e_i = \mathbb{E}[X]$ and for $u = \sum_{i=1}^d u_i e_i$

$$q_X(u) = \sum_{i,j=1}^d u_i u_j \operatorname{Cov}(X_i, X_j).$$

Show that the characteristic function of X is

$$\mathbb{E}[\mathrm{e}^{\mathrm{i}\langle \xi, X\rangle_E}] = \exp\left\{ \mathrm{i}\langle m_X, \xi\rangle_E - \frac{1}{2} q_X(\xi) \right\}, \quad \xi \in E.$$

Exercise 2.8. The coordinates $(X_1, \dots, X_d)$ of a Gaussian vector X are independent if and only if the matrix $(\operatorname{Cov}(X_i, X_j))_{i,j=1}^d$ is diagonal, that is q_X is diagonal in the basis $(e_1, \dots, e_d)$.

To q_X is associated a positive symmetric endomorphism γ_X via

$$q_X(u) = \langle u, \gamma_X(u)\rangle_E.$$

(γ_X has matrix form $(\operatorname{Cov}(X_i, X_j))_{i,j=1}^d$ in the basis $(e_1, \dots, e_d)$).

Theorem 2.9 (Existence and properties of Gaussian vectors). *(i) For every positive symmetric endomorphism γ on E there is a Gaussian vector X such that $\gamma = \gamma_X$.*

(ii) Let X be a Gaussian vector. If $(\Sigma_1, \dots, \Sigma_d)$ is the basis diagonalizing γ_X: $\gamma_X \Sigma_j = \lambda_j \Sigma_j$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_d$ then

$$X = \sum_{i=1}^r Y_j \Sigma_j$$

where Y_i, $i = 1, \dots, r$ are independent, $Y_i \sim \mathcal{N}(0, \lambda_i)$. In particular

$$\operatorname{supp}(\mathbb{P}_X) = \operatorname{span}(\Sigma_1, \dots, \Sigma_r).$$

(iii) The law of X is absolutely continuous with respect to the Lebesgue measure on E if and only if $r = d$. In this case X has a density

$$p_X(x) = \frac{1}{(2\pi)^{d/2}\sqrt{\det \gamma_X}} \exp\left\{ -\frac{1}{2}\langle x, \gamma_X^{-1}(x)\rangle_E \right\}.$$

Reminder 2.10. The support of a measure is defined as follows: Let S be a topological space equipped with its Borel σ-algebra $\mathcal{B}(S)$ (i.e., the σ-algebra generated by all open subsets). The support of a measure $\mu : \mathcal{B}(S) \to \mathbb{R} \cup \{+\infty\}$ is given by

$$\operatorname{supp}(\mu) := \{x \in S \mid \mu(N_x) > 0 \,\forall \text{ open neighbourhoods } N_x \text{ of } x\}.$$

The above set is closed as it is the complement of the union of all open sets A with $\mu(A) = 0$. This also means that it is measurable. If μ is a signed measure with Jordan-Hahn-decomposition $\mu = \mu_+ - \mu_-$ we define

$$\operatorname{supp}(\mu) = \operatorname{supp}(\mu_+) \cup \operatorname{supp}(\mu_-).$$

Proof of Theorem 2.9. *(i)*: Let $(\Sigma_1, \dots, \Sigma_d)$ be an orthonormal basis diagonalizing γ i.e., $\gamma\Sigma_i = \lambda_i\Sigma_i$, and Y_i independent, $Y_i \sim \mathcal{N}(0, \lambda_i)$ such that $X = \sum_{i=1}^d Y_i\Sigma_i$. Then one sees easily that for $u = \sum_{i=1}^d u_i\Sigma_i$

$$q_X(u) = \mathbb{E}\left[\left(\sum_{i=1}^d u_j Y_j\right)^2\right] = \sum_{i=1}^d \lambda_j u_j^2 = \langle u, \gamma u\rangle_E.$$

(ii): By definition of the basis $(\Sigma_1, \dots, \Sigma_d)$, the covariance matrix

$$(\operatorname{Cov}(X_i, X_j))_{i,j=1}^d = \operatorname{diag}(\lambda_1, \dots, \lambda_d).$$

For $i > r$, $\mathbb{E}[Y_i^2] = 0$, so $Y_i = 0$ almost surely. Using Exercise 2.8 we see that $(Y_i)_{i\le r}$ are independent. Since $X = \sum_{i=1}^r Y_i\Sigma_i$ almost surely , then $\operatorname{supp}(\mathbb{P}_X) \subseteq \operatorname{span}(\Sigma_1, \dots, \Sigma_r)$. On the other hand,

$$\mathbb{P}\left(X \in \prod_{i=1}^r [a_i, b_i]e_i\right) = \prod_{i=1}^r \mathbb{P}(Y_i \in [a_i, b_i]) > 0,$$

we see that $\operatorname{supp}(\mathbb{P}_X) \supseteq \operatorname{span}(\Sigma_1, \dots, \Sigma_r)$.

(iii): If $r < d$, $\operatorname{supp}(\mathbb{P}_X) \subseteq \operatorname{span}(\Sigma_1, \dots, \Sigma_r)$ which has 0 Lebesgue measure on E. If $r = d$, let $Y = (Y_1, \dots, Y_d)$ be a random vector in $\mathbb{R}^d$. The random vector X is then image of Y by a bijection $\varphi : \mathbb{R}^d \to E$ given by $\varphi(y_1, \dots, y_d) = \sum_{i=1}^d y_i\Sigma_i$. Then

$$\begin{aligned}\mathbb{E}[g(X)] = \mathbb{E}\left[g\big(\varphi(Y)\big)\right] &\overset{\text{Exercise 2.2 (1)}}{=} \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} g\big(\varphi(y)\big) \prod_{i=1}^d \frac{\mathrm{d}y_i}{\sqrt{\lambda_i}} \exp\left\{-\frac{1}{2}\frac{y_i^2}{\lambda_i}\right\} \\ &= \frac{1}{(2\pi)^{d/2}} \frac{1}{\sqrt{\det \gamma_X}} \int_{\mathbb{R}^d} g\big(\varphi(y)\big) \exp\left\{-\frac{1}{2}\langle\varphi(y), \gamma_X^{-1}\varphi(y)\rangle_E\right\} \mathrm{d}y_1 \dots \mathrm{d}y_d\end{aligned}$$

$$= \frac{1}{(2\pi)^{d/2}} \frac{1}{\sqrt{\det \gamma_X}} \int_E g(x) \exp\left\{-\frac{1}{2}\langle x, \gamma_X^{-1} x\rangle_E\right\} \mathrm{d}x$$

where in the last equality we used the fact that the Lebesgue measure on E is the image of the Lebesgue measure on $\mathbb{R}^d$ by φ. □

2.3. Gaussian spaces and processes

Definition 2.11. A *Gaussian space* is a closed sub-space of the space $L^2(\Omega, \mathcal{A}, \mathbb{P})$ formed by centred Gaussian random variables.

Example 2.12. If $(X_1, \dots, X_d) \in \mathbb{R}^d$ is a Gaussian vector, then $\operatorname{span}(X_1, \dots, X_d)$ is a Gaussian space. ♦

Definition 2.13. A collection $X = (X_t)_{t\in T}$ of real valued random variables on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ is called *Gaussian process* if every finite linear combination of X_t, $t \in T$ is a (centred) Gaussian random variable.

Lemma 2.14. *If $(X_t)_{t\in T}$ is a Gaussian process, then the closed subspace of $L^2(\Omega, \mathcal{A}, \mathbb{P})$ generated by the random variables X_t, $t \in T$ is a Gaussian space which is called* Gaussian space generated by X.

Proof. It is sufficient to observe that a limit in L^2 of Gaussian random variables is again a Gaussian random variable, by Lemma 2.3. □

2.3.1. Existence of Gaussian processes

Let $X = (X_t)_{t\in T}$ be a Gaussian process. Its *covariance function* $\Gamma : T \times T \to \mathbb{R}$ is given by

$$\Gamma(s,t) = \operatorname{Cov}(X_s, X_t) = \mathbb{E}[X_s X_t], \qquad s,t \in T.$$

The distribution of X is determined by Γ (in the non- centred case one needs in addition a function $m(t) = \mathbb{E}[X_t]$). Indeed, for every finite collection $\{t_1, \dots, t_k\} \subseteq T$, the vector $(X_{t_1}, \dots, X_{t_k})$ has a Gaussian distribution with covariance matrix $(\Gamma(t_i, t_j))_{i,j=1}^k$.

On the other hand, one may ask if every $\Gamma : T\times T \to \mathbb{R}$ is a covariance of a Gaussian process. Obviously, Γ needs to be symmetric, $\Gamma(t,s) = \Gamma(s,t)$, and *positivly definite* in the following sense: For any $k \geq 0$, $\{t_1, \dots, t_k\} \subseteq T$ and $\xi \in \mathbb{R}^k$

$$\sum_{i,j=1}^{k} \xi_i \Gamma(t_i, t_j) \xi_j = \mathbb{E}\left[\left(\sum_{i=1}^{k} \xi_i X_{t_i}\right)^2\right] \geq 0. \tag{2.3}$$

It turns out that those two conditions are sufficient:

Theorem 2.15. *If $\Gamma : T \times T \to \mathbb{R}$ is symmetric and positive definite (as in (2.3)), then there is a Gaussian process on T with covariance Γ.*

Proof. The proof is an easy application of Kolmogorov's extension theorem. We construct a probability $\mathbb{P}$ on the measurable space $(\mathbb{R}^T, \mathcal{B}(\mathbb{R})^{\otimes T})$ such that the canonical coordinate process $X_t(\omega) = \omega_t$ is a Gaussian process with covariance Γ. To this end, let for every $\{t_1, \dots, t_k\} \subseteq T$, $\mathbb{P}_{\{t_1,\dots,t_k\}}$ be the law of a Gaussian vector on $\mathbb{R}^{\{t_1,\dots,t_k\}} \cong \mathbb{R}^k$ with covariance $(\Gamma(t_i, t_j))_{i,j=1}^k$ existing by Theorem 2.9.

It is easy to check that $\mathbb{P}_{\{t_1,\dots,t_k\}}$ satisfy the assumptions of Kolmogorov's extension theorem, which yields the existence. □

It is, however, rather difficult to check that Γ is positive definite. We give two important examples when it is possible:

Example 2.16. Let E be at most countable and $(Z_t)_{t\geq 0}$ be a Markov chain on E with transition probabilities $\mathbb{P}(Z_t = y \mid Z_t = x) = p_t(x,y)$ such that $p_t(x,y) = p_t(y,x)$. Consider its Green function

$$g(x,y) = \mathbb{E}_x\left[\int_0^{+\infty} \mathbf{1}_{\{Z_t = y\}}\,\mathrm{d}t\right] = \int_0^{+\infty} p_t(x,y)\,\mathrm{d}t < +\infty.$$

Then for every $\xi \in \mathbb{R}^E$ with finite support

$$\begin{aligned}\sum_{x,y\in E} \xi_x g(x,y)\xi_y &= \sum_{x,y\in E}\int_0^{+\infty} \xi_x p_t(x,y)\xi_y\,\mathrm{d}t = \sum_{x,y,z\in E}\int_0^{+\infty} \xi_x p_{t/z}(x,y)\xi_y\,\mathrm{d}t \\ &= \int_0^{+\infty}\sum_{z\in E}\left(\sum_{y\in E} p_{t/z}(y,z)\xi_z\right)^2 \mathrm{d}t > 0\end{aligned}$$

(In fact, E does not need to be countable here). If Z is a Brownian Motion or random walk, the associated Gaussian process is sometimes called a *Gaussian free field.* ♦

Example 2.17. Let $T = \mathbb{R}$ and let μ be a finite symmetric measure i.e., $\mu(A) = \mu(-A)$ on $\mathbb{R}$. Set

$$\Gamma(s,t) = \int e^{i\lambda(s-t)} \, d\mu(\lambda). \tag{2.4}$$

Then for $\xi \in \mathbb{R}^{\mathbb{R}}$ with finite support, i.e., $\{x \in \mathbb{R} \mid \xi_x \neq 0\}$ is finite,

$$\sum_{\mathbb{R}\times\mathbb{R}} \xi_s \Gamma(s,t) \xi_t = \int \left| \sum_T \xi_s e^{i\lambda s} \right|^2 d\mu(\lambda) \geq 0$$

so by Theorem 2.15, there is a corresponding Gaussian process.

Moreover, as $\Gamma(s,t) = \Gamma(s-t)$, $(X_t)_t$ is stationary, i.e.,

$$\operatorname{law}(X_{t_1+t}, \dots, X_{t_k+t}) = \operatorname{law}(X_{t_1}, \dots, X_{t_k})$$

for any k finite and $t_1, \dots, t_k \in \mathbb{R}$.

The converse is also true. ♦

Theorem 2.18 (Bochner). *If $(X_t)_{t\in\mathbb{R}}$ is a stationary Gaussian process then there is a symmetric finite measure μ such that (2.4) holds.*

Proof. See [6, 5], Theorem 24.9. □

2.3.2. Orthogonality and Independence

Similarly as in 2.8 this is a connection between orthogonality and independence:

Theorem 2.19. *Let H be a Gaussian space and $(H_i)_{i\in I}$ its subspaces. Then H_i, $i \in I$ are independent if and only if the σ-algebras $\sigma(H_i)$, $i \in I$ are independent.*

Remark 2.20. It is crucial that the H_i's are subspaces of the *same* Gaussian space. To see this, let $X \sim \mathcal{N}(0,1)$ and ε be an independent Bernoulli random variable, meaning $\mathbb{P}(\varepsilon = \pm 1) = \frac{1}{2}$. Set $Y = \varepsilon X$. Then

$$\langle X, Y \rangle_{L^2} = \mathbb{E}[XY] = \mathbb{E}[\varepsilon]\,\mathbb{E}[X^2] = 0,$$

so X and Y are orthogonal in L^2. However it holds that $|X| = |Y|$ so the cannot be independent. Of course, here (X, Y) is *not* a Gaussian vector. ♦

Proof of Theorem 2.19. " $\Longleftarrow$ ": If $\sigma(H_i)$, $i \in I$ are independent σ-algebras, then for $X_i \in H_i$ and $i \neq j$ we obtain

$$\mathbb{E}[X_i X_j] = \mathbb{E}[X_i]\,\mathbb{E}[X_j] = 0,$$

i.e., the H_i's are orthogonal.

" $\Longrightarrow$ ": We only sketch this direction, the reader may fill in the details as an exercise.

Step 1: It is sufficient to check that for every finite $\{i_1, \dots, i_p\} \subseteq I$ and random variables $\xi_1^k, \dots, \xi_{j_k} \in H_{i_k}$, $1 \leq k \leq p$ the vectors $(\xi_1^k, \dots, \xi_{j_k}^k)$ for $1 \leq k \leq p$ are independent (use Dynkin's lemma here).

Step 2: For any $k \leq p$ find an orthonormal basis $\zeta_1^k, \dots, \zeta_{j_k}^k$ of $\operatorname{span}(\xi_1^k, \dots, \xi_{j_k}^k)$. Then the covariance matrix of

$$(\zeta_1^1, \dots, \zeta_{j_1}^1, \dots, \zeta_1^p, \dots, \zeta_{j_p}^p)$$

is the identity matrix,so they are independent by Exercise 2.8. This implies the independence required in the first step. $\square$

Corollary 2.21. *Let H be a Gaussian space and K its closed subspace. If p_K denotes the orthogonal projection on K and $X \in H$, then*

(i) $\mathbb{E}\left[X \mid \sigma(K)\right] = p_K(X)$

(ii) Let $\sigma^2 = \mathbb{E}[(X - p_K(X))^2]$. Then for any $\Gamma \in \mathcal{B}(\mathbb{R})$ it holds that

$$\mathbb{P}\left(X \in \Gamma \mid \sigma(K)\right)(\omega) = Q(\omega, \Gamma)$$

where $Q(\omega, \cdot)$ is the distribution $\mathcal{N}(p_K(X)(\omega), \sigma^2)$.

Remark 2.22. Recall that for $X \in L^2$, $\mathbb{E}\left[X \mid \sigma(K)\right] = p_{L^2(\Omega, \sigma(K), \mathbb{P})}$ is also an orthogonal projection, but on a larger space. In the Gaussian setting, we may project to a "smaller" Gaussian space. ♦

Proof of Corollary 2.21. *(i)*: Let $Y = X - p_K(X)$. Then Y is orthogonal to X and thus independent of X by Theorem 2.19, hence

$$\mathbb{E}\left[X \mid \sigma(K)\right] = \mathbb{E}\left[p_K(X) \mid \sigma(K)\right] + \mathbb{E}\left[Y \mid \sigma(K)\right] = p_K(X) + \mathbb{E}[Y] = p_K(X).$$

(ii): By construction $Y \sim \mathcal{N}(0,\sigma^2)$. Moreover, for $f : \mathbb{R} \to \mathbb{R}_+$ measurable,

$$\mathbb{E}\left[f(X)\,|\,\sigma(K)\right] = \mathbb{E}\left[f(p_K(X)+Y)\,|\,\sigma(K)\right] = \int f(p_K(X)+y)\mathbb{P}_Y(\mathrm{d}y),$$

which implies the claim (Here we use: if Z is $\mathcal{G}$ measurable, Y independent of $\mathcal{G}$, then $\mathbb{E}\left[g(Y,Z)\,|\,\mathcal{G}\right] = \int g(y,Z)\mathbb{P}_Y(\mathrm{d}y)$). $\square$

Exercise 2.23. Let $X = (X_1,\dots,X_d)$ be a Gaussian vector such that $\mathbb{E}[X_i] = m_i$ and $\mathrm{Cov}(X_i,X_j) = \Sigma_{i,j}$. Let $\mathcal{G} = \sigma(X_{k+1},\dots,X_d)$ for some $k \in \{1,\dots,d-1\}$. Then

$$\mathbb{P}\left((X_1,\dots,X_k) \in \Gamma\,|\,\mathcal{G}\right)(\omega) = Q(\Gamma,\omega), \qquad \Gamma \in \mathcal{B}(\mathbb{R}^k)$$

and $Q = \mathcal{N}(\tilde m, \tilde\Sigma)$ with $\tilde m = m_A + \Sigma_{A,B}\Sigma_{B,B}^{-1}(X_B - m_B)$ and $\tilde\Sigma = \Sigma_{A,A} - \Sigma_{A,B}\Sigma_{B,B}^{-1}\Sigma_{B,A}$ where

$$m = (\underbrace{m_1 \quad \dots \quad m_k}_{=m_A} \quad \underbrace{m_{k+1} \quad \dots \quad m_d}_{=m_B})^\top, \quad \text{and} \quad \Sigma = \left(\begin{array}{c|c} \Sigma_{A,A} & \Sigma_{A,B} \\ \hline \Sigma_{B,A} & \Sigma_{B,B} \end{array}\right) \begin{array}{l} \}k \text{ lines} \\ \}d-k \text{ lines} \end{array}$$

2.4. Gaussian measures

We have seen that checking if a function Γ is a positive definite quadratic form is hard, see for example Example 2.16 and Example 2.17. But, in the situation where T is a vector space with an inner product, $p(s,t) = \langle s,t\rangle_E$ is positive definite, e.g., Gramm matrix.

Definition 2.24. Let $(E,\mathcal{E},\mu)$ be a σ-finite measure space. A *Gaussian measure with intensity* μ is an isometry G of $L^2(E,\mathcal{E},\mu)$ with a Gaussian space.
That is for $f,g \in L^2(E,\mathcal{E},\mu)$, $G(f)$ and $G(g)$ are centred Gaussian random variables with

$$\mathbb{E}[G(f)G(g)] = \langle f,g\rangle_{L^2(E,\mathcal{E},\mu)} \quad \text{and} \quad \mathbb{E}[G(f)^2] = \|G(f)\|^2_{L^2(\mathbb{P})} = \|f\|^2_{L^2(E,\mathcal{E},\mu)}.$$

If $A \in \mathcal{E}$, $\mu(A) < +\infty$ then $G(A) := G(\mathbf{1}_A) \sim \mathcal{N}(0,\mu(A))$. Further if $A_i \in \Sigma$, $i = 1,\dots,k$ are disjoint with $\mu(A_i) < +\infty$ then the vector $(G(A_i))_{i=1}^k$ has diagonal covariance matrix since for $i \neq j$, $\mathbb{E}[G(A_i)G(A_j)] = \langle \mathbf{1}_{A_i},\mathbf{1}_{A_j}\rangle_{L^1(\mu)} = 0$. Hence, by Exercise 2.8, $G(A_i)$, $i = 1,\dots,k$ are independent. Let $A = \bigcup_{i=1}^{+\infty} A_i$ for A_i disjoint, i.e., $\mathbf{1}_A = \sum_{i=1}^{+\infty}\mathbf{1}_{A_i}$ and assume $\mu(A) < +\infty$. By isometry then $G(A) = \sum_{i=1}^{+\infty} G(A_i)$ in $L^2(\mathbb{P})$ and also $\mathbb{P}$-a.s.. Hence the properties of $A \mapsto G(A)$ are similar to those of a signed measure (dependent on ω). In general, however, $A \mapsto G(A)(\omega)$ is not a measure for a fixed $\omega \in \Omega$.

Proposition 2.25 (Existence). *Let $(E, \mathcal{E}, \mu)$ be a σ-finite measure space. Then there exists a Gaussian measure with intensity μ.*

Proof. Let $(f_i)_{i\in I}$ be a total orthonormal system in $L^2(E, \mathcal{E}, \mu)$. Hence every $f \in L^2(\mu)$ can be written as

$$f = \sum_{i\in I} \alpha_i f_i \quad \text{with } \alpha_i = \langle f, f_i\rangle_{L^2} \text{ and } \sum_{i\in I} \alpha_i^2 = \|f\|_{L^2}^2 < +\infty.$$

We now consider a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ with an i.i.d. collection $(X_i)_{i\in I}$ of standard normal random variables. This means also $X_i \perp X_j$ in $L^2(\mathbb{P})$ for $i \neq j$. We define $G(f) := \sum_{i\in I} \alpha_i X_i$. This series then converges in $L^2(\mathbb{P})$, i.e., G is an element of the Gaussian space generated by $(X_i)_{i\in I}$.

The isometry property is obvious. □

Remark 2.26. When $L^2(E, \mathcal{E}, \mu)$ is separable (e.g. when $E = \mathbb{R}$, μ =Lebesgue), the total orthonormal system $(f_i)_{i\in I}$ of the last proof is actually a countable orthonormal basis of L^2. ♦

Proposition 2.27 (Quadratic variation of Gaussian measures). *Let G be a Gaussian measure on $(E, \mathcal{E}, \mu)$ and $A \in \mathcal{E}$ with $\mu(A) < +\infty$. Assume that there is a sequence $(A_1^n, \dots, A_{k_n}^n)_{n\in\mathbb{N}}$ of partitions of A i.e., $A = A_1^n \dot\cup \dots \dot\cup A_{k_n}^n$ with "step" tending to 0, that is*

$$\lim_{n\to+\infty} \left(\sup_{1\le j\le k_n} \mu(A_j^n) \right) = 0.$$

Then

$$\lim_{n\to+\infty} \sum_{j=1}^{k_n} G(A_j^n)^2 = \mu(A) \quad \text{in } L^2(\mathbb{P}).$$

Proof. For $n \in \mathbb{N}$ fixed, the random variables $(G(A_j^n))_{j=1}^{k_n}$ are independent, and $\mathbb{E}[G(A_j^n)^2] = \mu(A_j^n)$. Hence,

$$\mathbb{E}\left[\left(\sum_{j=1}^{k_n} G(A_j^n)^2 - \mu(A) \right)^2 \right] = \sum_{j=1}^{k_n} \operatorname{Var}\left(G(A_j^n)^2 \right) = 2 \sum_{j=1}^{k_n} \mu(A_j^n)^2,$$

since for $X \sim \mathcal{N}(0, \sigma^2)$, $\mathrm{Var}(X^2) = \mathbb{E}[X^4] - \sigma^4 \overset{\text{Exercise 2.1}}{=} 3\sigma^4 - \sigma^4 = 2\sigma^4$. Finally

$$2 \sum_{j=1}^{k_n} \mu(A_j^n)^2 \leq 2 \underbrace{\sup_{1 \leq j \leq k_n} \mu(A_j^n)}_{\to 0} \underbrace{\sum_{j=1}^{k_n} \mu(A_j^n)}_{=\mu(A) < +\infty} \xrightarrow{n \to +\infty} 0,$$

proving the L^2 convergence. □

3. Brownian Motion

3.1. Pre-Brownian Motion

Definition 3.1. Let G be a Gaussian measure on $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$ whose intensity is the Lebesgue measure. Then the process $(B_t)_{t\in\mathbb{R}_+}$ given by

$$B_t = G(\mathbf{1}_{[0,t]}) \tag{3.1}$$

is called *pre-Brownian motion.*

Proposition 3.2. *The process $(B_t)_{t\geq 0}$, given by (3.1), is a Gaussian process whose covariance $K(s,t) := \mathrm{Cov}(X_s, X_t)$ satisfies*

$$K(s,t) = \min\{s,t\} := s \wedge t.$$

Proof. By definition B_t, $t \in \mathbb{R}_+$ are elements of the same Gaussian space. That is $(B_t)_{t\geq 0}$ is a Gaussian process (Definition 2.24). Moreover,

$$\mathrm{Cov}(B_s, B_t) = \mathbb{E}[B_s B_t] = \mathbb{E}[G(\mathbf{1}_{[0,s]})G(\mathbf{1}_{[0,t]})] = \int_{\mathbb{R}_+} \mathbf{1}_{[0,s]}(r)\mathbf{1}_{[0,t]}(r)\,\mathrm{d}r = s \wedge t. \square$$

We now give several characterisations of pre-Brownian Motion:

Proposition 3.3. *A process $X = (X_t)_{t\geq 0}$ is a pre-Brownian motion if and only if*

(i) $X_0 = 0$ *a.s.*

(ii) *For every $0 \leq s < t$ the random variable $X_t - X_s$ is independent of $\sigma(X_r)_{r\leq s}$ and has $\mathcal{N}(0, t-s)$-distribution.*

Proof. $\Longrightarrow$: If X_t is a pre-Brownian motion, then $X_0 = G(\mathbf{1}_{[0,0]})$ so $X_0 = 0$ almost surely. To show (ii), let H_s be the Gaussian space generated by X_r, $r \leq s$, and $\tilde{H}_s$

the Gaussian space generated by $(X_{s+k} - X_s; k > 0)$. Then S_s and $\tilde{H}_s$ are orthogonal, since $\mathbb{E}[X_r(X_{s+n} - X_s)] = \mathbb{E}[G(\mathbf{1}_{[0,r]})G(\mathbf{1}_{[s,s+n]})] = 0$. Moreover, H_s and $\tilde{H}_s$ are parts of the same Gaussian space so they are independent by Theorem 2.19. In particular $X_t - X_s$ is independent of $\sigma(H_s) = \sigma(X_r)_{r\leq s}$. Finally, $X_t - X_s = G(\mathbf{1}_{(s,t]}) \sim \mathcal{N}(0, t-s)$.

$\Longleftarrow$: Let $0 = t_0 < t_1 < \ldots < t_n$. By (ii), $X_{t_i} - X_{t_{i-1}}$, $i \leq n$ are independent and $X_{t_i} - X_{t_{i-1}} \sim \mathcal{N}(0, t_i - t_{i-1})$. It follows that X is a Gaussian process. Further, for $f = \sum_{i=1}^n \lambda_i \mathbf{1}_{(t_{i-1},t_i]}$, $\lambda_i \in \mathbb{R}$, we set

$$G(f) = \sum_{i=1}^n \lambda_i \left(X_{t_i} - X_{t_{i-1}} \right).$$

If g is another such step-function we use from (ii) that

$$\mathbb{E}[G(f)G(g)] = \int_{\mathbb{R}_+} f(r)g(r)\,\mathrm{d}r.$$

Using the density of step-functions in $L^2(\mathbb{R}_+, \mathrm{d}x)$, we see that $f \mapsto G(f)$ can be extended to an isometry of $L^2(\mathbb{R}_+, \mathrm{d}x)$ and the Gaussian space is generated by X. Moreover $G([0,t]) = X_t - X_0 = X_t$ by (i), i.e., X is a pre-Brownian motion. □

Proposition 3.4. *A process $X = (X_t)_{t\geq 0}$ is a pre-Brownian Motion if and only if $X_0 = 0$ a.s. and for any choice $0 = t_0 < t_1 < \ldots < t_n$, the random variables $X_{t_i} - X_{t_{i-1}}$, $i \leq n$ are independent and $X_{t_i} - X_{t_{i-1}} \sim \mathcal{N}(0, t_i - t_{i-1})$.*
In particular the vector $(X_{t_i})_{i=1}^n$ has density

$$\frac{1}{\sqrt{(2\pi)^n \prod_{i=1}^n (t_i - t_{i-1})}} \exp\left(-\frac{1}{2} \sum_{i=1}^n \frac{(y_i - y_{i-1})^2}{t_i - t_{i-1}} \right), \qquad y_0 = 0.$$

Exercise 3.5. Show Proposition 3.4.

3.2. Invariances of pre-Brownian Motion

Proposition 3.6. *Let $(B_t)_{t\geq 0}$ be a pre-Brownian Motion. Then the following statements hold:*

(1) $-B$ is a pre-Brownian motion.

(2) *For any* $\lambda > 0$, $B_t^\lambda := \frac{1}{\lambda} B_{\lambda^2 t}$ *is a pre-Brownian Motion (This property is called* scaling invariance*).*

(3) $\beta_t := tB_{1/t}$ *for* $t > 0$ *and* $\beta_0 := 0$ *is a pre-Brownian Motion (time -inversion invariance).*

(4) *For any* $s \geq 0$, $B_t^{(s)} := B_{t+s} - B_t$, $t \geq 0$ *is a pre-Brownian Motion (shift invariance, simple Markov property).*

Exercise 3.7. Prove the converse to Proposition 3.2 i.e., if $X = (X_t)_{t\geq 0}$ is a Gaussian process with covariance $\mathbb{E}[X_s X_t] = s \wedge t$, then X is a pre-Brownian Motion. Then prove Proposition 3.6.

Notation 3.8. Let $B = (B_t)_{t\geq 0}$ be a pre-Brownian Motion and G the associated Gaussian measure. For $f \in L^2(\mathbb{R}_+, \mathrm{d}x)$ we set

$$G(f) =: \int_0^{+\infty} f(s)\,\mathrm{d}B_s \sim \mathcal{N}(0, \|f\|_{L^2}^2) \quad \text{and} \quad G(f\mathbf{1}_{[0,t]}) =: \int_0^t f(s)\,\mathrm{d}B_s.$$

This is justified by

$$\int_u^v \mathrm{d}B_s = G((u,v]) = B_v - B_u, \quad 0 \leq u < v < +\infty.$$

The map $f \mapsto \int_0^{+\infty} f(s)\,\mathrm{d}B_s$ is called *Wiener integral* with pre-Brownian motion B. Since the Gaussian measure is not a true measure for a a given $\omega \in \Omega$, the Wiener integral is not a true integral. We will, in this lecture, extend the definition to f which may depend on ω. We will generalize this notion of the integral later in the lecture where we will be able to replace B_t by a semi-martingale X_t.

3.3. Continuity of trajectories

Let $B = (B_t)_{t\geq 0}$ be a pre-Brownian Motion. The map $t \mapsto B_t(\omega)$ for a fixed $\omega \in \Omega$ is called the *trajectory* of the pre-Brownian Motion. At this point we cannot say almost anything about it. Is is even not clear (or even not true in general) that $t \mapsto B_t(\omega)$ is measurable (say for $\mathbb{P}$-a.e. $\omega \in \Omega$). We are now going to modify B such that its trajectories are continuous.

Definition 3.9. Let $(X_t)_{t\in T}$, $(Y_t)_{t\in T}$ be two stochastic processes with the same indices in T. Y is called *modification* of X if

$$\mathbb{P}(X_t = Y_t) = 1 \quad \text{for any } t \in T.$$

Remark 3.10. If Y is a modification of X, then they have the same law on $(\mathbb{R}^T, \mathcal{B}(\mathbb{R})^{\otimes T})$, since $\mathcal{B}(\mathbb{R})^{\otimes T}$ is generated by finite dimensional cylinders events and the vectors $(X_{t_i})_{i=1}^n$ and $(Y_{t_i})_{i=1}^n$ have the same law for any $\{t_1, \dots, t_n\} \subset T$. In particular if X is a pre-Brownian motion, then Y is a pre-Brownian motion. ♦

Remark 3.11. The processes X and Y may be rather different. X for example might be continuous a.s. and Y a.s. discontinuous everywhere. ♦

Exercise 3.12. Find an example illustrating Remark 3.11 with $T = \mathbb{R}_+$.

Definition 3.13. Two stochastic processes $X = (X_t)_{t\in T}$ and $Y = (Y_t)_{t\in T}$ are called *indistinguishable* if there exists $\hat{\Omega} \in \mathcal{A}$ with

$$\hat{\Omega} \subseteq \{X_s = Y_s \, \forall s \in T\}$$

satisfying $\mathbb{P}(\hat{\Omega}) = 1$. Or if $\mathcal{A}$ is complete with respect to $\mathbb{P}$ we can simply require

$$\mathbb{P}(X_s = Y_s \, \forall s \in T) = 1.$$

Obviously if X and Y are indistinguishable, then Y is a modification of X. The other implication is false in general.

Lemma 3.14. *If T is separable and X and Y have a.s. continuous trajectories, then: X and Y are indistinguishable if and only if X is a modification of Y.*

Proof. $\Longrightarrow$: Is obvious.

$\Longleftarrow$: Let D be a countable dense set in T. Then $\mathbb{P}(X_s = Y_s \forall s \in D) = 1$ (because the condition is a countable union) and thus $\mathbb{P}(X_s = Y_s \forall s \in T) = 1$ by continuity. □

We now introduce the important result which ensures the existence of continuous modifications:

Theorem 3.15 (Kolmorov's continuity theorem). *Let $X = (X_t)_{t\in I}$ be a stochastic process, indexed by a bounded interval $I \subset \mathbb{R}$ taking values in a complete metric space (S, d).*
Assume that there exist some fixed $q, \varepsilon > 0$ such that

$$\mathbb{E}\left[d(X_s, X_t)^q\right] \lesssim |t-s|^{1+\varepsilon} \quad \text{for all } t, s \in I. \tag{3.2}$$

Then there exists a modification $\tilde{X} = (\tilde{X}_t)_{t\in I}$ of X which is α-Hölder continuous for any $\alpha \in (0, \frac{\varepsilon}{q})$, that is

$$d\big(\tilde{X}_s(\omega), \tilde{X}_t(\omega)\big) \lesssim_{\alpha,\omega} |t-s|^{\alpha} \quad \text{for any } t, s \in I \text{ and for all } \omega \in \Omega.$$

Proof. Without loss of generality we assume that $I = [0, 1]$ and denote by D the sets all dyadic rationals in $[0, 1)$,

$$D = \left\{ \sum_{i=1}^{p} \varepsilon_i 2^{-i} \,\middle|\, p \in \mathbb{N}, \varepsilon_i \in \{0, 1\} \right\}.$$

Observe also, that it is sufficient to show the result for a $\alpha \in (0, \frac{\varepsilon}{q})$ fixed. Indeed, we may then take a sequence $\alpha_k \nearrow \frac{\varepsilon}{q}$ and observe that the modification obtained for those α_k are indistinguishable by Lemma 3.14.

Step 1: X is a.s. α-Hölder on D. This means we need to show that

$$d\big(X_s(\omega), X_t(\omega)\big) \lesssim_{\alpha,\omega} |t-s|^{\alpha} \quad \text{for any } t, s \in D \text{ and for } \mathbb{P}\text{-a.e. } \omega \in \Omega. \tag{3.3}$$

To see this, observe that the assumption of the theorem (3.2) implies for $a > 0$, $s, t \in I$,

$$\mathbb{P}\big(d(X_s, X_t) \geq a\big) \leq \frac{1}{a^q}\,\mathbb{E}[d(X_s, X_t)^q] \lesssim \frac{1}{a^q}|t-s|^{1+\varepsilon}. \tag{3.4}$$

Taking $s = \frac{i-1}{2^n}$ and $t = \frac{i}{2^n}$ with $i \in \{1, \dots, 2^n\}$ and $a = \frac{1}{2^{n\alpha}}$ we obtain, using the above formula,

$$\mathbb{P}\left(d\left(X_{\frac{i-1}{2^n}}, X_{\frac{i}{2^n}}\right) \geq \frac{1}{2^{n\alpha}}\right) \lesssim 2^{n\alpha q-(1+\varepsilon)n}$$

hence,

$$\mathbb{P}\left(\bigcup_{i=1}^{2^n}\left\{d\left(X_{\frac{i-1}{2^n}}, X_{\frac{i}{2^n}}\right) \geq \frac{1}{2^{n\alpha}}\right\}\right) \lesssim 2^{(\alpha q-\varepsilon)n}.$$

For $\alpha \in (0, \frac{\varepsilon}{q})$ this is summable in $n \geq 0$ so by the Borel-Cantelli lemma we find that for $\mathbb{P}$-a.e. $\omega \in \Omega$ there exists $n_0(\omega)$ such that for any $n \geq n_0(\omega)$ and $i \in \{1, \dots, 2^n\}$, $d(X_{(i-1)2^{-n}}, X_{i2^{-n}}) \leq 2^{-n\alpha}$, that is

$$K_\alpha(\omega) := \sup_{n\in\mathbb{N}} \sup_{i=1,\dots,2^n} \left(\frac{1}{2^{n\alpha}} d\left(X_{\frac{i-1}{2^n}}, X_{\frac{i}{2^n}}\right)\right) < +\infty, \quad \text{for } \mathbb{P}\text{-a.e. } \omega \in \Omega.$$

We now show that (3.3) holds. To this end take $s, t \in D$ such that $s < t$. Let p be the smallest integer such that $2^{-p} < t - s$. Then for some $k, \ell, m \in \mathbb{N}$, we write s, t as

$$\begin{aligned} s &= k2^{-p} - \varepsilon_{p+1}2^{p-1} - \dots - \varepsilon_{p+\ell}2^{-p-\ell} \\ t &= k2^{-p} + \varepsilon'_p 2^{-p} + \dots + \varepsilon'_{p+m}2^{-p-m} \end{aligned} \qquad \varepsilon_i, \varepsilon'_j \in \{0, 1\}$$

We define

$$\begin{aligned} s_i &= k2^{-p} - \varepsilon_{p+1}2^{-p-1} - \dots - \varepsilon_{p+i}2^{-p-i}, \quad && 0 \leq i \leq \ell, \\ t_j &= k2^{-p} + \varepsilon'_p 2^{-p} + \dots + \varepsilon_{p+j}2^{-p-j}, && 0 \leq j \leq m \end{aligned}$$

Then, by triangle inequality, for $\mathbb{P}$-a.e. $\omega \in \Omega$,

$$\begin{aligned} d\left(X_s(\omega), X_t(\omega)\right) &= d\left(X_{s_\ell}(\omega), X_{t_m}(\omega)\right) \\ &\leq \left(d\left(X_{s_0}, X_{t_0}\right) + \sum_{i=1}^{\ell} d\left(X_{s_{i-1}}, X_{s_i}\right) + \sum_{j=1}^{m} d\left(X_{t_{j-1}}, X_{t_j}\right)\right)(\omega) \\ &\leq K_\alpha(\omega) \left(2^{p\alpha} + \sum_{i=1}^{\ell} 2^{-(p+i)\alpha} + \sum_{j=1}^{m} 2^{-(p+j)\alpha} \right) \\ &\leq 2K_\alpha(\omega) \cdot \frac{2^{-p\alpha}}{1 - 2^{-\alpha}} \leq 2K_\alpha(\omega) \cdot \frac{|t-s|^\alpha}{1 - 2^{-\alpha}}, \end{aligned}$$

where we used $|t - s| \geq 2^{-p}$ in the last inequality. This shows (3.3).

Step 2: Show the theorem. By Step 1, $t \mapsto X_t(\omega)$ is a.s. α-Hölder on D, so it is a.s. uniformely continuous on D. Since (S, d) is complete, there is a.s. a unique continuous extension of this trajectory to I, namely

$$\tilde{X}_t = \lim_{\substack{s \to t \\ s \in D}} X_s, \qquad t \in [0, 1], \text{ on } \{K_\alpha < +\infty\}.$$

This extension is also α-Hölder by density of $D \subseteq I$. If $K_\alpha(\omega) = +\infty$ we set $\tilde{X}_t(\omega) = x_0$ for all $t \geq 0$ and some fixed $x_0 \in S$. By construction $t \mapsto \tilde{X}_t(\omega)$ is continuous for any $\omega \in \Omega$.

To see that $\tilde{X}$ is a modification of X, observe that by (3.4),

$$X_s \xrightarrow[s\to t]{\mathbb{P}} X_t \quad \text{for all } t \in [0,1].$$

By construction we have $\tilde{X}_t = \lim_{D\ni s\to t} X_s$ a.e., which then implies that $X_t = \tilde{X}_t$ a.s. (using that convergence a.e. implies convergence in measure on finite measure spaces and the uniqueness of the limit). This concludes the proof. □

Remark 3.16. When the intervall I is not bounded, e.g., $I = \mathbb{R}_+$ on can apply the theorem on $[0,1], [1,2], \dots$ to find a modification that is *locally* α-Hölder continuous for any $\alpha \in (0, \frac{\varepsilon}{q})$. ♦

The following is an important corollary to Kolmogorov's theorem.

Corollary 3.17. *Let B be a pre-Brownian Motion. Then B has a modification which is locally α-Hölder (i.e., also continuous) for every $\alpha \in (0, \frac{1}{2})$.*

Proof. For $s < t$, $B_t - B_s \sim \mathcal{N}(0, t-s)$,that is $B_t - B_s \overset{\text{law}}{=} \sqrt{t-s} \cdot N$, where $N \sim \mathcal{N}(0,1)$. Hence, for any $q > 0$,

$$\mathbb{E}\left[|B_t - B_s|^q\right] = (t-s)^{q/2}\,\mathbb{E}[|N|^q] = C_q(t-s)^{q/2}$$

If $q > 2$, we may apply the theorem with $\varepsilon = \frac{q}{2} - 1$ to find a modification which is α-Hölder for $\alpha < \frac{q-2}{2q}$. Taking $q \nearrow +\infty$ proves the claim. □

3.4. Brownian Motion

We now introduce one of the central objects of this lecture.

Definition 3.18. A process $B = (B_t)_{t\geq 0}$ is a *Brownian Motion* if

(1) B is a pre-Brownian Motion,

(2) The trajectories of B i.e., the maps $t \mapsto B_t(\omega)$ are continuous for any $\omega \in \Omega$.

The existence of Brownian Motion follows from Corollary 3.17. Indeed, any modification of a pre-Brownian Motion is again a pre-Brownian Motion.

Remark 3.19. Proposition 3.6 remains true if replace all instances of pre-Brownian motion by Brownian motion. Indeed, the continuity of the trajectories is (almost) immediate with the exception at 0 (see the next exercise). ♦

Exercise 3.20. Show that $\beta_t = tB_{1/t}$ is continuous at 0.

Remark 3.21. In Proposition 3.3 and Proposition 3.4 one should, in addition, require the continuity of the trajectories. E.g., Proposition 3.3 then becomes: X is a Brownian Motion if and only if $X_0 = 0$, $t \mapsto X_t(\omega)$ is continuous for every $\omega \in \Omega$ and for all $0 < s \leq t$, $X_t - X_s \sim \mathcal{N}(0, t-s)$ and $X_t - X_s$ is independent of $\sigma(X_r, r \leq s)$. ♦

Wiener measure Let $C = C^0(\mathbb{R}_+, \mathbb{R})$ and $\mathcal{F}$ the smallest σ-field making the canonical coordinates $X_t : C \to \mathbb{R}$, $\omega \mapsto X_t(\omega) = \omega(t)$ measurable (This σ-field coincides with the Borel-σ-field of the topology of uniform convergence on compacts). Let $B = (B_t)_{t\geq 0}$ be a Brownian motion on some probability space $(\Omega, \mathcal{A}, \mathbb{P})$. B induces a measurable map

$$\Omega \to C^0(\mathbb{R}_+, \mathbb{R}), \qquad \omega \mapsto [t \mapsto B_t(\omega)].$$

The *Wiener measure* is the image of $\mathbb{P}$ by this map, $W := B_\#\mathbb{P}$, that is, W is the law of Brownian motion on C.

By Theorem 3.15, for $0 = t_0 < t_1 < \ldots < t_n$, $A_0, \ldots, A_n \in \mathcal{B}(\mathbb{R})$, $x_0 = 0$,

$$\begin{aligned} &W(\{\omega : \omega(t_i) \in A_i, i = 1, \ldots, n\}) \\ &= \mathbf{1}_{A_0}(0) \int\limits_{A_1\times\ldots\times A_n} \frac{\mathrm{d}y_1 \ldots \mathrm{d}y_n}{\sqrt{(2\pi)^n t_1 \cdot \prod_{i=2}^n (t_i - t_{i-1})}} \exp\left\{-\frac{1}{2}\sum_{i=1}^n \frac{(y_i - y_{i-1})^2}{t_i - t_{i-1}}\right\}. \end{aligned} \tag{3.5}$$

We now that finite-dimensional distributions characterize measures on $(C, \mathcal{C})$, that is (3.5) defines W uniquely. In particular the Wiener measure does not depend on which Brownian Motion we use for its construction.

If one takes the probability space $\Omega = C$, $\mathcal{A} = \mathcal{F}$ and $\mathbb{P} = W$ then the canonical process $X_t(\omega) = \omega(t)$ is a Brownian Motion. This is the *classical construction* of Brownian Motion.

d-dimensional Wiener measure The d-dimensional Wiener measure is constructed similar to the one-dimensional one. Let $C = C^0(\mathbb{R}_+, \mathbb{R}^d)$ and let $X_t : C \to \mathbb{R}^d$ be the canonical coordinates i.e., $X_t(\omega) = \omega(t)$. Taking $B^{(1)}, \ldots, B^{(d)}$, d copies of Brownian motion on $(\Omega, \mathcal{A}, \mathbb{P})$ and setting $W := B_\#\mathbb{P}$ for

$$\omega \overset{B}{\mapsto} [t \mapsto (B_t^{(1)}, \ldots, B_t^{(d)})].$$

Equation (3.5) remains valid as it is with $A_i \in \mathcal{B}(\mathbb{R}^d)$.

3.5. Lévy-Ciesielski construction of Brownian Motion*

One can construct a Brownian motion without Donsker's or Kolmogorov's theorem by taking a particular suitable basis in $L^2(\mathbb{R}_+, \mathrm{d}x)$ for the construction of the Gaussian measure corresponding to the pre-Brownian Motion:

We define

$$\varphi_\ell = \mathbf{1}_{[\ell,\ell+1]}, \quad \varphi_{n,k} := 2^{m/2}\left(\mathbf{1}_{[\frac{k}{2^m}, \frac{k+\frac{1}{2}}{2^m})} - \mathbf{1}_{[\frac{k+\frac{1}{2}}{2^m}, \frac{k+1}{2^m})}\right), \quad \ell, n, k \in \mathbb{N}.$$

be a complete orthonormal basis of $L^2(\mathbb{R}_+, \mathrm{d}x)$. For $f \in L^2(\mathbb{R}_+, \mathrm{d}x)$ let

$$G(f) := \sum_\ell \xi_\ell \langle f, \varphi_\ell\rangle + \sum_{m,k} \xi_{m,k}\langle f, \varphi_{m,k}\rangle$$

where $\xi_\ell, \xi_{n,k}$ are i.i.d $\mathcal{N}(0,1)$. G is a Gaussian measure and we set

$$B_t = G(\mathbf{1}_{[0,t]}) = \sum_\ell \xi_\ell \Phi_\ell(t) + \sum_{m,k} \xi_{m,k}\Phi_{m,k}(t)$$

where

$$\Phi_\ell(t) = \int_0^t \varphi_\ell(s)\,\mathrm{d}s, \quad \text{and} \quad \Phi_{m,k}(t) = \int_0^t \varphi_{m,k}(s)\,\mathrm{d}s.$$

$(B_t)_{t\geq 0}$ is a pre-Brownian Motion by definition. Moreover, if $n_0 \in \mathbb{N}$, $t \leq n_0$,

$$B_t^n := \sum_{\ell\leq n} \xi_\ell \Phi_\ell(t) + \sum_{m\leq n}\sum_{k\leq n_0 2^m} \xi_{m,k}\Phi_{m,k}(t),$$

then one can show that B_t^n converges uniformly on $[0, n_0]$ to B_t. Since B_t^n are continuous in t, this yields another proof of the existence of Brownian Motion.

3.6. (Quadratic) variation of Brownian Motion trajectories

Proposition 3.22. *Let $0 = t_0^n < \ldots < t_{k_n}^n = t$ be a sequence of sub-divsions of* $[0, t]$ *with step size tending to* 0*, i.e.,* $\sup_i t_i^n - t_{i-1}^n \xrightarrow{n\to+\infty} 0$*. Then*

$$\lim_{n\to+\infty} \sum_{i=1}^{k_n} (B_{t_i^n} - B_{t_{i-1}^n})^2 = t \quad \textit{in } L^2.$$

Proof. Recall that $B_{t_i^n} - B_{t_{i-1}^n} = G((t_{i-1}^n, t_i^n])$. The claim is then a consequence of Proposition 2.27. □

Corollary 3.23. *The trajectories $t \mapsto B_t$ have $\mathbb{P}$-a.s. infinite variation on every non-trivial interval.*

Proof. Without loss of generality we consider intervalls of the form $[0, t]$ with $t \in \mathbb{Q}$. Recall that the variation of $w \in C$ over $[0, t]$ is given by

$$\text{Variation}_t(w) = \sup_{\substack{0=t_0<\ldots<t_n=t\\ t_i\in\mathbb{Q}}} \sum_{i\le n} |w(t_i) - w(t_{i-1})|.$$

Assume that $\mathbb{P}(\text{Variation}_t(B) < +\infty) > 0$. Then

$$\sum_{i=1}^{k_n} (B_{t_i^n} - B_{t_{i-1}^n})^2 \le \underbrace{\sup_{i\le k_n} |B_{t_i^n} - B_{t_{i-1}^n}|}_{\to 0} \cdot \underbrace{\sum_{i=1}^{k_n} |B_{t_i^n} - B_{t_{i-1}^n}|}_{\le B_t(B)} \xrightarrow{n\to+\infty} 0$$

on $\{\text{Variation}_t(B) < +\infty\}$. This is a contradiction to Proposition 3.22. □

3.7. Simple Markov property of the Wiener measure

Let $C = C^0(\mathbb{R}_+, \mathbb{R}^d)$, $X_t : C \to \mathbb{R}^d$ be the canonical coordinates, $\mathcal{F} = \sigma(X_t)_{t\ge0}$ and W the Wiener measure. Let further $x \in \mathbb{R}^d$ and W_x be the law of a *Brownian Motion started from* x given by the image of W under the map $w \in C \mapsto w(\cdot) + x \in C$ and $\mathbb{E}_x$ the corresponding expectation. Then by (3.5),

$$\mathbb{E}_x[h(X_{t_0}, \ldots, X_{t_n})] = \mathbb{E}^W[h(X_{t_0} + x, \ldots, X_{t_n} + x)]$$

$$= \int_{\mathbb{R}^d} h(x, y_1, \dots, y_n) \prod_{i=1}^{n} \frac{\mathrm{d}y_i}{\sqrt{2\pi(t_i - t_{i-1})}} \exp\left\{-\frac{(y_i - y_{i-1})^2}{2(t_i - t_{i-1})}\right\}$$

for $0 = t_0 < \dots < t_n$, $h : \mathbb{R}^d \to \mathbb{R}$ bounded and measurable with $y_0 = x$. Let θ_s, $s \geq 0$, be shifts on C that is $\theta_s(w)(\cdot) = w(s + \cdot) \in C$. Finally let $\mathcal{F}_t = \sigma(X_s)_{s \leq t}$ and $\mathcal{F}_t^+ := \bigcap_{\varepsilon > 0} \mathcal{F}_{t+\varepsilon}$.

$\mathcal{F}_t$ describes the past before t and $\mathcal{F}_t^+$ peaks infinitesimally into the future.

Example 3.24. Consider the event "the trajectory leaves its static point immediately" which is given by

$$\bigcap_{n \in \mathbb{N}} \left(\bigcup_{r \in \mathbb{Q} \cap [0,n]} \{X_r \neq X_0\} \right).$$

This event is in $\mathcal{F}_0^+$ but not in $\mathcal{F}_0$. ♦

We now extend Proposition 3.6 (4).

Theorem 3.25 (Simple Markov Property). *Let $Y \in b\mathcal{F}$, $s \geq$ and $x \in \mathbb{R}^d$. Then*

(1) $\mathbb{E}_x\left[Y \circ \theta_s \,|\, \mathcal{F}_s^+\right] = \mathbb{E}_{X_s}[Y]$ W_x-a.s.

(2) Under W_x, $(X_{s+n} - X_s)_{n \geq 0}$ is a Brownian motion independent of $\mathcal{F}_s^+$.

Proof. **Step 1:** We leave the proof of the fact that $y \mapsto \mathbb{E}_y[Y]$ is Borel-measurable for any bounded and $\mathcal{F}$-measurable random variable Y as an exercise. This is necessary, otherwise $\mathbb{E}_{X_s}[Y]$ is not a random variable.

Step 2: From Proposition 3.6, (4) and the definition of W_x we deduce that for any $A \in \mathcal{F}_s$, $g : \mathbb{R}^{(n+1)d} \to \mathbb{R}$ bounded and Borel-measurable and $s \geq 0$, $0 = t_0 < \dots < t_n$,

$$\mathbb{E}_x[\mathbf{1}_A g(X_{s+t_0}, \dots, X_{s+t_n})] = \mathbb{E}_x[\mathbf{1}_A \, \mathbb{E}_{X_s}[g(X_{t_0}, \dots, X_{t_n})]]. \tag{3.6}$$

Step 3: We claim that for any $A \in \mathcal{F}_s^+$, $Y \in b\mathcal{F}$, $s \geq 0$

$$\mathbb{E}_x[\mathbf{1}_A Y \circ \theta_s] = \mathbb{E}_x[\mathbf{1}_A \, \mathbb{E}_{X_s}[Y]] \tag{3.7}$$

Indeed, let $g \in bC(\mathbb{R}^{(n+1)d})$. Then by the definition of W_x and by the dominated convergence theorem,

$$y \in \mathbb{R} \to \mathbb{E}_y[g(X_{t_0}), \dots, X_{t_n}] \in bC(\mathbb{R}).$$

Using step 2 with $s+\varepsilon$ in the place of s, $A \in \mathcal{F}_s^+ \subseteq \mathcal{F}_{s+\varepsilon}$. Letting $\varepsilon \searrow 0$ and by using dominated convergence once again, we see that (3.6) holds also for $g \in bC(\mathbb{R}^{d(n+1)})$, $s \geq 0$ and $A \in \mathcal{F}_s^+$ (here we also use that Brownian Motion has continuous trajectories (!), i.e., $X_{s+\varepsilon} \to X_s$ as $\varepsilon \searrow 0$).

By approximation, (3.6) then holds for $g(x_0, \dots, x_n) = \prod_{i=0}^n \mathbf{1}_{F_i}(x_i)$ where $F_i \subseteq \mathbb{R}^d$ closed, and thus, by Dynkin's lemma, (3.7) holds $Y = \mathbf{1}_{A'}$, $A' \in \mathcal{F}$. Another approximation step yields (3.7) for all $Y \in b\mathcal{F}$.

The claim of (1) follows directly from Step 3.

Step 4: It remains to prove (2). Observe that $(X_{n+s} - X_s)_{n\geq 0}$ has continuous trajectories. Moreover

$$\begin{aligned}
\mathbb{E}_x\left[f((X_{s+t_i} - X_s)_{i=0}^n) \,\middle|\, \mathcal{F}_s^+\right] &= \mathbb{E}_x\left[f((X_{t_i} - X_0)_{i=1}^n) \circ \theta_s \,\middle|\, \mathcal{F}_s^+\right] \\
&\overset{(1)}{=} \mathbb{E}_{X_s}[f((X_{t_i} - X_0)_{i=1}^n)] \\
&= \mathbb{E}_0[f(X_{t_0}, \dots, X_{t_n})]
\end{aligned}$$

proving (2). □

As a corollary we obtain:

Theorem 3.26 (Blumenthal's 0-1 law). *For $x \in \mathbb{R}$, $A \in \mathcal{F}_0^+$ it holds that $W_x(A) \in \{0, 1\}$.*

Proof. This follows from

$$\begin{aligned}
\mathbf{1}_A &= \mathbb{E}_x\left[\mathbf{1}_A \,\middle|\, \mathcal{F}_0^+\right], && \text{because } A \in \mathcal{F}_0^+, \\
&= \mathbb{E}_x[\mathbf{1}_A \circ \theta_0 \mid \mathcal{F}_0^+], && \text{using} \theta = \mathrm{Id}, \\
&\overset{\text{a.s.}}{=} \mathbb{E}_{X_0}[\mathbf{1}_A] && \text{by the Markov property,} \\
&= \mathbb{E}_x[\mathbf{1}_A] = W_x(A).
\end{aligned}$$

□

Example 3.27. (1) Let

$$\tilde{H}_+ := \inf\{s > 0 X_s > 0\} \quad \text{and} \quad \tilde{H}_- := \inf\{s > 0 : X_s < 0\}$$

be the hitting times of $(0, +\infty)$ and $(-\infty, 0)$ respectively. Then W_0-a.s. $\tilde{H}_+ = \tilde{H}_0 = 0$. Indeed,

$$\{H_t = 0\} = \bigcap_{n\in\mathbb{N}} \left(\bigcup_{r\in[0,\frac{1}{n}]\cap\mathbb{Q}} \{X_r > 0\} \right) \in \mathcal{F}_0^+.$$

Further, for every $t > 0$, $W_0(X_t > 0) = \frac{1}{2}$. So $W_0(\tilde{H}_t \le t) \ge W_0(X_t > 0) = \frac{1}{2}$. Hence,

$$W_0(\tilde{H}_t = 0) = \lim_{t \searrow 0} W_0(H_t \le t) \ge \frac{1}{2},$$

i.e., $W_0(H_t = 0) = 1$ by Theorem 3.26.

(2) We claim the following: For any $\varepsilon > 0$,

$$\sup_{0 \le s \le \varepsilon} X_s > 0, \quad \text{and} \quad \inf_{0 \le s \le \varepsilon} X_s < 0 \quad \text{both } W_0\text{-a.s.}.$$

In order to prove this let $A := \bigcap_{n \in \mathbb{N}} \{\sup_{s \le \frac{1}{n}} X_s > 0\} \in \mathcal{F}_0^+$. By the same arguments as before $W_0(A) \ge \frac{1}{2}$, so $W_0(A) = 1$ by Theorem 3.26. Since $\{\sup_{s \le \varepsilon} X_s > 0\} \supseteq \{\sup_{s \le \frac{1}{n}} X_s > 0\}$ for some $n \in \mathbb{N}$, the claim follows.

(3) Let $H_a := \inf\{t \ge 0 : X_t = a\}$, $a \in \mathbb{R}$. Then W_0-a.s. $T_a < +\infty$ and thus $\limsup_{t \to +\infty} X_t = \liminf_{t \to +\infty} X_t = +\infty$.

Indeed, we have

$$1 = \mathbb{P}\left(\sup_{s \le 1} X_s > 0\right) = \lim_{\delta \searrow 0} \mathbb{P}\left(\sup_{s \le 1} X_s > \delta\right)$$

where the RHS limit is increasing. By scaling invariance (Proposition 3.6), we thus obtain

$$1 = \lim_{\delta \searrow 0} \mathbb{P}\left(\sup_{s \le \frac{1}{\delta^2}} X_s > 0\right) = \mathbb{P}\left(\sup_{s \ge 0} X_s > 1\right) = \mathbb{P}(H_1 < +\infty).$$

Using the scaling invariance again, this holds for $a > 0$ and by reflection invariance for $a \in \mathbb{R}$. The second claim follows by continuity.

♦

Exercise 3.28. (1) Let $d \ge 2$, $K \subseteq \mathbb{R}^d$ be an open cone and $(X_s)_{s \ge 0}$ a d-dimensional Brownian motion. Let $\tilde{H}_k := \inf\{s \ge 0 : X_s \in K\}$. Then W_0-a.s. $\tilde{H}_k = 0$.

(2) We now look at the case $d = 1$. Consider a sequence $(t_n)_{n \in \mathbb{N}}$ of real numbers with $t_n \searrow 0$, $t_n > 0$. Show that

$$\limsup_{n \to +\infty} \frac{X(t_n)}{\sqrt{t_n}} = +\infty \quad W_0\text{-a.s.}.$$

3.8. Filtration and Stopping Times

Recall that for a discrete filtered probability space $(\Omega, \mathcal{A}, (\mathcal{A}_n)_{n \in \mathbb{N}}, \mathbb{P})$ (i.e., for a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ with a sequence of σ-algebras $\mathcal{A}_0 \subseteq \mathcal{A}_1 \subseteq \ldots \subseteq \mathcal{A}$) a *stopping* *time* T is a $\mathbb{N} \cup \{+\infty\}$-valued random variable such that $\{T = n\} \in \mathcal{A}_n$ for any $n \in \mathbb{N}$. This means the decision to stop at time n is only based on the past up to n.

This notion naturally generalizes to continuous times:

Definition 3.29. Let $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0})$ be a *filtered probability space*, i.e., $(\mathcal{G}_t)_{t\geq 0}$ is a *filtration* that is a family of σ-algebras satisfying $\mathcal{G}_s \subseteq \mathcal{G}_t$ for any $s \leq t$.
A map $T : \Omega \to [0, +\infty]$ is called a $(\mathcal{G}_t)_{t\in\mathbb{R}_+}$-stopping time if

$$\{T \leq t\} \in \mathcal{G}_t \quad \text{for all } t \geq 0.$$

The σ-algebra of the past of T is given by

$$\mathcal{G}_T = \{A \in \mathcal{G} \mid A \cap \{T \leq t\} \in \mathcal{G}_t \, \forall t \geq 0\}.$$

Exercise 3.30. Show that $\mathcal{G}_T$ is a σ-algebra.

Example 3.31 (Entrance time of a closed set). Consider the space $C = C^0(\mathbb{R}_+, \mathbb{R}^d)$ with canonical coordinates $X_t : C \to \mathbb{R}^d$ and canonical filtration $\mathcal{F}_t = \sigma(X_s, s \leq t)$, $\mathcal{F} = \sigma(X_s, s \geq 0)$. Let $A \subseteq \mathbb{R}^d$ be a closed set. The entrance time of X into A is

$$H_A = \inf\{s \geq 0 : X_s \in A\}, \quad \inf \emptyset := +\infty.$$

Then H_A is a stopping time. Indeed, since a is closed and $w \in C$ are continuous, $\{s \geq 0 X_s(w) \in A\}$ is a closed subset of $[0, +\infty)$ containing $H_A(w)$ if $H_A(w) < +\infty$. Hence, $H_A > t$ iff for all $s \leq t$ $\operatorname{dist}(X_s(w), A) > 0$ iff $\inf_{s\in[0,t]} \operatorname{dist}(X_s(w), A) > 0$. Hence,

$$\{H_A > t\} = \bigcup_{n\in\mathbb{N}} \bigcap_{s\in[0,t]\cap\mathbb{Q}} \left\{ \operatorname{dist}(X_s(w), A) > \frac{1}{n} \right\} \in \mathcal{F}_t,$$

proving the claim. ◆

Example 3.32 (Entrance time of an open set). In the setting of the last example, replace A by an open set $O \subseteq \mathbb{R}^d$. $H_0 = \inf\{s \geq 0 : X_s \in O\}$. Observe that $\{H_0 = t\} \notin \mathcal{F}_t$. Here, the filtration $\mathcal{F}_t^+ = \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}$ comes to help, since it "peaks into the future". We claim H_0 is an $(\mathcal{F}_t^+)_{t\geq 0}$-stopping time. Indeed,

$$\{H_0 < s\} = \bigcup_{n\in[0,s)\cap\mathbb{Q}} \{X_n \in O\} \in \mathcal{F}_s \quad \text{for } s \geq 0$$

and

$$\{H_0 \leq s\} = \bigcap_{\varepsilon>0} \{H_0 < s + \varepsilon\} \in \mathcal{F}_s^+.$$

♦

Definition 3.33. A filtration $(\mathcal{G}_t)_{t\geq 0}$ is called right continuous if $\mathcal{G}_t = \bigcap_{\varepsilon>0} \mathcal{G}_{t+\varepsilon}$.

Exercise 3.34.
- If the filtration $(\mathcal{G}_t)_{t\geq 0}$ is right-continuous, then T is a stopping time if and only if $\{T < t\} \in \mathcal{G}_t$ for any t.
- $(\mathcal{F}_t^+)_{t\geq 0}$ is a right-continuous filtration.

Proposition 3.35. *Let $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0})$ be a filtered space.*

(1) If T is a stopping time then T is $\mathcal{G}_T$ measurable.

(2) If S and T are stopping times then $S \vee T$ and $S \wedge T$ are stopping times.

(3) Consider $(\Omega, \mathcal{G}) = (C, \mathcal{F})$. If T and S are $\mathcal{F}_t^+$ stopping times, so is

$$T + S \circ \theta_T = \begin{cases} T(w) + S(\theta_{T(w)}(w)) & \text{if } T(w) < +\infty \\ +\infty & \text{if } T(w) = +\infty \end{cases}$$

Exercise 3.36. Prove the first two points of Proposition 3.35.

Proof of Proposition 3.35. For the first to points see Exercise 3.36. By Exercise 3.34 $(\mathcal{F}_t^+)_{t\geq 0}$ is right continuous so we need to show that

$$\{T + S \circ \theta_T < t\} \in \mathcal{F}_t^+. \tag{3.8}$$

Note that

$$\{T + S \circ \theta_T < t\} = \bigcup_{\substack{u,v\in\mathbb{Q}\cap(0,+\infty) \\ u+v<t}} \{T < u, S \circ \theta_T < v\}.$$

We claim that when $\{T < u\} \neq \emptyset$, then

$$\theta_T : (\{T < u\}, \mathcal{F}_{u+v} \cap \{T < u\}) \to (C, \mathcal{F}_v) \quad \text{is measurable.} \tag{3.9}$$

Indeed, for $s \in [0, v]$, $X_s \circ \theta_T$ is measurable as a map from $(\{T < n\}, \mathcal{F}_{v+u}) \cap \{T < u\})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ as it follows from

$$X_s \circ \theta_T(w) = X_{S+T(w)}(w) = \lim_{n\to+\infty} \sum_{k=1}^{n} X_{s+\frac{k}{n}u}(w) \mathbf{1}_{T\in[\frac{k-1}{n}u, \frac{k}{n}u)};$$

the first summand ist $\mathcal{F}_{s+u} \cap \{T < u\}$, hence, $\mathcal{F}_{u+v} \cap \{T < u\}$ measurable and the second one is $\mathcal{F}_n^+ \cap \{T < u\}$ and thus $\mathcal{F}_{u+v} \cap \{T < u\}$-measurable. Since $\mathcal{F}_v$ is the smallest σ-algebra making X_s, $s \in [0, v]$ measurable, (3.9) follows.

Further, $\{s < v\} = \bigcap_{r<v, r\in\mathbb{Q}} \{S \leq r\} \in \mathcal{F}_v$, hence,

$$\begin{aligned}&\{T < u, S \circ \theta_T < v\}\\ &\qquad = \{w \in \{T < u\}, \theta_T(w) \in \{S < v\}\} \overset{(3.9)}{\in} \mathcal{F}_{u+v} \cap \{T < u\} \subseteq \mathcal{F}_{u+v},\end{aligned}$$

since $\{T < u\} \in \mathcal{F}_u^+$. Going back, this yields $\{T + S \circ \theta_T < t\} \subseteq \mathcal{F}_t \subseteq \mathcal{F}_t^+$ proving (3.8). This concludes the proof. □

Exercise 3.37. Consider a filtered space $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\in\mathbb{R}_+})$ and stopping times T and S. Prove the following statements:

(i) If $S \leq T$ then $\mathcal{G}_S \subseteq \mathcal{G}_T$.

(ii) Both $\{S < T\}$ and $\{S \leq T\}$ belong to $\mathcal{G}_S \cap \mathcal{G}_T$.

(iii) If $A \in \mathcal{G}_S$ then $\{S < T\} \cap A \in \mathcal{G}_{S\wedge T}$ and $\{S \leq T\} \cap A \in \mathcal{G}_{S\wedge T}$.

3.9. Strong Markov Property

We now generalize the simple Markov Property (Theorem 3.25) to stopping times. To this end we need a last definition:

Definition 3.38. When T is a $\mathcal{F}_t$ stopping time, we set

$$\mathcal{F}_t^+ := \{A \in \mathcal{F} \,:\, A \cap \{T \leq t\} \in \mathcal{F}_t^+ \;\forall t \geq 0\} \overset{(!)}{=} \{A \in \mathcal{F}_t \,:\, A \cap \{T < t\} \in \mathcal{F}_t^+ \;\forall t \geq 0\}$$

To see (!) observe that "⊆" is trivial and for "⊇" write

$$A \cap \{T \leq t\} = \bigcap_{n\in\mathbb{N}} \left(A \cap \left\{ T < t + \frac{1}{n} \right\} \right) \in \mathcal{F}_t^+.$$

We are now ready to state the Markov Property:

Theorem 3.39 (Strong Markov property of Wiener measure). *Let T be an $\mathcal{F}_t^+$ stopping time, Y a bounded and $\mathcal{F}$-measurable random variable and $x \in \mathbb{R}^d$. Then*

$$\mathbb{E}_x\left[Y \circ \theta_T \,\middle|\, \mathcal{F}_T^+\right] = \mathbb{E}_{X_T}[Y], \qquad W_x\text{-a.s. on } \{T < +\infty\}. \tag{3.10}$$

Remark 3.40. The claim of Theorem 3.39 contains implicitly that:

(i) $\theta_T : (\{T < +\infty\}, \mathcal{F} \cap \{T < +\infty\}) \to (C, \mathcal{F})$ is measurable.

(ii) The random variable $\mathbb{E}_{X_T}[Y]$ defined on $\{T < +\infty\}$ is in $\mathcal{F}_T^+ \cap \{T < +\infty\}$.

Moreover (3.10) is equivalent with

$$\mathbb{E}_x[Y \circ \theta_T \mathbf{1}_A] = \mathbb{E}_x[\mathbf{1}_A \, \mathbb{E}_{X_T}[Y]], \qquad \text{for every } A \in \mathcal{F}_+ \cap \{T < +\infty\}.$$

♦

We need a preparatory step to show the second point of Remark 3.40.

Definition 3.41. Given a filtered space $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0})$, an $\mathbb{R}^d$-valued process $(Z_u)_{u\geq 0}$ is called *$\mathcal{G}_t$-adapted* if $Z_t \in \mathcal{G}_t$ for all $t \geq 0$. It is called *progressively measurable* if the restriction of $Z_t(\omega)$ to $[0,t] \times \Omega$ is $\mathcal{B}([0,t]) \times \mathcal{G}_t$ measurable for any $t \geq 0$.

Example 3.42. If Z_t is adapted and right-continuous, then it is progressively measurable. Indeed, on $[0,t] \times \Omega$,

$$Z_s(\omega) = \lim_{n\to+\infty} \sum_{k=1}^{n} Z_{\frac{k}{n}t}(\omega) \mathbf{1}\left\{\frac{k-1}{n} t \leq s < \frac{k}{n} t\right\} + Z_t(\omega)\mathbf{1}\{s = t\}$$

and the summands are obviously $\mathcal{B}([0,t]) \times \mathcal{G}_t$ measurable. ♦

Lemma 3.43. *Let $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0})$ be a filtered space and $(Z_u)_{u\geq 0}$ progressively measurable and T a $(\mathcal{G}_t)_{t\geq 0}$ stopping time. Then, the map $Z_T : \omega \in \{T < +\infty\} \mapsto Z_{T(\omega)}(\omega) \in \mathbb{R}^d$ is $\mathcal{G}_T \cap \{T < +\infty\}$ measurable.*

Proof. By definition of the σ-algebra $\mathcal{G}_T$, it is sufficient to show

$$(\{T \leq t\}, \mathcal{G}_T \cap \{T \leq t\}) \xrightarrow{Z_T} (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)) \quad \text{is measurable for all } t \geq 0. \tag{3.11}$$

The above map is a composition of the following measurable maps:

$$\omega \in (\{T \leq t\}, \mathcal{G}_t \cap \{T \leq t\}) \mapsto (T(\omega), \omega) \in ([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{G}_t) \quad \text{and}$$

$$(u, \omega) \in ([0, t] \times \Omega, \mathcal{B}([0, t] \otimes \mathcal{G}_t)) \mapsto Z_u(\omega) \in (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)).$$

The first one is measurable because both maps

$$(\{T \leq t\}, \mathcal{G}_t \cap \{T \leq t\}) \xrightarrow{T} ([0, t], \mathcal{B}([0, t])) \quad \text{and}$$
$$(\{T \leq t\}, \mathcal{G}_t \cap \{T \leq t\}) \xrightarrow{\mathrm{Id}} (\Omega, \mathcal{G}_t)$$

are measurable. The second one is measurable because $(Z_u)_{u \geq 0}$ is progressively measurable. □

Proof of Theorem 3.39. **Step 1: (Remark 3.40, (i))** We show that $\theta_T : (\{T < +\infty\}, \mathcal{F} \cap \{T < +\infty\}) \to (C, \mathcal{F})$ is measurable. To this end, it is sufficient to show that for any $s \geq 0$,

$$X_s \circ \theta_T : (\{T < +\infty\}, \mathcal{F} \cap \{T < +\infty\}) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)) \quad \text{is measurable.} \tag{3.12}$$

The claim then follows as in Exercise 2.6. We see for $\omega \in \{T < +\infty\}$, using that Brownian motion is right-continuous

$$X_s \circ \theta_T(\omega) = X_{s+T(\omega)}(\omega) = \lim_{n \to +\infty} \sum_{k \in \mathbb{N}} X_{x+\frac{k}{n}}(\omega) \mathbf{1}\left\{\frac{k-1}{n} \leq T < \frac{k}{n}\right\}.$$

From this the measurability of (3.12) follows.

Step 2: We show that $X_T : (\{T < +\infty\}, \mathcal{F}_T^+ \cap \{T < +\infty\}) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is measurable. By Example 3.42, Brownian motion is progressively measurable. The claim then follows from Lemma 3.43.

Step 3: (Remark 3.40, (ii)) We show that

$$\mathbb{E}_{X_T}[Y] : (\{T < +\infty\}, \mathcal{F}_T^+ \cap \{T < +\infty\}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

is measurable. We have shown that the map $\mathbb{R}^d \ni y \mapsto \mathbb{E}_y[Y]$ is measurable in Exercise 3.7 (i.e., $(y \in \mathbb{R}^d, A \in \mathcal{F}) \to W_y(A)$ is a stochastic kernel). Combining this with the previous steps implies the claim.

Step 4: We show that (3.10) holds true if T takes value in an at most denumerable set of values in $[0, +\infty]$. Let $(a_n)_{n=0}^N$, $N \leq +\infty$ be the family of values of T in $[0, +\infty)$. For $A \in \mathcal{F}_T^+ \cap \{T < +\infty\}$, $0 = t_0 < \ldots < t_k$, and $f \in b\mathcal{B}(\mathbb{R}^{d(K+1)})$ we have

$$\begin{aligned}
\mathbb{E}_x[\mathbf{1}_A f(X_{t_0}, \ldots, X_{t_n}) \circ \theta_T] &= \mathbb{E}_x[\mathbf{1}_A f(X_{t_0+T}, \ldots, X_{t_k+T})] \\
&= \sum_{n \in \mathbb{Z} \cap [0,N]} \mathbb{E}_x[\mathbf{1}_{A \cap \{T=a_n\}} f(X_{t_0+a_n}, \ldots, X_{t_k+a_n})].
\end{aligned}$$

Since $A \cap \{T = a_n\}$ is in $\mathcal{F}^+_{a_n}$, by the simple Markov property the above is equal to

$$\sum_n \mathbb{E}_x \left[\mathbb{E}_{X_{a_n}}[f(X_{t_0}, \dots, X_{t_n})]\mathbf{1}_{\{T=a_n\}\cap A}\right] = \mathbb{E}_x \left[\mathbb{E}_{X_T}[f(X_{t_0}), \dots, X_{t_n}]\mathbf{1}_A\right]. \tag{3.13}$$

Dynkin's lemma then gives Remark 3.40, (ii) and thus (3.10).

Step 5: Proof of (3.10) for a general stopping time T. We use a discrete skeleton approximation of T:

$$T_n := \begin{cases} \sum_{k\geq 0} \frac{k+1}{2^n} \mathbf{1}\left\{\frac{k}{2^n} \leq T < \frac{k+1}{2^n}\right\}, & T < +\infty, \\ +\infty, & T = +\infty. \end{cases} \tag{3.14}$$

Observe that T_n is a $\mathcal{F}^+_t$ stopping time and $T_n \searrow T$ as $n \to +\infty$. Indeed, the second claim is obvious. For the first one for $k, n \geq 0$, $\{T_n \leq \frac{k+1}{2^n}\} = \{T < \frac{k+1}{2^n}\} \in \mathcal{F}^+_{\frac{k+1}{2^n}}$ and for $t \in [\frac{k}{2^n}, \frac{k+1}{2^n})$,

$$\{T_n \leq t\} = \{T_n \leq \frac{k}{2^n}\} \in \mathcal{F}^+_{\frac{k}{2^n}} \subseteq \mathcal{F}^+_t,$$

so T_n is an $\mathcal{F}^+_t$ stopping time. Moreover, from $T < T_n$ follows $\mathcal{F}^+_T \subseteq \mathcal{F}^+_{T_n}$.

Take now $A \in \mathcal{F}^+_T \cap \{T < +\infty\}$. Since $T_n = +\infty$ if and only if $T = +\infty$ we see that $A \in \{T_n < +\infty\} \cap \mathcal{F}^+_{T_n}$. Using step 4, we get for $0 = t_0 < \dots < t_k$ and $f_0, \dots, f_k \in C^0_b$ that

$$\mathbb{E}_x\left[\prod_{i=0}^k f_i(X_{t_i+T_n})\mathbf{1}_A\right] = \mathbb{E}_x\left[\mathbb{E}_{X_T}\left[\prod_{i=0}^k f_i(X_{t_i})\right]\mathbf{1}_A\right].$$

Using the fact that $t \mapsto X_t$ is (right) continuous and f_i are continuous, $y \mapsto \mathbb{E}[\prod_{i=0}^k f_i(X_{t_i})]$ is continuous (which was shown in Remark 3.10) and by the dominated convergence we get with $n \to +\infty$ that

$$\mathbb{E}_x\left[\prod_{i=0}^k f_i(x_{t+T_l})\mathbf{1}_A\right] = \mathbb{E}_x\left[\mathbb{E}_{X_T}\left[\prod_{i=0}^k f_i(X_{t_i})\right]\mathbf{1}_A\right].$$

This claim can be then extended to an arbitrary $Y \in b\mathcal{F}$ with the same argument as around Remark 3.11. $\square$

3.10. Some Applications of the Strong Markov Property

We use Theorem 3.39 to derive some interesting properties of Brownian motion.

Theorem 3.44 (Reflection principle for Brownian motion). *Let* $M_t := \sup_{0\le s\le t} X_s$, $a > 0$ *and* $b \le a$. *Then:*

(a) $W_0(X_t \le b, M_t \ge a) = W_0(X_t \ge 2a - b)$.

(b) $W_0(M_t \ge a) = 2W_0(X_t \ge a) = W_0(|X_t| \ge a)$

Lemma 3.45. *Let* T *be a* $\mathcal{F}_t^+$ *stopping time,* $h(w_1, w_2) \in b\mathcal{F} \otimes \mathcal{F}_T^+$ *and* $x \in \mathbb{R}^d$. *Then*

$$\mathbb{E}_x \left[h\big(\theta_{T(w)}(w), w\big) \,\Big|\, \mathcal{F}_t^+\right] \quad = \int h(w_1, w)\, \mathrm{d}W_{X_{T(w)}(w)}(w_1), \qquad W_x\text{-a.s. on } \{T < +\infty\}.$$

Proof. For $h = \mathbf{1}_{A_1}(w_1)\mathbf{1}_{A_2}(w_2)$, $A_1 \in \mathcal{F}$, $A_2 \in \mathcal{F}_T^+$, the strong Markov property yields

$$\mathbb{E}_x[h(\theta_T(w), w)\mathbf{1}_B] = \mathbb{E}_x\left[\mathbf{1}_B \int h(w_1, w)\, \mathrm{d}W_{X_{T(w)}(w)}(w_1)\right]$$

for all $B \in \mathcal{F}_T^+ \cap \{T < +\infty\}$. The extension to arbitrary h follows by the usual approximation arguments. □

Proof of Theorem 3.44. Let $H_a = \inf\{t \ge 0 : X_t = a\}$ be a $\mathcal{F}_t^+$-stopping time. Then

$$\begin{aligned} W_0(X_t \le t, M_t \ge a) &= W_0(H_a \le t, X_t \le b) \\ &= W_0(\{w \in C : H_a \le t, X_{(t-H_a(w))_+}(\theta_{H_a}(w) \le b)\}) \end{aligned}$$

We now apply Lemma 3.45 with $h(w_1, w_2) = \mathbf{1}\left\{X_{(t-H_a(w_2))_+}(w_1) \le b\right\}$ which is $\mathcal{F} \otimes \mathcal{F}_{H_a}^+$-measurable (see Exercises). Take $\tilde{X}$ and $\tilde{W}$, copies of W and X respectively, so

$$\begin{aligned} W_0(H_a \le t, X_t \le b) \quad &= \quad \mathbb{E}_0\left[H_a \le t, \tilde{W}_{X_{H_a}}(\tilde{X}_{(t-H_a)_+} \le b)\right] \\ &\overset{\text{symmetry}}{=} \mathbb{E}_0\left[H_a \le t, \tilde{W}_a(\tilde{X}_{(t-H_a)_+} \le 2a - b)\right] \\ &\overset{\substack{\text{Lemma 3.45} \\ \text{backwards}}}{=} \mathbb{E}_0\left[H_a \le t, X_t \ge 2a - b\right] \\ &= \quad \mathbb{E}_0[X_t \ge 2a - b] \end{aligned}$$

which concludes the proof of (a).

To prove (b), for $a > 0$,

$$W_0(M_t \ge a) = W_0(H_a \le t)$$

$$= W_0(H_a \le t, X_t > a) + \underbrace{W_0(H_a \le t, X_t \le a)}_{=W_0(X_t \ge a) \text{ by (a)}}$$

$$= 2W_0(X_t \ge a). \qquad \square$$

Exercise 3.46. (a) the law of H_a is absolutely continuous with respect to the Lebesgue measure and

$$W_0(H_a \in \mathrm{d}t) = \frac{|a|}{\sqrt{2\pi t^2}} \exp\left\{-\frac{a^2}{2t}\right\} \mathbf{1}_{(0,+\infty)}(t)\,\mathrm{d}t.$$

(b) The point density of the pair (X_t, M_t) for $t > 0$ is

$$W_0(X_t \in \mathrm{d}a, Mt \in \mathrm{d}b) = 2 \cdot \frac{2a-b}{\sqrt{2\pi t^3}} \exp\left\{-\frac{(2a-b)^2}{2t}\right\} \mathbf{1}_{a<b,b>0}\,\mathrm{d}a\,\mathrm{d}b.$$

We push Theorem 3.44, (b) a bit further:

Corollary 3.47. *Let $Y_t = M_t - X_t$, then for any $t \ge 0$ under W_0, Y_t, $|X_t|$ and M_t have the same distribution given by Theorem 3.44, (b).*

Proof. By Exercise 3.46, (b), $M_t \overset{\text{law}}{=} |X_t|$. For Y_t, obviously $Y_t \ge 0$ W_0-a.s. Morover the time renewal process $B_s = X_{t-s} - X_t$, $s \in [0,t]$ is a Brownian motion (the proof of this fact is left as an exercise to the reader; use the Gaussian process characteristic of Brownian motion), and

$$Y_t = \sup_{s \le t} X_s - X_t = \sup_{0 \le s \le t} B_s.$$

Hence, $Y_t \overset{\text{law}}{=} M_t$. This concludes the proof. $\square$

We can push Theorem 3.44, (b) even more:

Theorem 3.48. *Under W_0, both $(Y_t)_{t\ge0}$ and $(|X_t|)_{t\ge0}$ are $\mathcal{F}_t^+$-Markov processes with transition semigroup*

$$K_s(x, \mathrm{d}y) = \big(p_s(x,y) + p_s(x,-y)\big)\,\mathrm{d}y, \quad \text{where} \quad p_s(x,y) = \frac{1}{\sqrt{2\pi s}} \mathrm{e}^{\frac{(x-y)^2}{2s}}.$$

In particular, Y and $|X|$ have the same law under W_0.

Remark 3.49. Even if $M_t \overset{\text{law}}{=} |X_t| \overset{\text{law}}{=} Y_t$, M as a process has not the same law as $|X|$ or Y. Indeed M is increasing while $|X|$ and Y are not. In fact M_t is not even a Markov process. ♦

Remark 3.50. Theorem 3.48 claims that for $Z \in \{Y, |X|\}$,

$$W_0\left(Z_{t+s} \in \mathrm{d}y \,\middle|\, \mathcal{F}_t^+\right) = K_s(X_t, \mathrm{d}y).$$

♦

Proof of Theorem 3.48. First, consider the process $(|X_t|)_{t\geq 0}$. By the simple Markov property

$$\begin{aligned} W_0\left(|X_{t+s}| \in \mathrm{d}y \,\middle|\, \mathcal{F}_t^+\right) &= W_0\left(|X_{t+s}| \in \mathrm{d}y \,|\, X_t\right) \\ &= W_0\left(X_{t+s} \in \mathrm{d} \,|\, X_t\right) + W_0\left(-X_{t+s} \in \mathrm{d}y \,|\, X_t\right) \\ &= p_s(X_t, y)\,\mathrm{d}y + p_s(X_t, -y)\,\mathrm{d}y \\ &= K_s(X_t, y)\,\mathrm{d}y, \qquad W_0\text{-a.s.} \end{aligned}$$

Comparing this with Remark 3.50 implies that $(|X_t|)_{t\geq 0}$ is a $\mathcal{F}_t^+$-Markov process with transition semigroup $K_s(x, \mathrm{d}y)$.

For $(Y_t)_{t\geq 0}$ we write, with $s > 0$, $t \geq 0$, $b \geq 0 \vee a$

$$\begin{aligned} W_0\left(X_{t+s} \leq a, M_{t+s} \leq b \,\middle|\, \mathcal{F}_t^+\right) &= W_0\left(X_{t+s} \leq a, M_t \leq b, \max_{0\leq u\leq s} X_{t+u} \leq b \,\middle|\, \mathcal{F}_t^+\right) \\ &= \mathbf{1}_{\{M_t\leq b\}} W_0\left(X_{t+s} \leq a, \max_{0\leq u\leq s} X_{t+s} \leq b \,\middle|\, X_t\right) \end{aligned}$$

by the simple Markov property again. The last expression is measurable with respect to the σ-field generated by (M_t, X_t). This implies that the process $\{(X_t, M_t)_{t\geq 0}\}$ is $\mathcal{F}_t^+$-Markov.

Y_t is a function of (X_t, M_t) and thus

$$W_0\left(Y_{t+s} \in A \,\middle|\, \mathcal{F}_t^+\right) = W_0\left(Y_{t+s} \in A \,|\, X_t, M_t\right) \qquad A \in \mathcal{B}(\mathbb{R}). \tag{3.15}$$

Let us determine the transition probabilities of $(X_t, M_t)_{t\geq 0}$: For $b > m \geq x$, $b \geq a$, $m \geq 0$ we have

$$\begin{aligned} &W_0\left(X_{t+s} \in \mathrm{d}a, M_{t+s} \in \mathrm{d}b \,|\, X_t = x, M_t = m\right) \\ &\quad= W_x(X_s \in \mathrm{d}a, M_s \in \mathrm{d}b) = W_0(X_s \in \mathrm{d}a - x, M_s \in \mathrm{d}b - x) \\ &\quad\overset{\text{Exercise 3.46, (b)}}{=} \frac{2(2b-a-x)}{\sqrt{2\pi s^3}} \exp\left\{-\frac{(2b-a-x)^2}{2s}\right\} \mathrm{d}a\,\mathrm{d}b \end{aligned} \tag{3.16}$$

and for $m \geq x$, $a \in \mathbb{R}$, $m \geq 0$ we have

$$\begin{aligned} &W_0\left(X_{t+s} \in \mathrm{d}a, M_{t+s} = m \,|\, X_t = x, M_t = m\right) \\ &\quad = W_x(X_s \in \mathrm{d}a, M_s \leq m) \\ &\quad = W_0(X_s \in \mathrm{d}a - x, M_s \leq m - x) \\ &\quad = \frac{1}{\sqrt{2\pi s}}\left(\exp\left\{-\frac{(a-x)^2}{2s}\right\} - \exp\left\{-\frac{(2m-a-w)^2}{2s}\right\}\right) \end{aligned} \tag{3.17}$$

where the last equality follows by integrating Exercise 3.46, (b).

From Equations (3.15) to (3.17) we get

$$\begin{aligned} &W_0\left(Y_{t+s} \in \mathrm{d}y \,|\, X_t = x, M_t = m\right) \\ &\quad = \int_{(m,+\infty)} W_0\left(X_{t+s} \in b - \mathrm{d}y, m_{t+s} \in \mathrm{d}b \,|\, X_t = x, M_t = m\right) \mathrm{d}b \\ &\qquad\qquad + W_0\left(X_{t+s} \in m - \mathrm{d}y, M_{t+s} = m \,|\, X_t = x, M_t = m\right) \\ &\quad \vdots \\ &\quad = K_s(m - x, \mathrm{d}y) = K_s(Y_t, \mathrm{d}y) \end{aligned}$$

Hence, $(Y_t)_{t\geq 0}$ is $\mathcal{F}_t^+$-Markov with the same transition semigroup as $(|X_t|)_{t\geq 0}$. The last claim follows from the fact that the semigroup determines the law of a Markov process. □

3.11. Properties of Brownian motion sample paths

Recall that we already know that the trajectories of Brownian motion are α-Hölder for every $\alpha \in \left(0, \frac{1}{2}\right)$ by Corollary 3.17 but otherwise rather "rough": they are of infinite variation (see Corollary 3.23) and not differentiable at 0 (Exercise 3.28). We now give another "roughness result".

Theorem 3.51 (law of iterated logarithm). *We assume $d = 1$. Then W_0-a.s.*

(i) $$\limsup_{t \searrow 0} \frac{X_t}{\sqrt{2t \log\log \frac{1}{t}}} = -\liminf_{t \searrow 0} \frac{X_t}{\sqrt{2t \log\log \frac{1}{t}}} = 1;$$

(ii) $$\limsup_{t \to +\infty} \frac{X_t}{\sqrt{2t \log\log t}} = -\liminf_{t \to +\infty} \frac{X_t}{\sqrt{2t \log\log t}} = 1.$$

Proof. Note that *(ii)* follows from *(i)* by the time inversion invariance of Brownian motion (see Proposition 3.6, (3)). Using the reflection invariance (Proposition 3.6, (1)) it is thus sufficient to show that $\limsup_{t\searrow 0} \frac{X_t}{\sqrt{2t\log\log\frac{1}{t}}} = 1$.

Step 1: Upper bound. Set $\varphi(t) := \sqrt{2t\log\log\frac{1}{t}}$. We approximate φ by piecewise constant functions and estimate X against those. To this end fix $\delta > 0$ and $q \in (0,1)$ such that

$$(1+\delta)^2 q > 1 \quad \text{i.e., } q \nearrow 1 \text{ as } \delta \searrow 0. \tag{3.18}$$

For $n \in \mathbb{N}$ set $t_n := q^n$ (i.e., $t_n \searrow 0$) and

$$A_n := \{w \in C \mid \max\{X_t : t \in [t_{n+1}, t_n]\} > (1+\delta)\varphi(t_{n+1})\}.$$

As $\varphi(t)$ is non-decrasing on some small intervall $[0,T]$, the upper bound follows if we show that for any δ, A_n occurs W_0-a.s. only finitely many times.

To this end with n large,

$$\begin{aligned} W_0(A_n) &\leq W_0\left(\sup_{s\leq t_n} X_s > (1+\delta)\varphi(t_{n+1})\right) \\ &\overset{\text{Theorem 3.44}}{=} 2W_0(X_{t_n} \geq (1+\delta)\varphi(t_{n+1})) \\ &\leq \sqrt{\frac{2}{\pi}} \cdot \frac{1}{x_n}\exp\left\{-\frac{x_n^2}{2}\right\}, \quad \text{where } x_n = \frac{(1+\delta)\varphi(t_{n+1})}{\sqrt{t_n}}. \end{aligned}$$

Note that

$$x_n = (1+\delta)\left\{2q\log((n+1)\log\frac{1}{q})\right\}^{1/2} = \left(2\log\left\{(\alpha(n+1))^{\lambda}\right\}\right)^{1/2}$$

with $\alpha = \log\frac{1}{q}$, $\lambda = q(1+\delta)^2 > 1$ by (3.18). Hence, for large $n \in \mathbb{N}$, it holds that

$$W_0(A_n) \leq \sqrt{\frac{2}{\pi}} \cdot \frac{1}{(n+1)^{\lambda}\alpha^{\lambda}}$$

which is summable. We thus deduce by the Borel-Cantelli Lemma that the upper bound holds.

Step 2: Lower bound. We fix $q \in (0,1)$ and $\varepsilon \in (0,\frac{1}{2})$ small and set $t_n = q^n$. We will use the Gaussian estimate

$$\mathbb{P}\left(\mathcal{N}(0,1) \geq x\right) > \frac{x}{1+x^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad x > 0. \tag{3.19}$$

Setting $x_n = (1-\varepsilon)\varphi(t_n)(t_n - t_{n+1})^{-1/2}$, we find for n large that

$$
\begin{aligned}
W_0\Big(X_{t_n} - X_{t_{n+1}} > (1-\varepsilon)\varphi(t_n)\Big) &\overset{\text{scaling}}{=} W_0(W_1 > x_n) \\
&\overset{(3.19)}{\geq} (2\pi)^{-1/2} x_n (1+x_n^2)^{-1} \exp\left\{-\frac{x_n^2}{2}\right\}.
\end{aligned}
\tag{3.20}
$$

Moreover

$$
x_n = \frac{1-\varepsilon}{\sqrt{1-q}}\sqrt{2\log(n\log\frac{1}{q})} = \sqrt{\beta\log(\alpha n)}
$$

with $\alpha = \log\frac{1}{q}$, $\beta = 2(1-\varepsilon)^2/(1-q)$. Hence, for n large $\frac{x_n}{1+x_n^2} \geq \frac{1}{2x_n}$. Assuming $q < \frac{\varepsilon^2}{4}$ i.e., $\beta < 2$ we get

$$
W_0(X_{t_n} - X_{t_{n+1}} > (1-\varepsilon)\varphi(t_n)) \geq \frac{c}{(\log n)^{1/2}} n^{-\beta/2}
$$

which is not summable and the events on the LHS of (3.20) are independent as n varies. Therefore, by the second Borel-Cantelli Lemma, W_0-a.s., for infinitely many $n \in \mathbb{N}$, $X_{t_n} - X_{t_{n+1}} \geq (1-\varepsilon)\varphi(t_n)$.

From the upper bound applied to $-X$, we see that W_0-a.s. for n large $X_{t_n} \geq -(1-\varepsilon)\varphi(t_n) - (1+\varepsilon)\varphi(t_{n+1})$ and thus W_0-a.s. for infinitely many n

$$
\begin{aligned}
X_{t_n} &= X_{t_n} - X_{t_{n+1}} + X_{t_{n+1}} \\
&\geq (1-\varepsilon)\varphi(t_n) - (1+\varepsilon)\varphi(t_{n+1}) \\
&= \varphi(t_n)\left[1 - \varepsilon - (1+\varepsilon)\frac{\varphi(t_{n+1})}{\varphi(t_n)}\right].
\end{aligned}
$$

Using $q < \varepsilon^2/4$, we get $\lim_{n\to+\infty}\frac{\varphi(t_{n+1})}{\varphi(t_n)} = \sqrt{q} < \frac{\varepsilon}{2}$ so $X_{t_n} \geq \varphi(t_n)[1-2\varepsilon]$. Letting $\varepsilon \searrow 0$ implies the claimed bound. □

Remark 3.52. There are a few related results:

(1) **Lévy's modulus of continuity of Brownian motion** (Lévy 1937): W_0-a.s.,

$$
\limsup_{u\searrow 0}\left(2u\log\frac{1}{u}\right)^{-1/2} \sup_{\substack{0\leq s<t\leq 1 \\ |t-s|\leq u}} |X_t - X_s| = 1.
$$

The proof of this claim is similar to the proof of Theorem 3.51, see e.g. [3, p.114]. Observe that $\sqrt{2t\log\log\frac{1}{t}}$ is replaced by the "larger" $\sqrt{2t\log\frac{1}{t}}$. This is a consequence of taking the supremum over the starting points $s \in [0,1]$. For fixed s, Theorem 3.51 holds by the Markov property.

(2) **Law of iterated logarithms (LIL) for small values of** $\sup_{s\le t}|X_s|$ (Chung 1948):

$$\liminf_{t\to+\infty}\left(\frac{\log\log t}{t}\right)^{1/2}\sup_{s\le t}|X_s| = \frac{\pi}{2\sqrt{2}} < 1, \qquad W_0\text{-a.s..}$$

Observe also, by the law of iterated logarithms,

$$\limsup_{t\to+\infty}\left(\frac{\log\log t}{t}\right)^{1/2}\sup_{s\le t}|X_s| = 1 \qquad W_0\text{-a.s..}$$

(3) **Strassen theorem**(1964) gives a functional version of Theorem 3.51: The set of limit points of $\left(\frac{X_{ut}}{2t\log\log t}\right)_{0\le u\le 1}$ as $t\to+\infty$ coincides with

$$\left\{ f \in C^0([0,1];\mathbb{R}) \,\middle|\, f(u) = \int_0^u g(v)\,\mathrm{d}v,\, g \in L^2([0,1]), \|g\|_{L^2} \le 1 \right\}.$$

♦

Theorem 3.53 ("Converse" to Corollary 3.17). *Let $\alpha \ge \frac{1}{2}$. Then for $0 \le a < b < +\infty$*

$$\sup_{a\le s<t\le b}\frac{|X_s - X_t|}{|s-t|^{\alpha}} = +\infty$$

Proof. By Theorem 3.51 and the Markov property we have W_0-a.s. for all $s \in \mathbb{Q}\cap[0,+\infty)$,

$$\limsup_{u\searrow 0}\frac{|X_{s+u}-X_s|}{\sqrt{2u\log\log\frac{1}{u}}} = 1$$

Hence, W_0-a.s. for all $s \in \mathbb{Q}\cap[0,+\infty)$, $\limsup_{u\searrow 0}\frac{|X_{s+u}-X_s|}{u^{\gamma}}$ and the claim follows. $\square$

4. Time-continuous Martingales

4.1. Filtration and stopping times revisited

In this chapter we develop the theory of martingales in continuous time and of stochastic processes in general.

Definition 4.1. We say that a filtered probability space $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ *the usual conditions* if

(i) $\mathcal{G}_0$ contains all $\mathbb{P}$-null sets, i.e., $\Gamma \subseteq N \in \mathcal{G}, \mathbb{P}(N) = 0 \implies \Gamma \in \mathcal{G}_0$;

(ii) $(\mathcal{G}_t)_{t\geq 0}$ is right continuous, i.e., $\mathcal{G}_t = \bigcap_{\varepsilon>0} \mathcal{G}_{t+\varepsilon}$ for all $t \geq 0$.

Example 4.2. Consider the canonical Brownian motion $(C, \mathcal{F}, W_0)$. For $t \geq 0$, set

$$\bar{\mathcal{F}}_t := \{A \subseteq C \mid \exists B \in \mathcal{F}_t \text{ s.t. } A \triangle B \text{ is } W_0\text{-null}\} \tag{4.1}$$

where $A \triangle B = (A \setminus B) \cup (B \setminus A)$. ♦

Lemma 4.3. *$\bar{\mathcal{F}}_t$ from eq. (4.1) satisfies the following conditions:*

(i) $\mathcal{F}_t^+ \subseteq \bar{\mathcal{F}}_t$;

(ii) $(\bar{\mathcal{F}}_t)_{t\geq 0}$ is right continuous;

(iii) $(C, \mathcal{F}, (\bar{\mathcal{F}}_t)_{t\geq 0}, W_0)$ satisfies the usual conditions.

Proof. (i) Observe that for every $A \in \mathcal{F}$ there is $Y \in b\mathcal{F}_t$ such that $\mathbb{E}\left[\mathbf{1}_A \,|\, \mathcal{F}_t^+\right] = Y$, W_0-a.s.. Indeed, for

$$A = \left(\bigcap_{i=1}^{k} \{X_{t_i} \in D_i\}\right) \cap \left(\bigcap_{j=1}^{m} \{X_{s_j} \in D_{k+j}\}\right)$$

with $0 = t_0 < \ldots < t_k = t$, $t < s_1 < \ldots < s_m$, $D_i \in \mathcal{B}(\mathbb{R}^d)$, by the simple Markov property

$$\mathbb{E}_0[\mathbf{1}_A \mid \mathcal{F}_t^+] = \mathbf{1}\{X_{t_0} \in D_0, \ldots, X_{t_k} \in D_k\} \cdot W_{X_t}(X_{s_1-t} \in D_{k+1}, \ldots, X_{s_m-t} \in D_{k+m})$$

which is $\mathcal{F}_t$-measurable. Dynkin's type arguments then yield the claim for general $A \in \mathcal{F}$ (this can actually be seen as a generalization of the 0-1 law). Hence, for $A \in \mathcal{F}_t^+$ it holds that $\mathbf{1}_A = \mathbb{E}[\mathbf{1}_A \mid \mathcal{F}_t^+] = Y$, W_0-a.s. for some $Y \in b\mathcal{F}_t$. Hence, $\mathbf{1}_A = \mathbf{1}_{\{Y=1\}}$, W_0-a.s. with $\{Y = 1\} \in \mathcal{F}_t$ and (i) follows.

(ii) Let $\bar{\mathcal{F}}_t^+ = \bigcap_{\varepsilon>0} \bar{\mathcal{F}}_{t+\varepsilon}$ and let $A \in \bar{\mathcal{F}}_t^+$. We need to show that $A \in \bar{\mathcal{F}}_t$. By assumption, $A \in \bar{\mathcal{F}}_{t+\frac{1}{n}}$ for each $n \geq 1$. Hence, there is $B_n \in \mathcal{F}_{t+\frac{1}{n}}$ such that $W_0(A \vartriangle B_n) = 0$. Define $B \in \mathcal{F}_t^+$ via $\mathbf{1}_B = \limsup_{n\to+\infty} \mathbf{1}_{B_n}$ (Exercise: prove that $B \in \mathcal{F}_t^+$). Then $W_0(A \vartriangle B) = 0$. By (i), there is $C \in \mathcal{F}_t$ such that $W_0(C \vartriangle B) = 0$, and thus $W_0(A \vartriangle C) = 0$ i.e., $A \in \bar{\mathcal{F}}_t$ as required.

(iii) Definition 4.1, (i) follows from the definition of $\bar{\mathcal{F}}_t$, Definition 4.1, (ii) follows from (ii). □

From now on, we assume that $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ satisfies the *usual conditions*, if not said otherwise

If $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ satisfies the *usual conditions* the following theorem allows us to construct numerous stopping times:

Theorem 4.4. *Let $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions and let $A \subseteq \mathbb{R}_+ \times \Omega$ be a progressively measurable set (i.e., $X_t(\omega) = \mathbf{1}_A(t, \omega)$) is a progressively measurable process). Let further D_A be the debut of A*

$$D_A(\omega) = \inf\{t \geq 0 \mid (t, \omega) \in A\}.$$

Then D_A is a stopping time.

The proof uses the following difficult theorem of measure theory:

Theorem 4.5. *Let $(\Lambda, \mathcal{H}, \Pi)$ be a complete probability space and let $H \in \mathcal{B}(\mathbb{R}_+ \times \mathcal{H})$. Then its projection $p(H) = \{\omega \in \Lambda \mid \exists t \geq 0 (t, \omega) \in H\}$ is in $\mathcal{H}$.*

Proof of Theorem 4.4. Fix $t \geq 0$, take $(\Lambda, \mathcal{H}, \Pi) = (\Omega, \mathcal{G}_t, \mathbb{P})$ and $H = ([0, t) \times \Omega) \cap A$. Then $H \in \mathcal{B}(\mathbb{R}_+) \otimes \mathcal{H}$ since A is progressively measurable and thus

$$p(H) = \{\omega \in \Omega : \exists s < t, (s, \omega) \in A\} = \{D_A < t\} \in \mathcal{G}_t.$$

As $\mathcal{G}_t$ is right-continuous, D_A is a stopping time by Exercise 3.34. □

4.2. Martingales

Definition 4.6. An $\mathbb{R}$-valued process $(X_t)_{t\geq 0}$ on filtered probability space $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ is called $\mathcal{G}_t$*-martingale* if it is $\mathcal{G}_t$ adapted, $X_t \in L^1(\mathbb{P})$ for all $t \geq 0$ and for any $0 \leq s \leq t < +\infty$,

$$\mathbb{E}[X_t \mid \mathcal{G}_s] = X_s \quad \mathbb{P}\text{-a.s.}. \tag{4.2}$$

It is called *sub-/super-martingale* if (4.2) holds with "$\geq$"/"$\leq$" respectively.

Exercise 4.7. Let $Z = (Z_t)_{t\geq 0}$ be a process with *independent increments* with respect to $\mathcal{G}_t$, that is for $0 \leq s \leq t$, $Z_t - Z_s$ is independent of $\mathcal{G}_s$ (e.g. Brownian motion has independent increments with respect to $\mathcal{F}_t$, $\mathcal{F}_t^+$ or $\bar{\mathcal{F}}_t$). Show that

(i) If $Z_t \in L^1$ for any $t \geq 0$ then $\tilde{Z}_t := Z_t - \mathbb{E}[Z_t]$ is a martingale.

(ii) If $Z_t \in L^2$ for all $t \geq 0$, then $Z'_t := Z_t^2 - \mathbb{E}[Z_t^2]$ is a martingale.

(iii) If for $\theta \in \mathbb{R}$, $\mathbb{E}[\mathrm{e}^{\theta Z_t}] < +\infty$ for all $t \geq 0$, then

$$\bar{Z}_t := \frac{\mathrm{e}^{\theta Z_t}}{\mathbb{E}[\mathrm{e}^{\theta Z_t}]}$$

is a martingale.

Example 4.8. Using Exercise 4.7 it follows that if B_t is a Brownian motion, $\mathcal{G}_t = \sigma(B_s)_{s\leq t}$ then B_t, $B_t^2 - t$ and $\exp\{\theta B_t - \frac{1}{2}\theta^2 t\}$ are martingales.

Also for $f \in L^2_{\mathrm{loc}}(\mathbb{R}_+, \mathrm{d}t)$,

$$Z_t = \int_0^t f(s)\,\mathrm{d}B_s \quad \left(= G(f\mathbf{1}_{[0,t]})\right)$$

has independent increments which follows from the Gaussian measures properties. This means that

$$\int_0^t f(s)\,\mathrm{d}B_s, \quad \left(\int_0^t f(s)\,\mathrm{d}B_s\right)^2 - \int_0^t f(s)^2\,\mathrm{d}s \quad \text{and} \quad \mathrm{e}^{\theta \int_0^t f(s)\,\mathrm{d}B_s - \frac{\theta^2}{2}\int_0^t f(s)^2\,\mathrm{d}s}$$

are martingales. ♦

Exercise 4.9. Let N_t be the standard Poisson process, $\mathcal{G}_t = \sigma(N_s)_{s\leq t}$. Then N_t has independent increments and $N_t - t$ and $(N_t - t)^2 - t$ are martingales.

Exercise 4.10. (a) Let X_t be a $\mathcal{G}_t$-(sub-) martingale and $f : \mathbb{R} \to \mathbb{R}$ convex (increasing). If $f(X_t) \in L^1$ for any t then $f(X_t)$ is a sub martingale.

(b) Let X_t be a $\mathcal{G}_t$ -sub/super -martingale. Then $\sup_{s\leq t} \mathbb{E}[|X_s|] < +\infty$ for all $t \geq 0$.

We now generalize some results on discrete times to continuous martingales to continuous time.

Theorem 4.11 (Doob's inequality). *Let X be a right-continuous sub martingale. Then*

$$\lambda \mathbb{P}\left(\sup_{0\leq s\leq t} X_s \geq \lambda\right) \leq \mathbb{E}[X_t^+] := \mathbb{E}[X_t \vee 0], \quad \lambda > 0, t \geq 0.$$

Proof. Recall that a discrete-time sub martingale $(Y_n)_{n\in\mathbb{N}_0}$ satisfies the discrete version of this inequality

$$\lambda \mathbb{P}\left(\sup_{m\leq n} Y_m \geq \lambda\right) \leq \mathbb{E}[Y_n^+], \quad \text{for } \lambda > 0, m \in \mathbb{N}.$$

Observe also that for any $t_0 \leq t_1 \leq \ldots \leq t_k = t$ the process $Y_k := X_{t_k}$ is a discrete time sub martingale with respect to the filtration $\mathcal{H}_k := \mathcal{G}_{t_k}$. Hence, for any countable set $D \subset \mathbb{R}_+$ we can write as an increasing limit of such finite sequences,

$$\lambda \mathbb{P}\left(\sup_{s\in[0,t]\cap D} X_s \geq \lambda\right) \leq \mathbb{E}[X_t^+].$$

If one assumes that X is right continuous and takes D to be dense in $\mathbb{R}_+$, one obtains the claim of the theorem. □

In an analogous way, one proves:

Theorem 4.12 (Doobs L^p-inequality). *Let X be a right-continuous non-negative sub martingale. Then*

$$\mathbb{E}\left[\left(\sup_{s\leq t} X_s\right)^p\right]^{1/p} \leq \frac{p}{p-1}\mathbb{E}[X_t^p]^{1/p}, \quad 0 \leq t < +\infty, p \in (1, +\infty).$$

Exercise 4.13. Prove the theorem.

Upcrossings. For $f : T \to \mathbb{R}$ where $T \subseteq \mathbb{R}_+$ and $a < b$, let $U^f_{a,b}(T)$ be the number of upcrossings of $[a,b]$ by f in T, i.e., the maximal k such that there are $s_1 < t_1 < \ldots < s_k < t_k$ with $s_i, t_i \in T$, $f(s_i) < a$, $f(t_i) > b$.

For a discrete-time sub martingale Y_n we have

$$(b-a)\,\mathbb{E}\left[U^Y_{a,b}(\{0,\ldots,n\})\right] \le \mathbb{E}\left[(Y_n - a)_+\right]. \tag{4.3}$$

As a consequence:

Proposition 4.14. *Let $(X_t)_{t\ge0}$ be a sub martingale, D a countable subset of $\mathbb{R}_+$. Then*

$$(b-a)\,\mathbb{E}\left[M^X_{a,b}([0,t]\cap D)\right] \le \mathbb{E}\left[(X_t - a)_+\right]$$

Recall also the discrete-time martingale convergence theorems:

Theorem 4.15. *(a) If Y_n is a sub martingale and $\sup_{n\in\mathbb{N}_0} \mathbb{E}[Y_n^+] < +\infty$, then $Y_n \xrightarrow[n\to+\infty]{\mathbb{P}\text{-a.s.}} Y$.*

(b) If $(Y_n)_{n\in-\mathbb{N}}$ is a sub martingale indexed by $-\mathbb{N}$, $\mathbb{E}[Y_0^+] < +\infty$, then $Y_n \xrightarrow[n\to-\infty]{\mathbb{P}\text{-a.s.}} Y$. Moreover, if $\sup_n \mathbb{E}|Y_n| < +\infty$, then $Y_n \xrightarrow[n\to+\infty]{L^1} Y$.

4.3. Regularity of Martingales

We now show that martingale trajectories posses certain regularity.

Theorem 4.16. *Let $(X_t)_{t\ge0}$ be a sub martingale and $D \subset \mathbb{R}_+$ dense and countable.*

(i) For $\mathbb{P}$-a.e. $\omega \in \Omega$, the map $D \ni s \mapsto X_s(\omega)$ admits for any $t \in [0,+\infty)$ a finite limit from the right $X_{t+}(\omega)$ and for $t \in (0,+\infty)$ a finite limit from the left $X_{t-}(\omega)$.

(ii) For any $t \ge 0$, $X_{t+} \in L^1$ and

$$X_t \le \mathbb{E}[X_{t+} \mid \mathcal{F}_t]$$

with equality if $t \mapsto \mathbb{E}[X_t]$ is right-continuous in t (in particular when X is a *martingale). The process $(X_{t+})_{t\geq 0}$ is then a sub martingale with resepct to $\mathcal{G}_t^+$ (and a martingale, if X is a martingale).*

Proof. Let D be as in Theorem 4.16. Let N be a null-set, where (i) of Theorem 4.16 does not hold and set

$$Y_t(\omega) := \begin{cases} X_{t+}(\omega) & \text{if } \omega \notin N, \\ 0 & \text{if } \omega \in N. \end{cases}$$

Then Y has right continuous trajectories. Indeed, for $\omega \in N$, $t \geq 0$, $\varepsilon > 0$

$$\sup_{s\in[t,t+\varepsilon)} Y_s(\omega) \leq \sup_{s\in(t,t+\varepsilon]\cap D} X_s(\omega), \qquad \inf_{s\in[t,t+\varepsilon)} Y_s(\omega) \geq \inf_{s\in(t,t+\varepsilon]\cap D} X_s(\omega)$$

and

$$\lim_{\varepsilon\searrow 0}\left(\sup_{s\in(t,t+\varepsilon]\cap D} X_s(\omega)\right) = \lim_{\varepsilon\searrow 0}\left(\inf_{s\in(t,t+\varepsilon]\cap D} X_s(\omega)\right) = X_{t+}(\omega) = Y_t(\omega).$$

Similarly, one can show that X has left limits. Moreover, as $\mathcal{G}_t = \mathcal{G}_t^+$,

$$X_t = \mathbb{E}[X_{t+} \mid \mathcal{G}_t] = \mathbb{E}[X_{t+} \mid \mathcal{G}_t^+] = X_{t+} = Y_t \quad \mathbb{P}\text{-a.s.},$$

so Y is a modification of X. As $\mathcal{G}_t$ is completet $Y_t \in \mathcal{G}_t$ and it is a sub martingale. □

Theorem 4.17. *Assume that $(\Omega, \mathcal{G}, \mathcal{G}_t, \mathbb{P})$ satisfies the usual conditions. If $(X_t)_{t\geq 0}$ is a submartingale and $t \mapsto \mathbb{E}[X_t]$ is right-continuous, then X has a modifications which is also a $\mathcal{G}_t$-submartingale whose trajectories are cadlag.*

Proof. □

4.4. Stopping Theorems

This is analogous to Theorem 4.15, (a):

Exercise 4.18. If X is a right-continuous sub martingale and $\sup_{t\in\mathbb{R}_+} \mathbb{E}|X_t| < +\infty$, then there is a random variable $X_{\infty-} \in L^1$ such that $\lim_{t\to+\infty} X_t = X_{\infty-}$ a.s..

Proposition 4.19. *Let $(X_t)_{t\geq 0}$ be a right-continuous martingale. The following are equivalent:*

(i) There exists $X_\infty \in L^1$ such that $X_t = \mathbb{E}[X_\infty \mid \mathcal{G}_t]$ for $t \geq 0$.

(ii) X_t converges as $t \to +\infty$ almost surely and in L^1 to a random variable $X_{\infty-}$.

(iii) The process $(X_t)_{t\geq 0}$ is uniformly integrable.

Proof. "*(i)* $\Longrightarrow$ *(iii)*": We leave this part as an exercise to the reader.

"*(iii)* $\Longrightarrow$ *(ii)*": By Exercise 4.18 we have $\lim_{t\to+\infty} X_t = X_{\infty-}$ almost surely and by uniform integrability also in L^1.

"*(ii)* $\Longrightarrow$ *(i)*": We have $X_s = \mathbb{E}[X_t \mid \mathcal{G}_s]$. Take $t \to +\infty$. Then by *(ii)* also $X_s = \mathbb{E}[X_{\infty-} \mid \mathcal{G}_s]$ i.e., *(i)* holds with $X_\infty = X_{\infty-}$. □

Theorem 4.20. *Let X be a right-continuous martingale such that $X_t = \mathbb{E}[X_\infty \mid \mathcal{G}_t]$. Let S, T be stopping times with $S \leq T$. Then $X_S, X_T \in L^1$ and*

$$X_S = \mathbb{E}[X_T \mid \mathcal{G}_S]$$

where we use the convention $X_T = X_\infty$ on $\{T = +\infty\}$. In particular, $X_S = \mathbb{E}[X_\infty \mid \mathcal{G}_S]$.

Sketch of the proof. The result follows from the discrete-time optimal stopping theorem by a discretisation argument, i.e., by setting

$$T_n = \inf\left\{ t \in \frac{\mathbb{N}}{2^n} \;\middle|\; t > T \right\}, \quad \text{i.e.,} \quad T_n \searrow T.$$

We then conclude using the right-continuity and the backward martingale theorem. □

Corollary 4.21. *If X is a right continuous, uniformly integrable martingale and T a stopping time, then*

$$Y_t = X_{t\wedge T} =: X_t^T$$

is also a uniformly integrable martingale and $X_t^T = \mathbb{E}[X_T \mid \mathcal{G}_t]$.

Exercise 4.22. Prove Corollary 4.21.

4.5. Irregularity of continuous martingales

We have seen that martingale trajectories can be modified to be continuous. We will now show that in the continuous case they are quite irregular.

Recall that the variation of $w \in C$ in $[s,t]$ is given by

$$\mathrm{Variation}_{s,t}(w) = \sup_{\substack{s=t_0<\cdots<t_n=t\\ t_1,\dots,t_{n-1}\in\mathbb{Q}}} \sum_{i=1}^{n} |w(t_i) - w(t_{i-1})|. \tag{4.4}$$

Lemma 4.23. *Let X be a continuous martingale with $X_0 = 0$ and such that $s \mapsto X_s(\omega)$ has a finite variation on every $[0,t]$, $t \geq 0$, for almost any $\omega \in \Omega$. Then $X \equiv 0$ $\mathbb{P}$-a.s..*

Proof. Let S_n be stopping times given by

$$S_n = \inf\{s \geq 0 \mid |X_s| \geq n\} \wedge \inf\left\{ s \geq 0 \mid \mathrm{Variation}_{0,s}(X) \geq n \right\} \wedge n.$$

By continuity of X and the assumption of finite variation $S_n \nearrow +\infty$ as $n \to +\infty$. Moreover, $X_{t\wedge S_n}$ is a bounded martingale for every $n \geq 0$ (exercise). It is thus sufficient to show (4.4) for bounded continuous martingales with bounded variation.

For those, by the martingale property,

$$\begin{aligned}
\mathbb{E}[X_t^2] &= \mathbb{E}\left[\left(\sum_{k=1}^{2^\ell}\left(X_{t\frac{k+1}{2^\ell}} - X_{t\frac{k}{2^\ell}}\right)\right)^2\right] \\
&= \mathbb{E}\left[\sum_{k=1}^{2^\ell}\left(X_{t\frac{k+1}{2^\ell}} - X_{t\frac{k}{2^\ell}}\right)^2\right] \\
&\leq \mathbb{E}\left[\underbrace{\sup_{1\leq k\leq 2^\ell}\left|X_{t\frac{k+1}{2^\ell}} - X_{t\frac{k}{2^\ell}}\right|}_{\xrightarrow{\ell\to+\infty} 0 \text{ by continuity}} \cdot \underbrace{\sum_{i=1}^{2^\ell}\left|X_{t\frac{i+1}{2^\ell}} - X_{t\frac{i}{2^\ell}}\right|}_{\leq \mathrm{Variation}_{0,t}(X)<+\infty}\right] \xrightarrow{\ell\to+\infty} 0
\end{aligned}$$

It follows that $\mathbb{P}$-a.s. for all $t \in \mathbb{Q}$, $X_t \equiv 0$ so $X \equiv 0$ $\mathbb{P}$-a.s. by continuity. ☐

4.6. Functions and processes of finite variation

Let $a : [0,T] \to \mathbb{R}$ be of finite variation, i.e., $\mathrm{Variation}_{0,T}(a) < +\infty$. From measure theory it follows that there exists a finite signed measure μ on $([0,T], \mathcal{B}([0,T]))$

associated with a which satisfies

$$\mu([0,t]) = a(t), \quad \text{for any } t \in [0,T]. \tag{4.5}$$

The measure μ is the *Lebesgue-Stieltjes measure of a*. We can write $\mu = \mu_+ - \mu_-$ for two finite positive measure with disjoint supports. This decomposition is unique (Jordan-Hahn decomposition). Consequently, a is a difference of two increasing functions $t \mapsto \mu_+([0,t])$ and $t \mapsto \mu_-([0,t])$. If a is continuous and $a(0) = 0$, then μ_+ and μ_- have no atoms[1] and the above functions are also continuous. Finally, we denote the variation of μ as $|\mu| = \mu_+ + \mu_-$.

If $f : [0,T] \to \mathbb{R}$ is measurable and $\int_{[0,T]} |f| \, \mathrm{d}|\mu| < +\infty$ we set

$$\begin{aligned} \int_0^T f(s) \, \mathrm{d}a(s) &:= \int_{[0,T]} f(s) \mu(\mathrm{d}s), \\ \int_0^T f(s) |\mathrm{d}a(s)| &:= \int_{[0,T]} f(s) |\mu|(\mathrm{d}s) \end{aligned} \tag{4.6}$$

Observe that the map $t \mapsto \int_0^t f(s) \, \mathrm{d}a(s)$ is also of finite variation with associated Lebesgue-Stieltjes measure $\mu'(\mathrm{d}s) = f(s)\mu(\mathrm{d}s)$.

Exercise 4.24. Let $f : [0,T] \to \mathbb{R}$ be a continuous function and $0 = t_0^n < \cdots < t_{k_n}^n = T$ a sequence of subdivisions with step size tending to zero. Then

$$\int_0^T f(s) \, \mathrm{d}a(s) = \lim_{n \to +\infty} \sum_{i=1}^{k_n} f(t_{i-1}^n) \Big(a(t_i^n) - a(t_{i-1}^n) \Big).$$

Definition 4.25. A function $a : \mathbb{R}_+ \to \mathbb{R}$ is called *(locally) of finite variation* if it is of finite variation on every $[0,T]$ for any $T \geq 0$. In this case we define

$$\int_0^{+\infty} f(s) \, \mathrm{d}a(s)$$

for all f such that

$$\int_0^{+\infty} |f(s)| |\mathrm{d}a(s)| = \sup_{T \geq 0} \int_0^T |f(s)| |\mathrm{d}a(s)|.$$

In the following let $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions .

[1] An atom is a measurable set which has positive measure and contains no set of smaller positive measure.

Definition 4.26. A *process of finite variation* $(A_t)_{t\geq 0}$ is an adapted stochastic process whose trajectories are of finite variation as in Definition 4.25. The process A is called *increasing*, if its trajectories are increasing.

Remark 4.27. We will mostly consider *continuous* processes with finite variation and $A_0 \equiv 0$. In this case the associated (random) Lebesgue-Stieltjes measure has no atoms. ♦

Proposition 4.28 (Integral with resepct to A). *Let A be a process of finite variation and H a progressively measurable process such that for any $t \geq 0$ and $\omega \in \Omega$,*

$$\int_0^t |H_s(\omega)||\mathrm{d}A_s(\omega)| < +\infty.$$

Then the process $H \cdot A$ given by

$$(H \cdot A)_t(\omega) = \int_0^t H_s(\omega)\,\mathrm{d}A_s(\omega)$$

is also of finite variation and continuous if A is.

Proof. The fact that the trajectories of $H \cdot A$ are of finite variation (and continuous) follows directly from the definition, cf. (4.6). We only need to show that $H \cdot A$ is adapted, that is $(H \cdot A)_t \in \mathcal{G}_t$.

This is satisfied if $H_s(\omega) = \mathbf{1}_{(u,v]}\mathbf{1}_\Gamma(\omega)$ with $(u, v] \subseteq [0, t]$, $\Gamma \in \mathcal{G}_t$, thus by Dynkin's argument, for $H_s(\omega) = \mathbf{1}_G(s, w)$ with $G \in \mathcal{B}([0, t]) \otimes \mathcal{G}_t$.

General progressively measurable processes H can be written as a limit of step processes $H^n = \sum_{i=1}^{k_n} c_i \mathbf{1}_{G_i}$, $G_i \in \mathcal{B}([0, t]) \otimes \mathcal{G}_t$, $c_i \in \mathbb{R}$ with $|H^n| \leq H$, which ensures that $(H^n \cdot A)(\omega) \xrightarrow{n\to+\infty} (H \cdot A)(\omega)$ for any $\omega \in \Omega$ by dominated convergence. This proves that $H \cdot A$ is adapted. □

4.7. Local martingales

Let $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ satisfy the usual conditions. If T is a stopping time and X a continuous stochastic process, we define the *stopped process* X^T by

$$X_t^T := X_{t\wedge T}, \quad t \geq 0. \tag{4.7}$$

Definition 4.29. An adapted continuous process $(M_t)_{t\geq 0}$ with $M_0 = 0$ is called *(continuous) local martingale* if there exists an increasing sequence $(T_n)_{n\in\mathbb{N}}$ of stopping times such that $T_n \nearrow +\infty$ and for any $n \in \mathbb{N}$, M^{T_n} is a uniformly integrable martingale. If $M_0 \neq 0$, we say that M is a (continuous) local martingale, if $M_t = M_0 + N_t$ for any $t \geq 0$ where $(N_t)_{t\geq 0}$ is a locall martingale with $N_0 = 0$.
In both cases, we say that $(T_n)_{n\in\mathbb{N}}$ reduces M and in both cases $(M^{T_n})_{n\in\mathbb{N}}$ is a uniformly integrable martingale.

Remark 4.30. It is not necessary that $M_t \in L^1$ for a local martingale. ♦

Proposition 4.31 (Properties of local martingales).

(i) *A continuous martingale is a local martingale (take $T_n = n$).*

(ii) *In Definition 4.29 if the local martingale started at 0 we may replace "uniformly integrable martingale" by martingale (as by can replace T_n by $T_n \wedge n$).*

(iii) *If M is a local martingale and T a stopping time, then M^T is a local martingale.*

(iv) *If $(T_n)_{n\in\mathbb{N}}$ reduces M and S_n is a sequence of stopping times with $S_n \nearrow +\infty$ then $(S_n \wedge T_n)_{n\in\mathbb{N}}$ reduces M.*

(v) *The space of local martingales is a vector space.*

Proof. We leave the proof as an exercise to the reader. □

Exercise 4.32. Prove *(v)* by applying *(iv)*.

Proposition 4.33. (i) *A non-negative local martingale M with $M_0 \in L^1$ is a super martingale.*

(ii) *If M is a local martingale and $|M_t| \leq Z$ for any $t \geq 0$ with $Z \in L^1$, then M is a martingale.*

(iii) *If M is a local martingale, $M_0 = 0$, then $T_n = \inf\{t \geq 0 \mid |M_t| \geq n\}$ reduces M.*

Proof. *(i)*: Let $M_t = M_0 + N_t$ and let $(T_n)_{n\in\mathbb{N}}$ be a stopping time reducing N. Then for $s < t$ and $n \in \mathbb{N}$,

$$N_{s\wedge T_n} = \mathbb{E}[N_{t\wedge T_n} \mid \mathcal{G}_s]$$

and thus, adding M_0 which is in L^1,

$$M_{s\wedge T_n} = \mathbb{E}[M_{t\wedge T_n} \mid \mathcal{G}_s]$$

As $M \geq 0$, by Fatou's lemma for conditional expectation,

$$M_s \geq \mathbb{E}[M_t \mid \mathcal{G}_s].$$

Taking $s = 0$, we also see that $\mathbb{E}[M_t] \leq \mathbb{E}[M_0]$, i.e., $M_t \in L^1$ and M is thus a super martingale.

(ii): Follows by the same steps, using dominated convergence instead of Fatou's lemma.

(iii): The fact that M^{T_n} is a uniformly integrable martingale follows from *(ii)*, $T_n \nearrow +\infty$ follows from the continuity. $\square$

Lemma 4.34. *Let M be a continuous local martingale with $M_0 = 0$. If M is a process of finite variation, then M is indistinguishable from* 0.

Proof. This is a small extension of Lemma 4.23. $\square$

4.8. Quadratic variation of local martingales

Recall that in Proposition 3.22 we proved that quadratic variation of Brownian motion is t. Moreover by Example 4.8, $B_t^2 - t$ is a martingale. We now prove that for arbitrary continuous local martingales M there is an increasing process $\langle M \rangle$ which has similar properties as the process "$t \mapsto t$" in the case of Brownian motion. This process will play a big role when developing the stochastic integral.

Theorem 4.35. *Let $(M_t)_{t\geq 0}$ be a (continuous) local martingale on $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. Then there exists an essentially unique (i.e., defined modulo indistinguishability) increasing process denoted $\langle M \rangle = (\langle M \rangle_t)_{t\geq 0}$ with $\langle M \rangle_0 = 0$ such that $M_t^2 - \langle M \rangle_t$ is a local martingale. Moreover, for any $T > 0$, if $0 = t_0^n < t_1^n < \cdots < t_{k_n}^n = T$ is a sequence of refining partitions with mesh tending to zero,*

$$\langle M \rangle_t = \lim_{n\to+\infty} \sum_{i=1}^{k_n} (M_{t_i^n} - M_{t_{i-1}^n})^2 \quad \textit{in probability.} \tag{4.8}$$

The process $\langle M \rangle$ is called the quadratic variation of the local martingale M.

Proof. **Step 1: Uniqueness.** Let A, B two adapted continuous increasing processes with $A_0 = B_0 = 0$ and $M_t^2 - A_t$, $M_t^2 - B_t$ being local martingales. Then $(M_t^2 - A_t) - (M_t^2 - B_t)$ is again a continuous local martingale. Moreover, it is of finite variation, so by Lemma 4.34, $B_t - A_t \equiv 0$ $\mathbb{P}$-a.s., yielding the essential uniqueness.

Step 2: Existence. We are going to use Equation (4.8) to construct $\langle M \rangle$. To this end, we fix $T > 0$ and $((t_i^n)_{i=1}^{k_n})_{n\in\mathbb{N}}$ as in the theorem and define

$$I_t^n := \sum_{i=1}^{k_n} M_{t_{i-1}^n} \left(M_{t_i^n \wedge t} - M_{t_{i-1}^n \wedge t} \right), \qquad n \in \mathbb{N}, t \in [0, T].$$

Case 1: $M_0 = 0$, M is a bounded martingale. The role of I^n is the following: First, for any $t \in [0, T]$ and $n \in \mathbb{N}$,

$$\begin{aligned} M_t^2 - 2I_t^n &= \left(\sum_{i=1}^{\ell} (M_{t_i^n} - M_{t_{i-1}^n}) + (M_t - M_{t_\ell^n}) \right)^2 - 2 \sum_{i=1}^{\ell} M_{t_{i-1}^n} (M_{t_i^n} - M_{t_{i-1}^n}) \\ &\quad - 2M_{t_\ell^n}(M_t - M_{t_\ell^n}) \\ &\overset{M_0=0}{=} \sum_{i=1}^{\ell} (M_{t_i}^n - M_{t_{i-1}^n})^2 + (M_t - M_{t_\ell^n})^2 \end{aligned} \tag{4.9}$$

where ℓ is such that $t_\ell^n \leq t < t_{\ell+1}^n$, which limits I^n with (4.8). Moreover, for any n, $(I_t^n)_{t\in[0,T]}$ is a continuous martingale which makes ist suitable for computations. To sree that, let $s \leq t \leq T$, $A \in \mathcal{G}_s$, and □

4.9. Propoerties of quadratic variation

Lemma 4.36. *Let M be a local martingale and T a stopping time. Then*

$$\langle M^T \rangle_t = \langle M \rangle_{t\wedge T} = \langle M \rangle_t^T.$$

Proof. Observe that $M_{t\wedge T}^2 - \langle M \rangle_{t\wedge T}$ is a local martingale by Proposition 4.31, *(iii)*, i.e., $\langle M \rangle_{t\wedge T}$ satisfies the defining property of quadratic variation of M^T. □

For an increasing process $(A_t)_{t\geq 0}$ we define $A_\infty := \lim_{t\to+\infty} A_t$. The following theorem links the properties of $\langle M \rangle$ to those of M.

Theorem 4.37. *(i) If M is a martingale, bounded in L^2, then $\mathbb{E}[\langle M\rangle_\infty] < +\infty$ and $M_t^2 - \langle M\rangle_t$ is a uniformly integrable martingale.*

(ii) If $\mathbb{E}[\langle M\rangle_\infty] < +\infty$, then M is a martingale bounded in L^2.

(iii) The following are equivalent:

(a) M is a martingale with $\mathbb{E}[M_t^2] < +\infty$ for any $t \in \mathbb{R}_+$;

(b) $\mathbb{E}[\langle M\rangle_t] < +\infty$ for any $t \in \mathbb{R}_+$.

In this case $M_t^2 - \langle M\rangle_t$ is a martingale.

Remark 4.38. There are L^2 bounded local martingales which are not true martingale (we will see one later), so in *(i)* one should really assume that M is martingale. On the technical side, Doob's inequality does not hold for local martingales. ♦

Proof of Theorem 4.37. *(i)*: If M is an L^2 bounded martingale, Doob's inequality yields

$$\mathbb{E}\left[\sup_{t\geq 0} M_t^2\right] \leq 4 \sup_{t\geq 0} \mathbb{E}[M_t^2] < +\infty.$$

Hence, M is uniformly integrable. By Proposition 4.19, there is $M_\infty = \lim_{t\to+\infty} M_t$ almost surely and in L^2.

For $n \in \mathbb{N}$, let $S_n := \inf\{t \geq 0 \mid \langle M\rangle_t \geq 0\}$. Then S_n is a stopping time and $\langle M\rangle_{t\wedge S_n} \leq n$. This means, that the local martingale $M_{t\wedge S_n}^2 - \langle M\rangle_{t\wedge S_n}$ is dominated by the integrable random variable $\sup_{t\geq 0} M_t^2 + n$. Hence, by Proposition 4.33, it is a true martingale, and thus

$$\mathbb{E}[\langle M\rangle_{S_n}] = \mathbb{E}[\langle M^2\rangle_{S_n}].$$

Using the same arguments for $n \to +\infty$ we find

$$\mathbb{E}[\langle M\rangle_\infty] = \mathbb{E}[M_\infty^2] < +\infty.$$

In particular, the local martingale $M_t^2 - \langle M\rangle_t$ is dominated by the integrable random variable $\sup_{t\geq 0} M_t^2 + \langle M\rangle_\infty$, so it is uniformly integrable by Proposition 4.33 again.

(ii): Assume that $\mathbb{E}[\langle M\rangle_\infty] < +\infty$. Let $T_n = \inf\{t \mid |M_t| \geq n\}$, i.e., M^{T_n} is a bounded martingale. The local martingale $M_{t\wedge T_n}^2 - \langle M\rangle_{t\wedge T_n}$ is dominated by $n + \langle M\rangle_\infty$, i.e., it is uniformly integrable by Proposition 4.33. Using the stopping theorem, Theorem 4.20, for a stopping time S which is a.s. finite,

$$\mathbb{E}[M_{S\wedge T_n}^2] = \mathbb{E}[\langle M\rangle_{S\wedge T_n}]$$

and thus by Fatou's lemma and monotone convergence theorem

$$\mathbb{E}[M_S^2] \leq \mathbb{E}[\langle M\rangle_S] \leq \mathbb{E}[\langle M\rangle_\infty] < +\infty. \tag{4.10}$$

This means that the family

$$\{M_S \mid S \text{ is a finite stopping time}\}$$

is uniformly integrable. We may thus pass to the L^1-limit as $n \to +\infty$ in the equality $\mathbb{E}[M_{t\wedge T_n} \mid \mathcal{F}_s] = M_{s\wedge T_n}$, $s \leq t$ to deduce $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$, i.e., M is a martingale. It is bounded in L^2 by (4.10).

(iii): Let $a > 0$. By *(i)* and *(ii)*, $\mathbb{E}[\langle M\rangle_a] < +\infty$ if and only if M_t^a is an L^2-bounded true martingale. This proves the equivalence of *(a)* and *(b)*. Point *(i)* then implies that $M_{t\wedge a}^2 - \langle M\rangle_{t\wedge a}$ is a martingale and the last claim follows. □

Corollary 4.39. *Let M be a local martingale with $M_0 = 0$. Then $\langle M\rangle_t = 0$ a.s. for all $t \geq 0$ if and only if M is indistinguishable from* 0.

4.10. Bracket process of two local martingales

Definition 4.40. If M and N are local martingales we define

$$\langle M, N\rangle_t := \frac{1}{2}\Big(\langle M+N\rangle_t - \langle M\rangle_t - \langle N\rangle_t\Big)$$

Remark 4.41. With this notation, $\langle M, M\rangle_t = \langle M\rangle_t$. Some authors thus prefer to write $\langle M, M\rangle$ instead of $\langle M\rangle$. ♦

Proposition 4.42.

(i) the process $\langle M, N\rangle$ is the (esentially) unique process of finite variation started from 0 *such that $M_tN_t - \langle M, N\rangle_t$ is a local martingale.*

(ii) The map $(M, N) \mapsto \langle M, N\rangle$ is bilinear and symmetric.

(iii) If $0 = t_0^n < \cdots < t_{k_n}^n$ is a sequence of refining partitions of $[0, t]$ with meshsize

tending to 0, then

$$\lim_{n\to+\infty}\sum_{i=1}^{k_n}\left(M_{t_i^n}-M_{t_{i-1}^n}\right)\left(N_{t_i^n}-N_{t_{i-1}^n}\right)=\langle M,N\rangle_t \quad \textit{in probability.}$$

(iv) For any stopping time T,

$$\langle M^T,N^T\rangle_t=\langle M^T,N\rangle_t=\langle M,N\rangle_{t\wedge T}.$$

Proof. *(i)*: uniqueness follows as in Theorem 4.35 and the stated property follows directly from the analogous statement in Theorem 4.35.

(iii): This follows from the analogous statement in Theorem 4.35 as well.

(ii): This point is a consequence of *(iii)*.

(iv): From *(iii)* we can see that a.s.,

$$\begin{aligned}\langle M^T,N^T\rangle_t=\langle M^T,N\rangle_t&=\langle M,N\rangle_t && \text{on } \{T\geq t\} \text{ and}\\ \langle M^T,N^T\rangle_t-\langle M^T,N^T\rangle_s=\langle M^T,N\rangle_t-\langle M^T,N\rangle_s&=0, && \text{on } \{T\leq s\leq t\},\end{aligned}$$

which yields *(iv)*. □

Definition 4.43. Two local martingales M,N are called orthogonal if $\langle M,N\rangle=0$, in which case MN is a local martingale.

Example 4.44. Let B be a $\mathcal{G}_t$-Brownian motion (meaning it is $\mathcal{G}_t$-adapted and B_t-B_s is independent of $\mathcal{G}_s$ for $s\leq t$). Then B is a (local) martingale and $\langle B\rangle_t=t$. If B' is another $\mathcal{G}_t$ Brownian motion, independent of B, then B and B' are orthogonal. To see this observe that $\frac{1}{\sqrt{2}}(B_t+B'_t)$ is again a Brownian motion. ♦

Proposition 4.45 (Kunita-Watanabe inequality). *Let M,N be two local martingales and H,K be two measurable processes. Then $\mathbb{P}$-a.s.,*

$$\int_0^{+\infty}|H_s||K_s||\mathrm{d}\langle M,N\rangle_s|\leq\left(\int_0^{+\infty}H_s^2\,\mathrm{d}\langle M\rangle_s\right)^{1/2}\left(\int_0^{+\infty}K_s^2\,\mathrm{d}\langle N\rangle_s\right)^{1/2}. \tag{4.11}$$

Here $|\mathrm{d}\langle M,N\rangle_s|$ denotes the total variation of the signed measure $\mathrm{d}\langle M,N\rangle_s$.

Proof. For $s < t$ define $\langle M,N\rangle_s^t = \langle M,N\rangle_t - \langle M,N\rangle_s$. Then $\mathbb{P}$-a.s. for all $\lambda \in \mathbb{Q}$ and $s < t$, $s,t \in \mathbb{Q}$,

$$\langle M+\lambda N\rangle_s^t = \langle M\rangle_s^t + 2\lambda\langle M,N\rangle_s^t + \lambda^2\langle N\rangle_s^t \geq 0.$$

Taking discriminant in λ we deduce that $\mathbb{P}$-a.s. for any $s,t \in \mathbb{Q}$,

$$|\langle M,N\rangle_s^t| \leq \left(\langle M\rangle_s^t\langle N\rangle_s^t\right)^{1/2}. \tag{4.12}$$

By continuity this extends to all $s,t \in \mathbb{R}_+$ with $s \leq t$. From this we get

$$\int_s^t |\mathrm{d}\langle M,N\rangle_u| \leq \left(\langle M\rangle_s^t\langle N\rangle_s^t\right)^{1/2} = \left(\int_s^t \mathrm{d}\langle M\rangle_u \int_s^t \mathrm{d}\langle M\rangle_u\right)^{1/2}. \tag{4.13}$$

Indeed for any subdivision $s = t_0 < \cdots < t_n = t$,

$$\begin{aligned}\sum_{i=1}^n \left|\langle M,N\rangle_{t_{i-1}}^{t_i}\right| &\overset{(4.12)}{\leq} \sum_{i=1}^n \left(\langle M\rangle_{t_{i-1}}^{t_i}\langle N\rangle_{t_{i-1}}^{t_i}\right)^{1/2} \\ &\overset{\mathrm{CS}}{\leq} \left(\sum_{i=1}^n \langle M\rangle_{t_{i-1}}^{t_i}\right)^{1/2}\left(\sum_{i=1}^n \langle N\rangle_{t_{i-1}}^{t_i}\right)^{1/2} \\ &\leq \left(\langle M\rangle_s^t\langle M\rangle_s^t\right)^{1/2}\end{aligned}$$

and (4.13) follows from the properties of Lebesgue-Stieltjes integral, e.g., Exercise 4.24. Equation (4.13) implies that for any Borel set $A \subseteq \mathbb{R}_+$, by similar arguments,

$$\int_A |\mathrm{d}\langle M,N\rangle_u| \leq \left(\int_A \mathrm{d}\langle M\rangle_u \int_A \mathrm{d}\langle N\rangle_u\right)^{1/2}. \tag{4.14}$$

Let now $h = \sum_i \lambda \mathbf{1}_{A_i}$ and $k = \sum_i \mu_i \mathbf{1}_{A_i}$ be two positive step functions, then by Cauchy-Schwartz inequality again,

$$\begin{aligned}\int h(s)k(s)|\mathrm{d}\langle M,N\rangle_s| &= \sum_i \lambda_i\mu_i \int_{A_i} |\mathrm{d}\langle M,N\rangle_s| \\ &\leq \left(\sum_i \lambda_i^2 \int_{A_i} \mathrm{d}\langle M\rangle_s\right)^{1/2}\left(\sum_i \mu_i^2 \int_{A_i} \mathrm{d}\langle N\rangle_s\right)^{1/2} \\ &\leq \left(\int h^2(s)\,\mathrm{d}\langle M\rangle_s\right)^{1/2}\left(\int k^2(s)\,\mathrm{d}\langle N\rangle_s\right)^{1/2}\end{aligned}$$

for a.e. ω, where (4.14) holds. It remains to write any measurable $H_s(\omega)$, $K_s(\omega)$ as a difference of two limits of sequences of positive step functions. $\square$

4.11. Continuous semi martingales

Definition 4.46. A process $(X_t)_{t\geq 0}$ is called *continuous semi martingale* if it can be written as

$$X_t = X_0 + M_t + A_t,$$

where $(M_t)_{t\geq 0}$ is a local martingale (with $M_0 = 0$) and A is a process of finite variation.

This above decomposition is essentially unique. This follows from Lemma 4.34 using the standard arguments.

If $Y = Y_0 + N_t + B_t$ is another continuous semi martingale with the corresponding decomposition, we define

$$\langle X, Y\rangle_t = \langle M, N\rangle_t.$$

In particular, $\langle X\rangle_t = \langle M\rangle_t$.

Proposition 4.47. *Let* $\left((t_i^n)_{i=1,\dots,k_n}\right)_{n\in\mathbb{N}}$ *be a refining sequence of partitions of* $[0, t]$ *with mesh size tending to 0. Then*

$$\lim_{n\to+\infty}\sum_{i=1}^{k_n}\left(X_{t_i^n} - X_{t_{i-1}^n}\right)\left(Y_{t_i^n} - Y_{t_{i-1}^n}\right) = \langle X, Y\rangle_t, \quad \textit{in probability.}$$

Proof. We consider only $X = Y$, the general result follows by polarisation.

$$\begin{aligned}&\sum_{i=1}^{k_n}\left(X^n_{t_i} - X_{t_{i-1}^n}\right)^2\\ &= \sum_i\left(M_{t_i^n} - M_{t_{i-1}^n}\right)^2 + \sum_i\left(A_{t_i^n} - A_{t_{i-1}^n}\right)^2 + 2\sum_i\left(M_{t_i^n} - M_{t_{i-1}^n}\right)\left(A_{t_i^n} - A_{t_{i-1}^n}\right)\end{aligned} \tag{4.15}$$

The first term converges to $\langle M\rangle_t$ by Theorem 4.35 and $\langle M\rangle = \langle X\rangle$.

On the other hand

$$\sum_i\left(A_{t_i^n} - A_{t_{i-1}^n}\right)^2 \leq \underbrace{\left(\sup_i|A_{t_i^n} - A_{t_{i-1}^n}|\right)}_{\xrightarrow{n\to+\infty}0}\underbrace{\sum_i|A_{t_i^n} - A_{t_{i-1}^n}|}_{\leq\int_0^t|\mathrm{d}A_s|<+\infty} \xrightarrow[n\to+\infty]{\text{Monotone Convergence}} 0.$$

The same arguments also prove that the last term converges to 0 as $n \to +\infty$. $\square$

5. Stochastic Integral

We now develop the theory of the stochastic integral with respect to martingales, and later semimargingales. We know that these processes are a.s. of infinite variation which preludes the definition of the "Stieltjes-type" integral with respect to $\mathrm{d}M_t(\omega)$ path wise. We will see that the big role will be played by the quadratic variation process $\langle M \rangle$ constructed in the last chapter.

The stochastic integral was first constructed by Itô (1942) for Brownian motion, and extended to general martingales by Kunita and Watanabe (1967).

In the whole chapter we assume that $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ satisfies the *usual conditions*.

5.1. Stochastic Integrals for Martingales Bounded in L^2

Definition 5.1. We denote by $\mathbb{H}^2$ the space of continuous martingales M which are bounded in L^2,

$$\sup_{t\geq 0} \|M_t\| < +\infty$$

with $M_0 = 0$.

By Theorem 4.37, if $M \in \mathbb{H}^2$, then $\mathbb{E}\langle M \rangle_\infty < +\infty$, and thus if $M, N \in \mathbb{H}^2$ then $\mathbb{E}|\langle M, N\rangle_\infty| < +\infty$. By Proposition 4.45,

$$\mathbb{E}|\langle M, N\rangle_\infty| \leq \mathbb{E}\left[\int_0^{+\infty} |\mathrm{d}\langle M, N\rangle_s|\right] \leq \sqrt{\mathbb{E}[\langle M\rangle_\infty]}\sqrt{\mathbb{E}[\langle N\rangle_\infty]} \tag{5.1}$$

We can thus define a scalar product on $\mathbb{H}^2$ by

$$(M, N)_{\mathbb{H}^2} = \mathbb{E}[\langle M, N\rangle_\infty]. \tag{5.2}$$

By Corollary 4.39, $(M, M)_{\mathbb{H}^2} = 0$ if and only if $M \equiv 0$ (identifying indistinguishable processes). We also define a norm on $\mathbb{H}^2$ by

$$\|M\|_{\mathbb{H}^2} = \sqrt{(M,M)_{\mathbb{H}^2}} = \sqrt{\mathbb{E}[\langle M\rangle_\infty]}. \tag{5.3}$$

Proposition 5.2. *The space $\mathbb{H}^2$ endowed with the scalar product $(\cdot,\cdot)_{\mathbb{H}^2}$ is a Hilbert-space.*

Proof. We want to show that $\mathbb{H}^2$ is complete. Let $(M^n)_{n\in\mathbb{N}}$ be a Cauchy sequence in $\mathbb{H}^2$. Then, by definition of $(\cdot,\cdot)_{\mathbb{H}^2}$,

$$\lim_{m,n\to+\infty} \mathbb{E}[(M^n_\infty - M^m_\infty)^2] = \lim_{m,n}[\langle M^n - M^m\rangle_\infty] = 0.$$

Thus, by Doob's inequality,

$$\lim_{m,n\to+\infty} \mathbb{E}\left[\sup_{t\geq 0}(M^n_t - M^m_t)^2\right] = 0. \tag{5.4}$$

Hence, along some subsequence $(n_k)_{k\in\mathbb{N}}$ (using that convergence in L^2 implies convergence a.e. up to subsequence),

$$\lim_{\ell,k\to+\infty} \sup_{t\geq 0}|M^{n_k}_t - M^{n_\ell}_t| = 0, \quad \mathbb{P}\text{-a.s.}. \tag{5.5}$$

Therefore, for $\mathbb{P}$-a.e. ω there exists a limit $M_t(\omega) = \lim_{k\to+\infty} M^{n_k}_t(\omega)$ and by the uniform convergence (5.5) said limit is continuous in t.

Setting $M\equiv 0$ on the $\mathbb{P}$-negligible set where M is not defined, we see from (5.4) that $M^{n_k}_t \xrightarrow[k\to+\infty]{} M_t$ in L^2 also. Passing to the limit in the martingale property of M^{n_k}, we see that M is a martingale. Moreover, since $(M^n_t)_{t\geq 0}$ is uniformly bounded in L^2 for any n the same is true for $(M_t)_{t\geq 0}$. Finally,

$$\lim_{k\to+\infty} \mathbb{E}\left[\langle M^{n_k} - M\rangle_\infty\right] = \lim_{k\to+\infty} \mathbb{E}\left[\left(M^{n_k}_\infty - M_\infty\right)^2\right] = 0,$$

so M^{n_k} converges to M also in $\mathbb{H}^2$. □

Recall that a process H is *progressively measurable* if the map

$$([0,T]\times\Omega, \mathcal{B}([0,t])\otimes\mathcal{G}_t) \ni (t,\omega) \mapsto H_t(\omega) \in (\mathbb{R},\mathcal{B}(\mathbb{R}))$$

is measurable for all $T\geq 0$, and that $A\in\mathcal{B}(\mathbb{R}_+)\otimes\mathcal{G}$ is called *progressively measurable* if $(t,\omega)\mapsto \mathbf{1}_A(t,\omega)$ is a progressively measurable process.

We define the *progression σ-algebra* Prog as the smallest σ-algebra containing all progressively measurable sets.

Exercise 5.3. H is progressively measurable if and only if $(\mathbb{R}_+ \times \Omega, \mathrm{Prog}) \ni H_t(\omega) \in (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable.

For $M \in \mathbb{H}^2$ we define

$$L^2(M) = L^2(\mathbb{R}_+ \times \Omega, \mathrm{Prog}, \mathrm{d}\mathbb{P}\,\mathrm{d}\langle M\rangle_s). \tag{5.6}$$

That is $L^2(M)$ is the space of progressively measurable processes H such that

$$\mathbb{E}\left[\int_0^{+\infty} H_s^2 \,\mathrm{d}\langle M\rangle_s\right] < +\infty.$$

As every L^2-space, $L^2(M)$ is a Hilbert space with scalar product

$$(H, K)_{\mathbb{H}^2(M)} = \mathbb{E}\left[\int_0^{+\infty} H_s K_s \,\mathrm{d}\langle M\rangle_s\right].$$

Definition 5.4. A process H is called *simple* if

$$H_s(\omega) = \sum_{i=0}^{n-1} H_{(i)}(\omega)\mathbf{1}_{(t_i, t_{i+1}]}(s),$$

where $n \in \mathbb{N}$, $0 = t_0 < \cdots < t_n$, and $H_{(i)}$ is a bounded $\mathcal{G}_{t_i}$-measurable random variable. We will use $\mathcal{E}$ to denote the vector space of all simple processes.

Proposition 5.5. *For every $M \in \mathbb{H}^2$, $\mathcal{E}$ is dense in $L^2(M)$.*

Proof. It is sufficient to show that if $K \in L^2(M)$ is orthogonal to $\mathcal{E}$ then $K = 0$. Assume that K is orthogonal to $\mathcal{E}$. Let $0 \le s < t$ and let Z be a $\mathcal{G}_s$ measurable and bounded random variable. Then, by orthogonality

$$0 = \left(K, Z\mathbf{1}_{(s,t]}\right)_{\mathbb{H}^2(M)} = \mathbb{E}\left[Z \int_s^t K_u \,\mathrm{d}\langle M\rangle_u\right]. \tag{5.7}$$

Let now

$$X_t = \int_0^t K_u \,\mathrm{d}\langle M\rangle_u, \qquad t \ge 0. \tag{5.8}$$

By Cauchy-Schwartz inequality,

$$X_t \le \sqrt{\int_0^t K_u^2 \,\mathrm{d}\langle M\rangle_u}\sqrt{\int_0^t \mathrm{d}\langle M\rangle_u} < +\infty, \quad \text{a.s.},$$

as $K \in L^2(M)$ and $M \in \mathbb{H}^2$, so X_t is a.s. well defined and also $X_t \in L^1$ for any $t \geq 0$. Equation (5.7) implies that $\mathbb{E}\left[Z(X_t - X_s)\right] = 0$ so X is a martingale. By (5.8), $X_0 = 0$ and X has finite variation.

So by Lemma 4.23, $X = 0$. Hence,

$$\int_0^t K_u \, \mathrm{d}\langle M\rangle_u = 0 \quad \text{for all } t \geq 0 \text{ a.s.}$$

which implies $K_u = 0$ $\mathrm{d}\langle M\rangle$-a.e. $\mathbb{P}$-a.s. which means $K = 0$ in $L^2(M)$. □

Theorem 5.6. *Let $M \in \mathbb{H}^2$ For $H \in \mathcal{E}$ of the form*

$$H_s(\omega) = \sum_{i=0}^{n-1} H_{(i)}(\omega) \mathbf{1}_{(t_i, t_{i+1}]}(s)$$

we define $H \cdot M \in \mathbb{H}^2$ by

$$(H \cdot M)_t = \sum_{i=0}^{n-1} H_{(i)} \left(M_{t_{i+1} \wedge t} - M_{t_i \wedge t} \right). \tag{5.9}$$

The map $H \mapsto H \cdot M$ can be extended to an isometry of $L^2(M)$ into $\mathbb{H}^2$. Moreover $H \cdot M$ is characterized by the relation

$$\langle H \cdot M, N\rangle = H \cdot \langle M, N\rangle \quad \text{for all } N \in \mathbb{H}^2, \tag{5.10}$$

and if T is a stopping time, then

$$\left(\mathbf{1}_{[0,T]} H\right) \cdot M = (H \cdot M)^T = H \cdot M^T. \tag{5.11}$$

Notation 5.7. The process $H \cdot M$ is called the stochastic integral of H with respect to M and is often denoted as

$$(H \cdot M)_t = \int_0^t H_s \, \mathrm{d}M_s.$$

Remark 5.8.

- The formula (5.10) is called *martingale characterization* of the stochastic integral and is used to define it in some books.

  Observe that the integral on the RHS of (5.10) is with respect to a process os finite variation that we know well already.

- The isometry of Theorem 5.6 is called *Itô's isometry*.

♦

Proof of Theorem 5.6. **Step 1: We show that $\mathcal{E} \ni H \mapsto H \cdot M$ is an isometry $L^2(M) \to \mathbb{H}^2$.** Define $M_t^i = H_{(i)}(M_{t_{i+1}\wedge t} - M_{t_i \wedge t})$, so $H \cdot M_t = \sum_{i=0}^{n-1} M_t^i$. It is trivial to verify that M^i is a martingale which is continuous and bounded in L^2, so $M^i \in \mathbb{H}^2$ and thus $H \cdot M \in \mathbb{H}^2$.

The linearity of the map $H \mapsto H \cdot M$ is obvious. Moreover, M^i, $i = 0, \dots, n-1$ are orthogonal (i.e., $\langle M^i, M^j \rangle = 0$ for all $i \neq j$, see Definition 4.43) and

$$\langle M^i \rangle_t = H_{(i)}^2 \Big(\langle M \rangle_{t \wedge t_{i+1}} - \langle M \rangle_{t \wedge t_i} \Big),$$

hence,

$$\langle H \cdot M \rangle_t = \sum_{i=0}^{n-1} H_{(i)}^2 \Big(\langle M \rangle_{t \wedge t_{i+1}} - \langle M \rangle_{t \wedge t_i} \Big)$$

and thus

$$\begin{aligned} \|H \cdot M\|_{\mathbb{H}^2}^2 &= \mathbb{E}\left[\sum_{i=0}^{n-1} H_{(i)}^2 \Big(\langle M \rangle_{t_{i+1}} - \langle M \rangle_{t_i} \Big) \right] \\ &= \mathbb{E}\left[\int_0^t H_s^2 \,\mathrm{d}\langle M \rangle_s \right] \\ &= \|H\|_{L^2(M)}^2, \end{aligned}$$

proving the isometry property

Step 2: We show the existence of an extension. By the first step , $H \mapsto H \cdot M$ is an isometry $\mathcal{E} \to \mathbb{H}^2$. As $\mathcal{E}$ is dense in $L^2(M)$, this map extends uniquely to an isometry of $L^2(M)$ into H^2, as claimed.

Step 3: We prove that $H \cdot M$ satisfies (5.10). Let $H \in \mathcal{E}$ as above. Then $\langle H \cdot M, N \rangle = \sum_{i=0}^{n-1} \langle M^i, N \rangle$. Moreover, $\langle M^i, N \rangle_t = H_{(i)}(\langle M, N \rangle_{t \wedge t_{i+1}} - \langle M, N \rangle_{t \wedge t_i})$. Hence, we have

$$\langle H \cdot M, N \rangle_t = \sum_{i=0}^{n-1} H_{(i)} \Big(\langle M, N \rangle_{t \wedge t_{i+1}} - \langle M, N \rangle_{t \wedge t_i} \Big) = \int_0^t H_s \,\mathrm{d}\langle M, N \rangle_s,$$

proving (5.10) for $H \in \mathcal{E}$.

In order to extend the above result to all $H \in L^2(M)$ we prove that $X \mapsto \langle X, N \rangle_\infty$ is a continuous map from $\mathbb{H}^2$ to L^1. Indeed,

$$\mathbb{E}[|\langle X, N \rangle_\infty|] \leq \sqrt{\mathbb{E}[\langle X, X \rangle_\infty]} \sqrt{\mathbb{E}[\langle N, N \rangle_\infty]} = \|X\|_{\mathbb{H}^2} \|N\|_{\mathbb{H}^2},$$

proving the continuity. Therefore: if $H^n \in \mathcal{E}$ with $H_n \xrightarrow[n\to+\infty]{} H$ in $L^2(M)$, then

$$\begin{aligned}\langle H\cdot M, N\rangle_\infty &\overset{\text{Definition}}{=} \lim_{n\to+\infty} \langle H^n\cdot M, N\rangle_\infty \\ &\overset{(5.10)\text{ on }\mathcal{E}}{=} \lim_{n\to+\infty} \left(H^n\cdot\langle H, N\rangle\right)_\infty \\ &= \left(H\cdot\langle M, N\rangle\right)_\infty,\end{aligned} \tag{5.12}$$

where the last inequality follows by Kunita-Watanabe inequality (Proposition 4.45). Replacing N by N^+ in Equation (5.12), using Proposition 4.42, *(iv)*, we find (5.10) for $H \in L^2(M)$ arbitrary.

Step 4: We verify that (5.10) determines $H\cdot M$. Assume that $X \in \mathbb{H}^2$ satisfies $\langle X, N\rangle_t = \int_0^t H_s\,\mathrm{d}\langle M, N\rangle_s$ for all $N \in \mathbb{H}^2$. Then $\langle H\cdot M - X, N\rangle = 0$ and in particular taking $N = H\cdot M - X$ we find, using Corollary 4.39, $X = H\cdot M$.

Step 5: Lastly, we prove (5.11) Let T be a stopping time. Using Proposition 4.42, *(iv)* for $N \in \mathbb{H}^2$,

$$\begin{aligned}\langle (H\cdot M)^T, N\rangle_t &= \langle H\cdot M, N\rangle_{t\wedge T} \\ &\overset{(5.10)}{=} \left(H\cdot\langle M, N\rangle\right)_{t\wedge T} \\ &= \left(H\mathbf{1}_{[0,T]}\cdot\langle M, N\rangle\right)_t \\ &= \langle (H\mathbf{1}_{[0,T]}\cdot M, N)\rangle\end{aligned}$$

and thus by (5.10), $(H\cdot M)^T = (H\mathbf{1}_{[0,T]})\cdot M$. For the second equality we write

$$\langle H\cdot M^T, N\rangle = H\cdot\langle M^T, N\rangle = H\cdot\langle M, N\rangle^T = H\mathbf{1}_{[0,T]}\cdot\langle M, N\rangle$$

completing the proof. □

Proposition 5.9. *If $K \in L^2(M)$ and $H \in L^2(K\cdot M)$, then $HK \in L^2(M)$ and*

$$(HK)\cdot M = H\cdot(K\cdot M).$$

Proof. By (5.10),

$$\langle K\cdot M, K\cdot M\rangle_s = K\cdot\langle M, K\cdot M\rangle_s = K^2\cdot\langle M, M\rangle_s$$

and thus

$$\int_0^{+\infty} H_s^2 K_s^2\,\mathrm{d}\langle M\rangle_s = \int_0^{+\infty} H_s^2\,\mathrm{d}\langle K\cdot M\rangle_s < +\infty$$

implying the first claim. For the second one, observe that for $N \in \mathbb{H}^2$ we have the equalities

$$\begin{aligned}\langle (HK)\cdot M, N\rangle &= HK \cdot \langle M, N\rangle \\ &= H\cdot (K\cdot \langle M,N\rangle) \\ &= H\cdot \langle K\cdot M, N\rangle \\ &= \langle H\cdot (K\cdot M), N\rangle. \quad \square\end{aligned}$$

Remark 5.10. "Informally", we may write Proposition 5.9 as

$$\int_0^t H_s(K_s\,\mathrm{d}M_s) = \int_0^t H_sK_s\,\mathrm{d}M_s.$$

Similarly, (5.10) could be written (still informally) as

$$\left\langle \int_0^\cdot H_s\,\mathrm{d}M_s, N\right\rangle_t = \int_0^\cdot H_s\,\mathrm{d}\langle M,N\rangle_s.$$

In particular,

$$\left\langle \int_0^\cdot H_s\,\mathrm{d}M_s\right\rangle_t = \int_0^t H_s^2\,\mathrm{d}\langle M\rangle_s.$$

Finally, since if $M, N \in \mathbb{H}^2$, $H \in L^2(M)$, $K \in L^2(N)$, then $H\cdot M, K\cdot N \in \mathbb{H}^2$, so

$$\mathbb{E}\left[\int_0^t H_u\,\mathrm{d}M_u \,\middle|\, \mathcal{G}_s\right] = 0$$

$$\mathbb{E}\left[\left(\int_s^t H_u\,\mathrm{d}M_u\right)\left(\int_s^t K_u\,\mathrm{d}K_u\right)\,\middle|\,\mathcal{G}_s\right] = \mathbb{E}\left[\int_s^t H_sK_s\,\mathrm{d}\langle M,N\rangle_s \,\middle|\, \mathcal{G}_s\right]$$

These relations will not be necessarily valid for extensions of the stochastic integral given below. ♦

5.2. Stochastic integral for local martingales

We now extend the definition of $H\cdot M$ to continuous local martingales. Let M be a (continuous) local martingale with $M_0 = 0$. We define

$$L^2_{\mathrm{loc}}(M) = \left\{ H \,\middle|\, \begin{array}{c} H \text{ progressively measurable process s.t.} \\ \text{for all } t \geq 0,\ \int_0^t H_s^2\,\mathrm{d}\langle M\rangle_s < +\infty \text{ a.s.}\end{array}\right\}, \quad \text{and}$$

$$L^2(M) = \left\{ H \,\middle|\, \begin{array}{c} H \text{ progressively measurable process s.t.} \\ \mathbb{E}\left[\int_0^{+\infty} H_s^2\,\mathrm{d}\langle M\rangle_s\right] < +\infty\end{array}\right\}.$$

Theorem 5.11. *Let M be a local martingale with $M_0 = 0$. Then for any $H \in L^2_{\mathrm{loc}}(M)$ there is an essentially unique local martingale started from 0, denoted $H \cdot M$ such that for any local martingale N*

$$\langle H \cdot M, N\rangle = H \cdot \langle M, N\rangle. \tag{5.13}$$

Then property (5.11) holds, and if $M \in \mathbb{H}^2$, $H \in L^2(M)$, then this definition coincides with the one of Theorem 5.6.

Proof. Let

$$T_n = \inf\left\{ t \geq 0 \,\middle|\, \int_0^t (1 + H_s^2)\, \mathrm{d}\langle M\rangle_s \geq n \right\}, \quad n \in \mathbb{N}.$$

Then T_n is an increasing sequence of stopping times tending to ∞ as $n \to +\infty$. Since $\langle M^{T_n}\rangle_t = \langle M\rangle_{T_n \wedge n} \leq n$, it follows from Theorem 4.37, *(ii)* that $M^{T_n} \in \mathbb{H}^2$. Also, $\int_0^{+\infty} H_s^n \,\mathrm{d}\langle M^{T_n}\rangle_s \leq n$, that is $H \in L^2(M^{T_n})$. Hence, for any $n \in \mathbb{N}$ the stochastic integral $H \cdot M^{T_n}$ is well defined.

Using Equations (5.10) and (5.11), one sees that for $m > n$,

$$H \cdot M^{T_n} = (H \cdot M^{T_m})^{T_n}.$$

This implies that there exists a unique process $H \cdot M$ such that

$$(H \cdot M)^{T_n} = H \cdot M^{T_n} \quad \text{for any } n \in \mathbb{N}.$$

Therefore also $(H \cdot M)^{T_n} \in \mathbb{H}^2$ and thus $H \cdot M$ is a local martingale.

Let N be a local martingale and without loss of generality assume $N_0 = 0$. Set

$$T_n' = \inf\{t \geq 0 \mid |N_t| \geq n\}, \quad S_n = T_N \wedge T_n'.$$

Then

$$\begin{aligned}
\langle H \cdot M, N\rangle^{S_n} &= \langle (H \cdot M)^{S_n}, N^{S_n}\rangle = \left\langle \left(H \cdot M^{T_n}\right)^{S_n}, N^{S_n} \right\rangle \\
&= \langle H \cdot M^{T_n}, N^{S_n}\rangle \overset{(5.10)}{=} H \cdot \langle M^{T_n}, N^{S_n}\rangle = H \cdot \langle M, N\rangle^{S_n} \\
&= \big(H \cdot \langle M, N\rangle\big)^{S_n}
\end{aligned}$$

and thus $\langle H \cdot M, N\rangle = H \cdot \langle M, N\rangle$. The fact that this equality characterises $H \cdot M$ is proved as in Theorem 5.6.

Equation (5.11) is proved as before, as ins proof only uses (5.10) which we already extended. Finally, for $M \in \mathbb{H}^2$, $H \in L^2(M)$, (5.13) shows $\langle H \cdot M\rangle = H^2 \cdot \langle M\rangle$ that is $H \cdot M \in \mathbb{H}^2$, so by Equations (5.10) and (5.13) the two definitions coincide. □

Remark 5.12. Moment formulas of Remark 5.10 need an additional assumption: If M is a local martingale and $H \in L^2_{\text{loc}}(M)$, $t \in \mathbb{R}_+ \cup \{+\infty\}$, then if

$$\mathbb{E}\left[\langle H \cdot M\rangle_t\right] = \mathbb{E}\left[\int_0^t H_s^2 \, \mathrm{d}\langle M\rangle_s\right] < +\infty$$

then

$$\mathbb{E}\left[\int_0^t H_s \, \mathrm{d}M_s\right] = 0, \quad \text{and} \quad \mathbb{E}\left[\left(\int_0^t H_s \, \mathrm{d}M_s\right)^2\right] = \mathbb{E}\left[\int_0^t H_s^2 \, \mathrm{d}\langle M\rangle_s\right].$$

♦

Finally, we extend the stochastic integral to semi martingales. We say that a progressively measurable process H is *locally bounded* if

$$\mathbb{P}\text{-a.s. for any } t \geq 0, \ \sup_{s \leq t} |H_s| < +\infty.$$

In particular, all adapted continuous processes are locally bounded. If H is locally bounded and A of finite variation, then

$$\mathbb{P}\text{-a.s. for any } t \geq 0, \ \int_0^t |H_s||\mathrm{d}A_s| < +\infty$$

and also, for every local martingale M, $H \in L^2_{\text{loc}}(M)$.

Definition 5.13. Let $X = X_0 + M + A$ be a continuous semi martingale and H a locally bounded progressively measurable process. The stochastic integral $H \cdot X$ is defined by

$$H \cdot X = H \cdot M + H \cdot A$$

and denoted

$$(H \cdot X)_t = \int_0^t H_s \, \mathrm{d}X_s.$$

We record some elementary properties of the integral whose proofs are left as an exercise to the reader.

(i) $(H, X) \mapsto H \cdot X$ is a bilinear map.

(ii) $H \cdot (K \cdot X) = (HK) \cdot X$ if H, K are locally bounded.

(iii) For any stopping time T, $H\mathbf{1}_{[0,T]} \cdot X = H \cdot X^T = (H \cdot X)^T$.

(iv) If X is a local martingale (or of finite variation), the same holds for $H \cdot X$.

(v) If

$$H_s(\omega) = \sum_{i=0}^{n-1} H_{(i)}(\omega)\mathbf{1}_{(t_i,t_{i+1}]}(s) \in \mathcal{E}$$

then

$$(H \cdot X)_t = \sum_{i=0}^{n-1} H_{(i)}\Big(X_{t_{i+1}\wedge t} - X_{t_i\wedge t} \Big).$$

We close this section by a useful approximation property.

Proposition 5.14. *Let X be a semi martingale and H adapted and continuous. Then for any sequence $((t_i^n)_{i=1,\dots,k_n})_{n\in\mathbb{N}}$ of refining partitions of $[0,t]$ with step tending to 0 we have*

$$\lim_{n\to+\infty} \sum_{i=0}^{n-1} H_{t_i^n}\Big(X_{i+1}^n - X_{t_i^n} \Big) = (H \cdot X)_t \quad \textit{in probability.}$$

Proof. Let $X = X_0 + M + A$ and set

$$H_s^n = \begin{cases} H_{t_i^n}^n, & \text{if } t_i^n < s \le t_{i+1}^n, \\ 0, & \text{if } s > t. \end{cases}$$

For the part of finite variation, for $\mathbb{P}$-a.e. ω,

$$\sum_{i=0}^{n-1} H_{t_i^n}(\omega)\Big(A_{t_{i+1}^n}(\omega) - A_{t_i^n}(\omega) \Big) = \int_0^t H_s^n(\omega)\,\mathrm{d}A_s(\omega) \xrightarrow[n\to+\infty]{\text{Dominated Convergence}} \int_0^t H_s(\omega)\,\mathrm{d}A_s(\omega)$$

For the local martingale part, define for $p \ge 1$,

$$T_p = \inf\{s \ge 0 \mid |H_s| + \langle M \rangle_s \ge p\}.$$

Then H, H^n and $\langle M \rangle$ are bounded on $[0, T_p)$. By L^2-theory of the stochastic integral we find

$$\left\| \Big(H^n \mathbf{1}_{[0,T_p)} \cdot M^{T_p} \Big)_t - \Big(H\mathbf{1}_{[0,T_p)} \cdot M^{T_p} \Big)_t \right\|_{L^2} = \mathbb{E}\left[\int_0^{t\wedge T_p} (H_s^n - H_s)^2\,\mathrm{d}\langle M \rangle_s \right]$$

which tends to 0 by dominated convergence theorem. Using (5.11) we deduce that

$$(H^n \cdot M)_{t\wedge T_p} \xrightarrow[n\to+\infty]{L^2} (H\cdot M)_{t\wedge T_p}.$$

As $\mathbb{P}(T_p > t) \nearrow 1$ as $p \to +\infty$ the claim follows. □

5.3. Itô's formula

Itô's formula is a principal tool of stochastic analysis. It replaces the usual deterministic formula

$$f\big(b(t)\big) = f\big(b(0)\big) + \int_0^t f'\big(b(s)\big)\,\mathrm{d}b(s), \tag{5.14}$$

which holds for $f \in C^1$ and b of finite variation. In differential form (5.14) is just the chain rule $\frac{\mathrm{d}}{\mathrm{d}t}(f\circ b)(t) = (f'\circ b)(t)\frac{\mathrm{d}}{\mathrm{d}t}b(t)$.

Itô's formula also shows that a sufficiently smooth function of a semi martingale is again a semi martingale and provides its decomposition.

Theorem 5.15 (Itô's formula (Itô 1948, Kunita-Watanabe 1967)). *Let $f : \mathbb{R}^K \to \mathbb{R}$ be a C^2 function and $X^1,\dots,X^K$ be continuous semi martingales. Then*

$$\begin{aligned} f(X^1_t,\dots,X^K_t) = f(X^1_0,\dots,X^K_0) &+ \sum_{i=1}^K \int_0^t \frac{\partial f}{\partial x_i}(X^1_s,\dots,X^K_s)\,\mathrm{d}X^i_s \\ &+ \frac{1}{2}\sum_{i,j=1}^K \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_j}(X^1_s,\dots,X^K_s)\,\mathrm{d}\langle X^i, X^j\rangle_s. \end{aligned}$$

Remark 5.16. The third term on the RHS is of finite variation, the second term can be then written as

$$\sum_{i=1}^K \int_0^t \frac{\partial f}{\partial x_i}(X^1_s,\dots,X^K_s)\,\mathrm{d}M_s + \sum_{i=1}^K \frac{\partial f}{\partial x_i}(X^1_s,\dots,X^K_s)\,\mathrm{d}A_s,$$

where $X^i_t = X^i_0 + M^i_t + A^i_t$ is the decomposition of the semi martingale X. From those two terms, the first is a local martingale and the second of finite variation. ♦

Proof of Theorem 5.15. We consider first the case $K = 1$ and write $X^1 = X$. Let $0 = t_0^n < \dots < t_{k_n}^n$ be a refining sequence of partitions of $[0,t]$. Then

$$f(X_t) = f(X_0) + \sum_{i=0}^{k_n-1} \left(f(X_{t_{i+1}^n}) - f(X_{t_i^n}) \right).$$

Using Taylor expansion

$$f(X_{t_{i+1}^n}) - f(X_{t_i^n}) = f'(X_{t_i^n}) \left(X_{t_{i+1}^n} - X_{t_i^n} \right) + \frac{\xi_{n,i}(\omega)}{2} \left(X_{t_{i+1}^n} - X_{t_i^n} \right)^2$$

where

$$\xi_{n,i}(\omega) = f''(s), \quad \text{for some } s \in \left[X_{t_i^n} \wedge X_{t_{i+1}^n}, X_{t_i^n} \vee X_{t_{i+1}^n} \right] \tag{5.15}$$

Using Proposition 5.14 with $H_s = f'(X_s)$ we see that

$$\lim_{n\to+\infty} \sum_{i=0}^{n-1} f'(X_{t_i^n}) \left(X_{t_{i+1}^n} - X_{t_i^n} \right) = \int_0^t f'(X_s)\, \mathrm{d}X_s, \quad \text{in probability.}$$

Hence, to complete the proof we show that

$$\lim_{n\to+\infty} \sum_{i=0}^{k_n-1} \xi_{n,i} \left(X_{t_{i+1}^n} - X_{t_i^n} \right)^2 = \int_0^t f''(X_s)\, \mathrm{d}\langle X \rangle_s, \quad \text{in probability.} \tag{5.16}$$

To show (5.16), observe that for $m < n$,

$$\begin{aligned} &\left| \sum_{i=0}^{k_n-1} \xi_{n,i} \left(X_{t_{i+1}^n} - X_{t_i^n} \right)^2 - \sum_{j=0}^{k_n-1} \xi_{m,j} \sum_{\substack{i=0 \\ t_j^n \le t_i^n < t_{j+1}^n}}^{k_n-1} \left(X_{t_{i+1}^n} - X_{t_i^n} \right)^2 \right| \\ &\qquad\qquad \le Z_{m,n} \left(\sum_{i=0}^{k_n-1} \left(X_{t_{i+1}^n} - X_{t_i^n} \right)^2 \right) =: \Phi_{m,n}, \end{aligned} \tag{5.17}$$

where

$$Z_{m,n} = \sup_{j=0,\dots,k_n} \sup_{\substack{i=0,\dots,k_n-1 \\ t_j^n \le t_i^n < t_{j+1}^n}} |\xi_{n,i} - \xi_{m,j}|.$$

From the continuity of f'' and (5.15) it follows that $Z_{m,n} \xrightarrow[m,n\to+\infty]{} 0$ a.s.. Moreover as $\sum_{i=0}^{k_n-1} (X_{t_{i+1}^n} - X_{t_i^n})^2 \xrightarrow[n\to+\infty]{} \langle X \rangle_t$ in probability, we see that for any $\varepsilon > 0$ we can choose m such that for all $n > m$,

$$\mathbb{P}(\Phi_{m,n} \ge \varepsilon) \le \varepsilon. \tag{5.18}$$

Fixing this value of m and using again the approximation of $\langle X\rangle$,

$$\lim_{n\to+\infty}\sum_{j=0}^{k_n-1}\xi_{m,j}\sum_{\substack{i=0\\ t_j^n\le t_i^n<t_{j+1}^n}}^{k_n-1}\left(X_{t_{i+1}^n}-X_{t_i^n}\right)^2=\sum_{j=0}^{k_n-1}\xi_{m,j}\left(\langle X\rangle_{t_{j+1}^n}-\langle X\rangle_{t_j^n}\right)$$
$$=\int_0^t h_m(s)\,\mathrm{d}\langle X\rangle_s,$$

where $h_m(s)=\xi_{m,j}$ if $t_j^m\le s<t_{j+1}^m$. Obviously $h_m(s)\xrightarrow[n\to+\infty]{}f''(X_s)$ a.s.. So for m large enough

$$\mathbb{P}\left(\left|\int_0^t H_m(s)\,\mathrm{d}\langle X\rangle_s-\int_0^t f''(s)\,\mathrm{d}\langle X\rangle_s\right|\ge\varepsilon\right)\le\varepsilon. \tag{5.19}$$

Combining Equations (5.17) to (5.19), for $n>m$ large enough,

$$\mathbb{P}\left(\left|\sum_{i=0}^{k_n-1}\xi_{n,i}\left(X_{t_{i+1}^n}-X_{t_i^n}\right)^2-\int_0^t f''(X_s)\,\mathrm{d}\langle X_s\rangle\right|\ge 3\varepsilon\right)\le 3\varepsilon,$$

proving (5.16).

We now show the general case $K>1$. Taylor expansion yields

$$f(X_{t_{i+1}^n}^1,\dots,X_{t_{i+1}^n}^K)-f(X_{t_i^n}^1,\dots,X_{t_i^n}^K)=\sum_{k=1}^K\frac{\partial f}{\partial x_k}(X_{t_i^n}^1,\dots,X_{t_i^n}^K)\left(X_{t_{i+1}^n}^k-X_{t_i^n}^k\right)$$
$$+\sum_{k,\ell=1}^K\frac{\xi_{m,i}^{k,\ell}}{2}\left(X_{t_{i+1}^n}^k-X_{t_i^n}^k\right)\left(X_{t_{i+1}^n}^\ell-X_{t_i^n}^\ell\right)$$

with $\xi_{n,i}^{k,\ell}=\frac{\partial f}{\partial x_k\partial x_\ell}(s)$ for some

$$s\in\left\{\left.\begin{pmatrix}\theta X_{t_i^n}^1+(1-\theta)X_{t_{i+1}^n}^1\\ \vdots\\ \theta X_{t_i^n}^K+(1-\theta)X_{t_{i+1}^n}^K\end{pmatrix}\right|\theta\in[0,1]\right\}.$$

Proposition 5.14 again gives the required result for the terms with first derivative and similar arguments as previously show that for all k,ℓ,

$$\lim_{n\to+\infty}\sum_{i=1}^{k_n}\xi_{n,i}^{k,\ell}\left(X_{t_{i+1}^n}^k-X_{t_i^n}^k\right)\left(X_{t_{i+1}^n}^\ell-X_{t_i^n}^\ell\right)=\int_0^t\frac{\partial f}{\partial x_k\partial x_\ell}(X_s^1,\dots,X_s^K)\,\mathrm{d}\langle X^k,X^\ell\rangle_s.\square$$

Next we record some direct consequences of Itô's formula.

- Integration by parts formula: If X, Y are semi martingales, then

$$X_t Y_t = X_0 Y_0 + \int_0^t X_s \, \mathrm{d}Y_s + \int_0^t Y_s \, \mathrm{d}X_s + \langle X, Y \rangle_t \tag{5.20}$$

- Taking $X = Y$ in (5.20) yields

$$X_t^2 = X_0^2 + 2 \int_0^t X_s \, \mathrm{d}X_s + \langle X \rangle_t. \tag{5.21}$$

Hence, if X is local martingale, we know that $X_t^2 - \langle X \rangle_t$ is also a local martingale and (5.21) shows that it equals $X_0^2 + 2 \int_0^t X_s \, \mathrm{d}X_s$. This is not surprising from the construction of $\langle X \rangle$ and the discrete approximation of the stochastic integral $\int X_s \, \mathrm{d}X_s$.

- For a $\mathcal{G}_t$-Brownian motion $(B_t)_{t \geq 0}$, possibly started not in 0, Itô's formula gives

$$f(B_t) = f(B_0) + \int_0^t f'(B_s) \, \mathrm{d}B_s + \frac{1}{2} \int_0^t f''(B_s) \, \mathrm{d}s \tag{5.22}$$

$$f(t, B_t) = f(0, B_0) + \int_0^t \frac{\partial f}{\partial x}(s, B_s) \, \mathrm{d}B_s + \int_0^t \left(\frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \right) (s, B_s) \, \mathrm{d}s, \tag{5.23}$$

and for d-dimensional Brownian motion $B = (B^1, \dots, B^d)$,

$$f(B_t) = f(B_0) + \sum_{i=1}^d \int_0^t \frac{\partial f}{\partial x_i}(B_s) \, \mathrm{d}B_s^i + \frac{1}{2} \int_0^t \Delta f(B_s) \, \mathrm{d}s \tag{5.24}$$

where Δ denotes the Laplacian operator which appears as $\langle B^i, B^j \rangle_t = \delta_{i,j} t$.

Remark 5.17. Equation (5.21) can actually be used to prove Itô's formula, see [6]. ♦

5.4. Applications of Itô's formula

5.4.1. Exponential martingales

Proposition 5.18. *Let M be a local martingale. Then for all $\lambda \in \mathbb{C}$,*

$$\mathcal{E}(\lambda M)_t := \exp\left\{\lambda M_t - \frac{\lambda^2}{2}\langle M\rangle_t\right\} \tag{5.25}$$

is a local martingale.

Proof. Define the function $f : \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ by $f(x,y) = \exp\left\{\lambda x - \frac{\lambda^2}{2}y\right\}$. Then

$$\frac{\partial f}{\partial y} + \frac{1}{2}\frac{\partial^2 f}{\partial x^2} = 0. \tag{5.26}$$

By Itô's formula for $(X^1, X^2) = (M, \langle M\rangle)$, noting that $\langle M\rangle$ is of finite variation and thus $\langle M, \langle M\rangle\rangle = 0 = \langle\langle M\rangle\rangle$,

$$\begin{aligned} f\big(M_t, \langle M\rangle_t\big) &= f(M_0, 0) + \int_0^t \partial_x f\big(M_s, \langle M\rangle_s\big)\,\mathrm{d}M_s + \int_0^t \partial_y f\big(M_s, \langle M\rangle_s\big)\,\mathrm{d}\langle M\rangle_s \\ &\quad + \frac{1}{2}\int_0^t \partial_x^2 f\big(M_s, \langle M\rangle_s\big)\,\mathrm{d}\langle M\rangle_s \\ &\overset{(5.26)}{=} f(M_0, 0) + \int_0^t \partial_x f\big(M_s, \langle M\rangle_s\big)\,\mathrm{d}M_s \end{aligned}$$

Since $\mathcal{E}(\lambda M) = f(M, \langle M\rangle)$, this shows that $\mathcal{E}(\lambda M)$ is a local martingale. □

It is important to know, e.g., for applications of Girsanov's theorem under which conditions $\mathcal{E}$ is a true martingale.

Theorem 5.19 (Novikov's condition). *Let M be a local martingale with $M_0 = 0$. Consider the following properties:*

(i) $\mathbb{E}\left[\exp\left\{\frac{1}{2}\langle M\rangle_\infty\right\}\right] < +\infty$;

(ii) M is uniformly integrable and $\mathbb{E}\left[\exp\left\{\frac{1}{2}M_\infty\right\}\right] < \infty$;

(iii) $\mathcal{E}(M)$, *defined by* (5.25), *is uniformly integrable.*

Then (i) $\implies$ *(ii)* $\implies$ *(iii).*

Proof. "*(i)* $\implies$ *(ii)*": By *(i)*, we have also $\mathbb{E}[\langle M\rangle_\infty] < +\infty$, so, by Theorem 4.37, M is a true martingale, bounded in L^2 and thus M is uniformly integrable. Moreover,

$$\exp\left\{\frac{1}{2}M_\infty\right\} = \mathcal{E}(M)_\infty^{1/2}\exp\left\{\frac{1}{2}\langle M\rangle_\infty\right\}^{1/2}$$

so by Cauchy-Schwartz inequality,

$$\mathbb{E}\left[\exp\left\{\frac{1}{2}M_\infty\right\}\right] \le \underbrace{\mathbb{E}[\mathcal{E}(M)_\infty]^{1/2}}_{\overset{\text{Proposition 4.33}}{\le}\ \mathbb{E}[\mathcal{E}(M)_0]^{1/2}<+\infty} \underbrace{\mathbb{E}\left[\exp\left\{\frac{1}{2}\langle M\rangle_\infty\right\}\right]^{1/2}}_{<+\infty \text{ by assumption}} < +\infty,$$

proving *(ii)*.

"*(ii)* $\implies$ *(iii)*": As M is uniformly integrable, we have $M_t = \mathbb{E}\left[M_\infty \,|\, \mathcal{G}_t\right]$, by Proposition 4.19. Hence, using Jensen's inequality,

$$\exp\left\{\frac{1}{2}M_t\right\} \le \mathbb{E}\left[\exp\left\{\frac{1}{2}M_\infty\right\}\middle|\mathcal{G}_t\right].$$

As a consequence, $\exp\{\frac{1}{2}M_t\} \in L^2$. Applying Jensen's inequality again, we see that $\exp\{\frac{1}{2}M_t\}$ is a submartingale, closed by its limit $\exp\{\frac{1}{2}M_\infty\}$. Using the stopping theorem for submartingales, we have

$$\exp\left\{\frac{1}{2}M_T\right\} \le \mathbb{E}\left[\exp\left\{\frac{1}{2}M_\infty\right\}\middle|\mathcal{G}_t\right], \quad \text{for all stopping times } T,$$

i.e., the family

$$\left\{\exp\left\{\frac{1}{2}M_T\right\}\,\middle|\, T \text{ stopping time}\right\} \tag{5.27}$$

is uniformly integrable.

Let now for $a \in (0,1)$, $Z_t^{(a)} = \exp\left\{\frac{aM_t}{1+a}\right\}$. Then

$$\mathcal{E}(aM)_t = \exp\left\{aM_t - \frac{a^2}{2}\langle M\rangle_t\right\}, \quad \text{and as } a = a^2 + \frac{a(1-a^2)}{1+a},$$

$$= \mathcal{E}(M)_t^{a^2} \left(Z_t^{(a)}\right)^{1-a^2}.$$

For $\Gamma \in \mathcal{G}_\infty$ and T a stopping time we have thus by Hölder inequality,

$$\begin{aligned}
\mathbb{E}\left[\mathbf{1}_\Gamma \mathcal{E}(aM)_t\right] &= \mathbb{E}\left[\mathcal{E}(M)_T\right]^{a^2} \mathbb{E}[\mathbf{1}_\Gamma Z_t^{(a)}]^{1-a^2} \quad (\mathcal{E}(M) \text{ is super martingale})\\
&\le \mathbb{E}\left[\mathbf{1}_\Gamma Z_T^{(a)}\right]^{1-a^2}\\
&= \mathbb{E}\left[\mathbf{1}_\Gamma Z_T^{(a)}\right]^{2a(1-a)\frac{1+a}{2a}}, \qquad \frac{1+a}{2a} > 1\\
&\overset{\text{Jensen}}{\le} \mathbb{E}\left[\mathbf{1}_\Gamma \exp\left\{\frac{1}{2} M_T\right\}\right]^{2a(1-a)}.
\end{aligned}$$

Equation (5.27) then implies that $\{\mathcal{E}(aM)_T \mid T \text{ stopping time}\}$ is uniformly integrable, in particular, $\mathcal{E}(aM)$ is an uniformly integrable martingale. Thus by evoking Hölder's inequality again

$$\begin{aligned}
1 = \mathbb{E}\left[\mathcal{E}(aM)_\infty\right] &\le \mathbb{E}\left[\mathcal{E}(M)_\infty\right]^{a^2} \mathbb{E}\left[Z_\infty^{(a)}\right]^{1-a^2}\\
&\le \mathbb{E}\left[\mathcal{E}(M)_\infty\right]^{a^2} \mathbb{E}\left[\exp\left\{\frac{1}{2} M_\infty\right\}\right]^{2a(1-a)}.
\end{aligned}$$

Letting $a \to 1$, this implies $\mathbb{E}[\mathcal{E}(M)_\infty] \ge 1$ and so $\mathbb{E}[\mathcal{E}(M)_\infty] = 1$, since $\mathcal{E}(M)$ is a super martingale. But this implies that $\mathcal{E}(M)$ is a uniformly integrable martingale by Proposition 4.19. □

5.4.2. Lévy characterisation of Brownian motion

Recall that a $\mathbb{R}^d$ valued process X with $X_0 = 0$ is called $\mathcal{G}_t$-Brownian motion if for all $0 \le s < t$, the increment $X_t - X_s$ is independent of $\mathcal{G}_s$ and $\mathcal{N}(0, (t-s)I)$.

Theorem 5.20 (Lévy). *Let $X = (X^1, \dots, X^d)$ be $\mathcal{G}_t$-adapted. Then the following are equivalent:*

(i) X is a d-dimensional $\mathcal{G}_t$-Brownian motion.

(ii) $X^1, \dots, X^d$ are continuous $\mathcal{G}_t$-local martingales with $X_0 = 0$ and

$$\langle X^i, X^j \rangle_t = \delta_{i,j} t \quad \text{for } 1 \le i, j \le d \text{ and } t \ge 0.$$

Proof. "(i) $\Longrightarrow$ (ii)": Is already known "*(ii)* $\Longrightarrow$ *(i)*": Let $\xi \in \mathbb{R}^d$. Then $\langle \xi \cdot X \rangle_t = \sum_{i,j} \xi_i \xi_j \langle X^i, X^j \rangle_t = |\xi|^2 t$, with $\cdot$ denoting the scalar product in $\mathbb{R}^d$. By Proposition 5.18 we thus see that

$$Z_t = \exp\left\{ \mathrm{i}\xi \cdot X_t + \frac{1}{2}|\xi|^2 t \right\}$$

is a local martingale. Hence, for $0 \le s \le t$, $\mathbb{E}\left[Z_t \,|\, \mathcal{G}_s \right] \stackrel{\text{a.s.}}{=} Z_s$. Since $|Z_t| > 0$ we can rewrite this as

$$1 \stackrel{\text{a.s.}}{=} \mathbb{E}\left[\frac{Z_t}{Z_s} \,\middle|\, \mathcal{G}_s \right] = \mathbb{E}\left[\exp\left\{ \mathrm{i}\xi \cdot (\mathrm{X_t} - \mathrm{X_s}) + \frac{|\xi|^2}{2}(\mathrm{t} - \mathrm{s}) \right\} \,\middle|\, \mathcal{G}_s \right].$$

This implies that $X_t - X_s$ is independent of $\mathcal{G}_s$ and $\mathcal{N}(0, (t-s)I)$, so X is a $\mathcal{G}_t$-Brownian motion. □

As an application of Lévy's theorem we now show that, "modulo time change", Brownian motion is the most general continuous local martingale.

Theorem 5.21 (Dubins-Schwarz, 1965). *Let M be a local martingale started from 0 and such that $\lim_{t\to+\infty}\langle M \rangle_t = +\infty$. For $t \ge 0$ define the stopping time*

$$\tau_t := \inf\{ s \,|\, \langle M \rangle_s > t \} \nearrow +\infty, \quad \text{as } t \to +\infty.$$

Set

$$\mathcal{H}_t = \mathcal{G}_{\tau_t} = \{ A \in \mathcal{G} \,|\, A \cap \{\tau_t \le s\} \in \mathcal{G}_s \}.$$

Then $B_t := M_{\tau_t}$ is a $\mathcal{H}_t$-Brownian motion. Moreover, for any fixed t, $\langle M \rangle_t$ is a $\mathcal{H}_t$ stopping time and

$$M_t = B\big(\langle M \rangle_t\big), \quad t \ge 0.$$

Remark 5.22. The hypothesis $\langle M \rangle_t \nearrow +\infty$ is unessential. If it does not hold, B_t is only defined up to the time $\langle M \rangle_\infty$, which is random. ♦

Proof of Theorem 5.21. M is continuous and progressively measurable, hence by Lemma 3.43, $B_t = M_{\tau_t}$ is $\mathcal{H}_t = \mathcal{G}_{\tau_t}$-measurable, i.e., B is $\mathcal{H}_t$-adapted. Observe also that $\mathcal{H}_t$ satisfies the usual conditions, which we leave for the reader to check. The function $t \mapsto \tau_t$ is non-decreasing and

$$\lim_{s \searrow t} \tau_s =: \tau_{t-} = \inf\{ s \ge 0 \,|\, \langle M \rangle_s = t \}.$$

To show the continuity of B we need

Lemma 5.23. *The intervals of constancy of M coincide a.s. with the intervals of constancy of $\langle M\rangle$, that is $\mathbb{P}$-a.s. for all $a < b$,*

$$M_t = M_a, \quad \text{for all } t \in [a, b] \iff \langle M\rangle_t = \langle M\rangle_a \quad \text{for all } t \in [a, b].$$

Proof of Lemma 5.23. " $\implies$ ": follows easily by approximation of $\langle M\rangle$ as usual.

" $\impliedby$ ": Consider local martingales $(0, b-a) \ni t \mapsto M_{a+t} - M_a$. the quadratic variation of this process is 0 by assumption, so it is constant by Corollary 4.39. □

Now observe that for $t \in [\tau_s-, \tau_s]$ we have $\langle M\rangle_t = s$ so $\mathbb{P}$-a.s. for $r \geq 0$,

$$\lim_{r \searrow s-} B_r = \lim M_{\tau_r} = M_{\tau_s-} = M_{\tau_s} = B_s,$$

so B is left-continuous. Their right continuity follows from this of τ.

Now show that B_t and $B_t^2 - t$ are $\mathcal{H}_t$-local martingales. For every $n \in \mathbb{N}$, the stopped martingale M^{τ_n}, $(M^{\tau_n})^2 - \langle M\rangle^{\tau_n}$ are true uniformly integrable martingales (using Theorem 4.37 and $\langle M^{\tau_n}\rangle_\infty = \langle M\rangle_{\tau_n} = n$). The stopping theorem then yields for $r \leq s \leq n$,

$$\mathbb{E}\left[B_s \,|\, \mathcal{H}_r\right] = \mathbb{E}\left[M^{\tau_n}_{\tau_s} \,\middle|\, \gamma_{\tau_n}\right] = M^{\tau_n}_{\tau_r} = B_r, \quad \text{and}$$
$$\mathbb{E}\left[B_s^2 - s \,\middle|\, \mathcal{H}_s\right] = \mathbb{E}\left[\left(M^{\tau_n}_{\tau_s}\right)^2 - \langle M^{\tau_n}\rangle_{\tau_s} \,\middle|\, \mathcal{G}_{\tau_r}\right] = (M^{\tau_n}_{\tau_r})^2 - \langle M^{\tau_n}\rangle_{\tau_r} = B_r^2 - r.$$

Using Lévy's theorem (Theorem 5.20) then implies that B is a $\mathcal{H}$-Brownian motion.

To see that $\langle M\rangle_t$ is a $\mathcal{H}$-stopping time, we write $\{\langle M\rangle_t \leq n\} = \{\tau_n > t\} \in \mathcal{H}_n$. The last claim follows by invoking the definition of B using Lemma 5.23. □

5.4.3. Bernstein's inequality

Another application of the exponential martingales yields the following inequality which also confirms that $\langle M\rangle$ is the "right clock" for M again

Theorem 5.24. *Let M_t be a constant local martingale with $M_0 =$ and $\langle M\rangle_t \leq ct$ for all $t \geq 0$ and some $c \in \mathbb{R}$. Then M is a martingale and*

$$\mathbb{P}\left(\sup_{0\leq s\leq t} M_s \geq a\right) \leq \exp\left\{-\frac{a^2}{2ct}\right\}, \quad \text{for all } a > 0,\ t > 0.$$

Proof. For $\lambda \in \mathbb{R}$, due to Proposition 5.18 and theorem 5.19, $Z_t = e^{\lambda M_t - \frac{\lambda^2}{2}\langle M\rangle_t}$ is a true martingale. For $\lambda > 0$, by Doob's inequality, using $\langle M\rangle_t \leq ct$,

$$\mathbb{P}\left(\sup_{s\leq t} M_s \geq a\right) \leq \mathbb{P}\left(\sup_{s\leq t} Z_s \geq e^{\lambda a - \frac{\lambda^2}{2}ct}\right)$$
$$= e^{-\lambda a + \lambda^2 ct/2} \underbrace{\mathbb{E}[Z_t]}_{=\mathbb{E}[Z_0]=1} .$$

Setting $\lambda = \frac{a}{ct}$ implies the first inequality. The second one follows by taking $-M$ instead of M. The fact that M is a martingale follows from (5.19) or from the second inequality. □

5.4.4. Burkholder-Davis-Gundy inequalities

The following important inequalities have a similar flavour.

Theorem 5.25. *For an $p > 0$ there are constants $c_p, C_p \in (0, +\infty)$ such that for any local martingale with $M_0 = 0$,*

$$c_p\,\mathbb{E}\left[\langle M\rangle_\infty^{\frac{p}{2}}\right] \leq \mathbb{E}\left[\left(\sup_{s\geq 0}|M_s|\right)^p\right] \leq C_p\,\mathbb{E}\left[\langle M\rangle_\infty^{\frac{p}{2}}\right].$$

Remark 5.26. By considering, for T a stopping time, the stopped local martingale M^T we obtain an analogous inequality:

$$c_p\,\mathbb{E}\left[\langle M\rangle_T^{\frac{p}{2}}\right] \leq \mathbb{E}\left[\left(\sup_{s\leq T}|M_s|\right)^p\right] \leq C_p\,\mathbb{E}\left[\langle M\rangle_T^{\frac{p}{2}}\right].$$

♦

Proof of Theorem 5.25. We only consider $p \geq 2$. For $p \in (0, 2)$ see [4, p. IV.4]

Step 1: Upper bound: By Itô's formula applied on $f(x) = |x|^p$ we obtain

$$|M_t|^p = \int_0^t p|M_s|^{p-1}\operatorname{sign} M_s\,\mathrm{d}M_s + \frac{1}{2}p(p-1)\int_0^t |M_s|^{p-2}\,\mathrm{d}\langle M\rangle_s \tag{5.28}$$

(Observe that for $p \geq 2$, $f \in C^2$).

Assume first that M and $\langle M\rangle$ are bounded. Then $M \in \mathbb{H}^2$, $p|M_s|^{p-1}\operatorname{sign} M_s \in L^2(M)$ and so the first term in (5.28) is a true martingale in $\mathbb{H}^2$. Hence, Doob's inequality yields

$$\begin{aligned}
\mathbb{E}\left[\left(\sup_{s\le t}|M_s|\right)^p\right] &\le \left(\frac{p}{p-1}\right)^p \mathbb{E}\left[|M_t|^p\right] \\
&\overset{(5.28)}{\lesssim_p} \mathbb{E}\left[\int_0^t |M_s|^{p-2}\,\mathrm{d}\langle M\rangle_s\right] \\
&\lesssim_p \mathbb{E}\left[\left(\sup_{s\le t}|M_s|\right)^{p-2}\langle M\rangle_t\right] \\
&\overset{\text{Hölder}}{\le} \mathbb{E}\left[\left(\sup_{s\le t}|M_s|\right)^p\right]^{1-\frac{2}{p}} \mathbb{E}\left[\langle M\rangle_t^{\frac{p}{2}}\right]^{\frac{2}{p}},
\end{aligned}$$

where we applied Hölder with "$\frac{1}{r} = 1 - \frac{2}{p}$" and "$\frac{1}{s} = \frac{2}{p}$".

Rearranging and taking $t \to +\infty$ gives the upper bound when M and $\langle M\rangle$ are bounded.

For a general local martingale M we proceed by localisation. Set

$$T_n = \inf\{s \ge 0 \mid |M_s| \ge n \text{ or } \langle M\rangle_s \ge n\}.$$

Then the upper bound holds for M^{T_n} and thus for M by monotone convergence theorem on both sides.

Step 2: Lower bound: By localisation we again assume that M and $\langle M\rangle$ are bounded. Consider

$$Y_t = \delta + \varepsilon\langle M\rangle_t + M_t^2 = \delta + (1+\varepsilon)\langle M_t\rangle + 2\int_0^t M_s\,\mathrm{d}M_s, \qquad \delta, \varepsilon > 0, t \ge 0.$$

By Itô's formula (Theorem 5.15) on $f(x) = x^m$, i.e., $f \in C^2([\delta, +\infty); \mathbb{R})$ for $m = \frac{p}{2}$,

$$\begin{aligned}
Y_t^m = \delta^m &+ m(1+\varepsilon)\int_0^t Y_s^{m-1}\,\mathrm{d}\langle M\rangle_s + 2m\int_0^t M_s Y_s^{m-1}\,\mathrm{d}M_s \\
&+ 2m(m-1)\int_0^t Y_s^{m-2}M_s^2\,\mathrm{d}\langle M\rangle_s.
\end{aligned}$$

Since M, $\langle M\rangle$ are bounded and $Y \ge \delta$, $2m\int_0^t M_s Y_s^{m-1}\,\mathrm{d}M_s$ is a bounded martingale so its expectation vanishes. Hence

$$\mathbb{E}\left[Y_t^m\right] = \delta^m + m(1+\varepsilon)\,\mathbb{E}\int_0^t Y_s^{m-1}\,\mathrm{d}\langle M\rangle_s + 2m(m-1)\,\mathbb{E}\int_0^t Y_s^{m-2}M_s^2\,\mathrm{d}\langle M\rangle_s.$$

The last term is non-negative. Taking $\delta \downarrow 0$, we obtain

$$\begin{aligned}\mathbb{E}\left[(\varepsilon\langle M\rangle_t + M_t^2)^m\right] &\geq m(1+\varepsilon)\,\mathbb{E}\left[\int_0^t (\varepsilon\langle M\rangle_s + M_s^2)^{m-1}\,\mathrm{d}\langle M\rangle_s\right] \\ &\geq m(1+\varepsilon)\varepsilon^{m-1}\,\mathbb{E}\int_0^t \langle M\rangle_s^{m-1}\,\mathrm{d}\langle M\rangle_s \\ &= (1+\varepsilon)\varepsilon^{m-1}\,\mathbb{E}\left[\langle M\rangle_t^m\right].\end{aligned}$$

For $m \geq 1$, x^n is convex on $[0, +\infty)$, so $2^{m-1}(x^m + y^m) \geq (x+y)^m$, hence the last inequality can be written as

$$\varepsilon^m\,\mathbb{E}\langle M\rangle_t^m + \mathbb{E}\left[|M_t|^{2m}\right] \geq (1+\varepsilon)\left(\frac{\varepsilon}{2}\right)^{m-1}\mathbb{E}\langle M\rangle_t^m$$

and thus

$$\mathbb{E}\left[|M_t|^{2m}\right] \geq \underbrace{\left[(1+\varepsilon)\left(\frac{\varepsilon}{2}\right)^{m-1} - \varepsilon^m\right]}_{=c(m)>0 \text{ for } \varepsilon \text{ small}}\mathbb{E}\left[\langle M\rangle_t^m\right].$$

Since $\mathbb{E}\left[|M_t|^{2m}\right] \leq \mathbb{E}\left[\left(\sup_{s\leq t}|M_s|^{2m}\right)^{2m}\right]$ this completes the proof after taking $t \to +\infty$. □

5.5. Brownian motion and harmonic functions

In this short section we derive two important properties of d-dimensional Brownian motion ($d \geq 2$). As we will see they can by proved by application of Itô's formula (see Equation (5.26)).

Definition 5.27. Let $U \subseteq \mathbb{R}^d$ be an non-empty open set. A function $f \in C^2(U;\mathbb{R})$ is called *harmonic* in U if

$$\Delta f(x) = \sum_{i=1}^d \partial_i^2 f(x) = 0$$

for all $x \in U$.

Harmonic functions play an important role in the study of Brownian motion.

Proposition 5.28. *When $d \geq 2$ and $x \in \mathbb{R}^d \setminus \{0\}$, then*

$$W_x\text{-a.s. } X_t \neq 0 \text{ for all } t \geq 0,$$

(here we work in the setting of canonical Brownian motion).

Remark 5.29. The above proposition implies that "Brownian motion does not hit a given point when $d \geq 2$". ♦

Proof. For $g \in C^2\big((0,+\infty);\mathbb{R}\big)$ we define the radial function

$$f : \mathbb{R}^d \setminus \{0\} \to \mathbb{R}, \quad x \mapsto g\big(|x|\big) = g\left(\sqrt{x_1^2 + \cdots + x_d^2}\right).$$

Then the following identity holds:

$$\Delta f(x) = g''(r) + \frac{d-1}{r} g'(r), \quad \text{where } r = |x| \neq 0 \in \mathbb{R}^d. \tag{5.29}$$

If $d \geq 3$ we choose $g(r) = r^{2-d}$ so that

$$g''(r) + g'(r) = (2-d)(1-d)r^{-d} + (d-1)(2-d)r^{-d} = 0$$

and thus

$$f(x) = \frac{1}{|x|^{d-2}}, \quad x \neq 0, \tag{5.30}$$

is harmonic on $\mathbb{R}^d \setminus \{0\}$. For $x \neq 0$ we fix $0 < a < |x| < b < +\infty$ and pick a smooth function f_a which satisfies $f_a = f$ on $\left\{x \in \mathbb{R}^d \,\middle|\, |x| \geq a\right\}$. We apply Itô's formula Theorem 5.15: For W_x-a.s. for $t \geq 0$,

$$f_a(X_t) = f_a(X_0) + \int_0^t \nabla f_a(X_s) \cdot \mathrm{d}X_s + \frac{1}{2}\int_0^t \Delta f_a(X_s)\,\mathrm{d}s$$

Introducing the stopping time

$$\tau = \inf\{s \geq 0 \mid |X_s| \geq b \text{ or } |X_s| \leq a\},$$

we see that W_x-a.s.

$$|X_{t\wedge\tau}|^{2-d} = f_a(X_{t\wedge t}) = |x|^{2-d} + \int_0^{t\wedge\tau} \nabla f_a(X_s) \cdot \mathrm{d}X_s + \frac{1}{2}\int_0^{t\wedge\tau} \Delta f_a(X_s)\,\mathrm{d}s.$$

As the last term vanishes since $\Delta f_a(X_s) = 0$ for $s \leq t \wedge \tau$ we see that $(|X_{t\wedge\tau}|^{2-d})_{t\geq 0}$ is a bounded local martingale and thus a martingale. Therefore,

$$\mathbb{E}\left[|X_{t\wedge\tau}|^{2-d}\right] = |x|^{2-d}, \quad \text{for all } t \geq 0.$$

Letting $t \to +\infty$, using dominated convergence theorem, we find

$$|x|^{2-d} = \mathbb{E}\left[|X_\tau|^{2-d}\right] = a^{2-d} W_x\big(|X_\tau| = a\big) + b^{2-d} W_x\big(|X_\tau| = b\big).$$

Then using $W_x\big(|X_\tau| = a\big) + W_x\big(|X_\tau| = b\big) = 1$ we obtain

$$W_x\big(|X_\tau| = a\big) = \frac{|x|^{2-d} - b^{2-d}}{a^{2-d} - b^{2-d}}. \tag{5.31}$$

Letting $a \to 0$, while keeping b fixed we see

$$W_x\left(H_{\{0\}} < H_{B_b(0)^\complement}\right) = 0, \quad \text{for} \quad H_A = \inf\{s \geq 0 \mid X_s \in A\}.$$

Then it follows that

$$W_x(X_t = 0 \text{ for some } t) = W_x\left(H_{\{0\}} < +\infty\right) = \lim_{b\to+\infty} W_x\left(H_{\{0\}} < H_{B_b(0)^\complement}\right),$$

and the proposition follows for $d \geq 3$.

For $d = 2$ one chooses $g(r) = \log\frac{1}{r}$, Again, we have that $f(x) = \log\frac{1}{|x|}$ is harmonic on $\mathbb{R}^2 \setminus \{0\}$. The same reasoning as above then yields

$$W_x\left(|X_\tau| = a\right) = \frac{\log\frac{b}{|x|}}{\log\frac{b}{a}}, \tag{5.32}$$

which then implies the proposition by sending $a \to 0$ and then $b \to +\infty$. $\square$

Similar ideas can be applied for discussing the recurrence and transience of Brownian motion.

Theorem 5.30 (Transience of Brownian motion for $d \geq 3$). *When $d \geq 3$, then for $x \in \mathbb{R}^d$, W_x-a.s. $\lim_{t\to+\infty}|X_t| = +\infty$.*

Proof. As for $x, z \in \mathbb{R}^d$ under W_x the process $(X_t + z)_{t\geq 0}$ is a Brownian motion started from $x + z$, it suffices to show Theorem 5.30 for some $x \neq 0$. As previously we see that (for $0 < a < |x|$) $(|X_{t\wedge H(B_a(0)^\complement)}|^{2-d})_{t\geq 0}$ is a continuous and bounded martingale under W_x. Using Fatou's lemma for conditional expectations, for $s \leq t$, W_x-a.s.

$$\mathbb{E}\left[|X_t|^{2-d} \,\middle|\, \mathcal{F}_s\right] = \mathbb{E}_x\left[\liminf_{n\to+\infty}\left|X_{t\wedge H(B_{1/n}(0))}\right|^{2-d} \,\middle|\, \mathcal{F}_s\right]$$

$$
\begin{aligned}
&\overset{\text{Fatou}}{\leq} \liminf_{n\to+\infty} \mathbb{E}_x \left[\left| X_{t\wedge H(B_{1/n}(0))} \right|^{2-d} \middle| \mathcal{F}_s \right] \\
&= \liminf_{n\to+\infty} \left| X_{s\wedge H(B_{1/n}(0))} \right|^{2-d} \\
&\overset{\text{Proposition 3.2}}{=} |X_s|^{2-d}.
\end{aligned}
$$

Hence $(|X_t|^{2-d})_{t\geq 0}$ is a continuous super martingale under W_x. Since this super martingale is non-negative, it follows from the super martingale convergence theorem that

$$
W_x\text{-a.s. } |X_t|^{2-d} \text{ has a finite limit as } t \to +\infty. \tag{5.33}
$$

On the other hand, by looking on one component of X we already know that

$$
W_x\text{-a.s. } \limsup_{t\to+\infty} |X_t| = +\infty. \tag{5.34}
$$

Equations (5.33) and (5.34) then together imply Theorem 5.30. ☐

We now return to the two-dimensional Brownian motion.

Theorem 5.31 (recurrence of Brownian motion in $\mathbb{R}^2$). *For any $x \in \mathbb{R}^2$, W_x-a.s., for any non-empty open set $O \subseteq \mathbb{R}^2$, the set $\{t \geq 0 \mid X_t \in O\}$ is unbounded.*

Proof. From (5.32) we see by letting $b \to +\infty$ that when $a < |x|$ then W_x-a.s. $H(B_b(0)) < +\infty$. This remains true when $|x| \leq a$, so

$$
\text{for any } x \in \mathbb{R}^2,\ a > 0,\ W_x\text{-a.s. } H(B_a(0)) < +\infty. \tag{5.35}
$$

We can then define a sequence of stopping times with respect to $\bar{\mathcal{F}}_t$:

$$
S_1 = H_{B_a(0)}, \quad S_2 = S_1 \circ \theta_{S_1+1} + S_1 + 1, \quad S_{i+1} = S_1 \circ \theta_{S_{i+1}} + S_i + 1
$$

for $i \in \mathbb{N}$. It then follows that $S_i \nearrow +\infty$. Using the strong Markov property, for $y \in \mathbb{R}^2$,

$$
\begin{aligned}
W_y(S_{i+1} < +\infty) &= W_y\left(S_i < +\infty, \theta^{-1}_{S_i+1}\{S_1 < +\infty\}\right) \\
&= \mathbb{E}_y\Big[S_i < +\infty, \underbrace{\mathbb{P}_{X_{S_i+1}}(S_1 < +\infty)}_{=1 \text{ by } (5.35)}\Big] \\
&= W_y(S_i < +\infty) \\
&\ \vdots \\
&= W_y(S_i < +\infty) = 1.
\end{aligned} \tag{5.36}
$$

Hence, by construction for $i \geq 1$, S_y-a.s. on $\{S_i < +\infty\}$, $X_{S_i} \in B_a(0)$ and thus, using (5.36) and $S_i \nearrow +\infty$, for any $y \in \mathbb{R}^2$, W_y-a.s. for any a, the set $\{t \geq 0 \mid X_t \in B_a(0)\}$ is unbounded.

Using that W_y is the law of $X_t + y$ under W_0 we get W_0-a.s. for all $z \in \mathbb{Q}^2$, $n \geq$, that

$$\left\{ t \geq 0 \,\middle|\, X_t \in B_{1/n}(z) \right\}$$

is unbounded, which proves the theorem for $x = 0$. The general case follows by translation. □

Remark 5.32. The last theorem implies that the trajectory of Brownian motion $\{X_t \mid t \geq 0\}$ is dense in $\mathbb{R}^2$. ♦

Exercise 5.33. Use martingale convergence theorem and the recurrence of Brownian motion in $d = 2$ to show Liouville's theorem, that is that every bounded harmonic function on $\mathbb{R}^2$ is constant.

5.5.1. Conformal invariance of Brownian motion

Itô's formula can be used to show the following striking property of Brownian motion paths in $\mathbb{R}^2$. In this section we identify $\mathbb{R}^2$ and $\mathbb{C}$ and use complex notation when convenient.

As a motivation consider the following examples:

Example 5.34.
(a) Let X be canonical Brownian motion in $d = 2$. For $\theta \in \mathbb{R}$ consider the rotation $f_\theta(x) = e^{i\theta}x$ of $\mathbb{C}$ and extend it to $C = C^0([0, +\infty); \mathbb{C})$ by $f_\theta(w)_t = f_\theta(w_t)$, $w \in C$. Then the image measure of W_0 by f_θ is again W_0, i.e., $f_{\theta\#}W_0 = W_0$ (we leave it as an exercise to the reader to verify this).

(b) Consider the map $\phi(z) = az+b$ for $a, b \in \mathbb{C}$, $a \neq 0$. Using the scaling property of Brownian motion, we obtain that, under W_0,

$$\phi(X_t) = B_{|a|^2 t}$$

where B_t is a Brownian motion on $\mathbb{R}^2$ distributed according to W_b.

In particular the image of the Brownian motion path under ϕ is again a time-change of Brownian motion path. We will now see that this is true even if $\phi(z) \sim az + b$ only locally.

♦

Recall that a function $\varphi : \mathbb{C} \to \mathbb{C}$ is called holomorphic in $U \subseteq \mathbb{C}$, where U is an open subset of $\mathbb{C}$, if the complex derivative

$$\lim_{z \to z_0} \frac{\varphi(z) - \varphi(z_0)}{z - z_0}$$

exists for any $z_0 \in U$. Every holomorphic function is analytic and, due to the Cauchy-Riemann equations both real and imaginary parts are harmonic when viewing them as functions $\mathbb{R}^2 \to \mathbb{C}$.

Theorem 5.35. *Let $U \subseteq \mathbb{C}$ be open, $z_0 \in U$ and $\varphi : U \to \mathbb{C}$ holomorphic and let B be a Brownian motion started from z_0. Set*

$$T_U = H_{U^\complement} = \inf\{s \geq 0 \mid B_s \notin U\} < +\infty$$

Then there exists a complex Brownian motion $\bar{B}$ such that for $t \in [0, \tau_n)$

$$\varphi(B_t) = \bar{B}_{C_t},$$

where

$$C_t = \int_0^t |\varphi'(B_s)|^2 \, \mathrm{d}s.$$

Proof. Write $\varphi = g + \mathrm{i}h$, so that g, h are harmonic on U. By Itô's formula applied to $g(B_t^1 + \mathrm{i}B_t^2)$ we get that for $t < T_U$,

$$g(B_t) = g(z_0) + \int_0^t \frac{\partial g}{\partial x}(B_s) \, \mathrm{d}B_s^1 + \int_0^t \frac{\partial g}{\partial y}(B_s) \, \mathrm{d}B_s^2$$

and, similarly,

$$h(B_t) = h(z_0) + \int_0^t \frac{\partial h}{\partial x} B_s \, \mathrm{d}B_s^1 + \int_0^t \frac{\partial h}{\partial y}(B_s) \, \mathrm{d}B_s^2$$

This shows that $M_t = g(B_t)$, $N_t = h(B_s)$ are continuous local martingales on the stochastic interval $[0, T_U)$.

The Cauchy-Riemann equations $\frac{\partial g}{\partial x} = \frac{\partial h}{\partial y}$, $\frac{\partial g}{\partial y} = -\frac{\partial h}{\partial x}$ imply then

$$\langle M \rangle_t = \int_0^t \left(\frac{\partial g}{\partial x}(B_s)\right)^2 + \left(\frac{\partial g}{\partial y}(B_s)\right)^2 \mathrm{d}s = \int_0^t |\varphi'(B_s)|^2 \, \mathrm{d}s = C_t,$$

and similarly, using the properties of the stochastic integral,

$$\langle N\rangle_t = C_t, \quad \text{and} \quad \langle M, N\rangle_t = 0.$$

Setting $\tau_t = \inf\{u \mid C_t > t\}$ it follows form Theorem 5.21 theorem that $\bar{B}^1_t = M_{\tau_t}$ and $\bar{B}^2_t = N_{\tau_t}$ are Brownian motions.

Moreover, $\langle M, N\rangle_t = 0$ and a slight extension of the argument of the last part of the proof of Theorem 5.21 implies that $\langle \bar{B}^1, \bar{B}^2\rangle = 0$, which by Lévy's theorem yields that $\bar{B} = (\bar{B}^1, \bar{B}^2)$ is a 2D Brownian motion and $\varphi(B_t) = \bar{B}_{C_t}$ on $[0, T_U)$.

Note that $\bar{B}^1$ and $\bar{B}^2$ are only defined on a stochastic interval $[0, C_{T_u})$ but the can be "arbitrarily" extended to the whole $\mathbb{R}_+$. $\square$

This next remark is an important counterexample:

Remark 5.36 (c.f. Theorem 4.37 and remark 4.38)**.** The previous theory provides us with an example of a local martingale M which is bounded in L^2, i.e., $\sup_{t\geq 0}\|M_t\|_{L^2(\mathbb{P})} < +\infty$, but is not a true martingale.

To this end fix $d = 3$ and consider a d-dimensional Brownian motion B_t. started from $x \neq 0$ and set $M_t = \frac{1}{|B_t|}$.

By arguments of the proof of Proposition 5.28, we know that M_t is a local martingale. Moreover, as

$$\mathbb{P}(B_t \in \mathrm{d}y) = \frac{1}{(2\pi t)^{3/2}} \mathrm{e}^{-\frac{(y-x)^2}{t}} =: p_t(x, y)$$

we have for $t \geq 0$, for an $\varepsilon < |x|$ small,

$$\begin{aligned}
\mathbb{E}\left[M_t^2\right] &\leq \frac{1}{2}\mathbb{P}(|B_t| > \varepsilon) + \int_{|y|\leq\varepsilon} p_t(x, y) y^{-2}\,\mathrm{d}y \\
&\leq \frac{1}{\varepsilon^2} + \sup_{|y|\leq\varepsilon} p_t(x, y) \underbrace{\int_{|y|\leq\varepsilon} y^{-2}\,\mathrm{d}y}_{<+\infty} \leq C < +\infty
\end{aligned}$$

for all $t \geq 0$.

Hence M is L^2 bounded and thus uniformly integrable. If M would be a martingale, then $M_t = \mathbb{E}\left[M_\infty \,|\, \mathcal{F}_t\right]$, but we already proved that $M_\infty = 0$ a.s., leading to a contradiction.

Observe that M is an uniformly integrable local martingale which is not a martingale. ♦

Exercise 5.37. Prove that $\mathbb{E}[\langle M\rangle_\infty] = +\infty$, c.f. Theorem 4.37, *(ii)*.

5.6. Change of measure and Girsanov transformation

In this section we study how semi-martingales on a filtered probability space $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$, satisfying the usual conditions, behave under different probability measures $\mathbf{P}$ on the same space. The measure $\mathbf{P}$ will be chosen absolutely continuous with respect to $\mathbb{P}$.

Lemma 5.38. *Let $\mathbf{P} \ll \mathbb{P}$ on $\mathcal{G}_\infty := \sigma\left(\bigcup_{t\geq 0} \mathcal{G}_t\right)$ and let*

$$D_t = \left.\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{G}_t}, \quad \text{for } t \in [0, +\infty]$$

be the Radon-Nikodym derivative of $\mathbf{P}$ with respect to $\mathbb{P}$ on $\mathcal{G}_t$. Then D is a uniformly integrable $\mathcal{G}_t$-martingale, which we can assure being cadlag. With this assumption, also $D_T = \left.\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}\mathbb{P}}\right|_T$ for any stopping time T. If $\mathbf{P}$ is equivalent to $\mathbb{P}$ on $\mathcal{G}_\infty$ i.e.,

$$\mathbf{P}(A) = 0 \iff \mathbb{P}(A) = 0 \quad \text{for any } A \in \mathcal{G}_t$$

then also

$$\mathbb{P}(\forall t \geq 0, D_t \text{ and } D_{t-} > 0) = 1.$$

Proof. For $A \in \mathcal{G}_t$ we have

$$\mathbf{P}(A) = \mathbf{E}[\mathbf{1}_A] = \mathbb{E}[\mathbf{1}_A D_\infty] = \mathbb{E}\left[\mathbf{1}_A \mathbb{E}\left[D_\infty \,|\, \mathcal{G}_t\right]\right].$$

As the Radon-Nikodym derivative is essentially unique, it follows that

$$D_t = \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbf{P}} = \mathbb{E}\left[D_\infty \,|\, \mathcal{G}_t\right].$$

Hence D is an uniformly integrable martingale closed by D_∞. The cadlag replacement is possible due to Theorem 4.17.

Further, for a stopping time T, where by Theorem 4.20 for $A \in \mathcal{G}_T$,

$$\mathbf{P}(A) = \mathbb{E}\left[\mathbf{1}_A D_\infty\right] = \mathbb{E}\left[\mathbf{1}_A D_T\right].$$

As D_T is $\mathcal{G}_T$ measurable, this implies that $D_T = \left.\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{G}_T}$.

To show the last claim, let $T = \inf\{t \mid D_t = 0 \text{ or } D_{t-} = 0\}$. We claim that

$$\mathbb{P}(D_t = 0 \forall t \geq T) = 1. \tag{5.37}$$

Indeed, set $T_\varepsilon = \inf\{t \mid D_t \le \varepsilon\}$. Then $T_\varepsilon \uparrow T$ as $\varepsilon \downarrow 0$. Thus, for $t \in \mathbb{Q}$, by the stopping theorem,

$$\mathbb{P}\left(D_{t\vee T_\varepsilon} > \sqrt{\varepsilon}\right) \le \frac{1}{\sqrt{\varepsilon}}\mathbb{E}\left[D_{t\vee T_\varepsilon}\right] = \frac{1}{\sqrt{\varepsilon}}\mathbb{E}\left[D_{T_\varepsilon}\right] \le \sqrt{\varepsilon}.$$

Letting $\varepsilon \downarrow 0$ implies that $\mathbb{P}(D_{t\vee T} = 0) = 1$ and thus $\mathbb{P}(D_t = 0 \forall t \in \mathbb{Q}, t \ge T) = 1$, which yields (5.37) by cadlag property. As $\mathbf{P}$ and $\mathbb{P}$ are equivalent we know that $\mathbb{P}(D_t = 0) = 0$. Comparing this with (5.37) then yields $\mathbb{P}(T = +\infty) = 1$ completing the proof. ☐

The next lemma links D from Lemma 5.38 to the exponential martingales of Proposition 5.18.

Lemma 5.39. *Let D be a* continuous *strictly positive local martingale. Then there is a unique continuous local martingale L such that*

$$D_t = \exp\left\{L_t - \frac{1}{2}\langle L\rangle_t\right\} = \mathcal{E}(L).$$

The process L is given by

$$L_t = \log D_0 + \int_0^t D_s^{-1}\,\mathrm{d}D_s.$$

Moreover,

$$D_t = D_0 + \int_0^t D_s\,\mathrm{d}L_s.$$

Proof. Uniqueness follow by standard arguments. If L and L' satisfy this defining relation, then $\langle L - L'\rangle_t = 0$.

To see the second property, we use $D_t > 0$ and apply Itô's formula on $\log D_t$.

$$\log D_t = \log D_0 + \int_0^t \frac{\mathrm{d}D_s}{D_s} - \frac{1}{2}\int_0^t \frac{\mathrm{d}\langle D\rangle_s}{D_s^2} = L_t - \frac{1}{2}\langle L\rangle_t.$$

The last property hold for every pair $L, D = \mathcal{E}(L)$. It follows directly from the proof of Proposition 5.18. ☐

Theorem 5.40 (Cameron-Martin 1944, Girsanov 1960). *Let $\mathbf{P}$ be equivalent with $\mathbb{P}$ on $\mathcal{G}_\infty$. Assume that the process D of Lemma 5.38 is continuous. Let L be as in Lemma 5.39. Then if M is $(\mathcal{G}_t, \mathbb{P})$-local martingale, then its Girsanov transform*

$$\tilde{M}_t = M_t - \langle M, L\rangle_t, \quad t \geq 0$$

is a $(\mathcal{G}_t, \mathbf{P})$-local martingale.

Proof. We first show the following:

Claim 5.41. *If T is a stopping time and X a continuous adapted process such that $(XD)^T$ is a $\mathbb{P}$-martingale, then X^T is a $\mathbf{P}$-Martingale.*

Proof of Claim 5.41. Observe

$$\mathbf{E}\left[|X_{t\wedge T}|\right] \overset{\text{Lemma 5.38}}{=} \mathbb{E}\left[|X_{t\wedge T} D_{t\wedge T}|\right] < +\infty,$$

since $(XD)^T$ is $\mathbb{P}$-martingale. That is, $X_t^T \in L^1(\mathbf{P})$ for all $t \geq 0$. Take $A \in \mathcal{G}_s$, $s < t$. Then $A \cap \{T > s\} \in \mathcal{G}_s$, so, as $(XD)^T$ is $\mathbb{P}$-martingale,

$$\mathbb{E}\left[\mathbf{1}_{A\cap\{T>s\}} X_{T\wedge t} D_{T\wedge t}\right] = \mathbb{E}\left[\mathbf{1}_{A\cap\{T>s\}} X_{T\wedge s} D_{T\wedge s}\right].$$

Since $D_{T\wedge s} = \frac{\mathrm{d}\mathbf{P}}{\mathrm{d}\mathbb{P}}\Big|_{\mathcal{G}_{T\wedge s}}$, $D_{T\wedge t} = \frac{\mathrm{d}\mathbf{P}}{\mathrm{d}\mathbb{P}}\Big|_{\mathcal{G}_{T\wedge t}}$ this implies

$$\mathbf{E}\left[\mathbf{1}_{A\cap\{T<s\}} X_{T\wedge t}\right] = \mathbf{E}\left[\mathbf{1}_{A\cap\{T>s\}} X_{T\wedge s}\right].$$

On the other hand we have

$$\mathbf{E}\left[\mathbf{1}_{A\cap\{T\leq s\}} X_{T\wedge t}\right] = \mathbf{E}\left[\mathbf{1}_{A\cap\{T\leq s\}} X_{T\wedge s}\right],$$

so $\mathbf{E}\left[\mathbf{1}_A X_t^T\right] = \mathbf{E}\left[\mathbf{1}_A X_s^T\right]$, proving the claim. ◇

As consequence of Claim 5.41 we see easily that if XD is a continuous $\mathbb{P}$-local martingale, then X is a continuous $\mathbf{P}$-local martingale. Taking now M a $\mathbb{P}$-local martingale, using $X = \tilde{M}$, Itô's formula yields

$$\begin{aligned}
\tilde{M}_t D_t &= M_0 D_0 + \int_0^t \tilde{M}_s \,\mathrm{d}D_s + \int_0^t D_s \,\mathrm{d}M_s - \int_0^t D_s \,\mathrm{d}\langle M, L\rangle_s + \langle M, D\rangle_t \\
&= M_0 D_0 + \int_0^t \tilde{M}_s \,\mathrm{d}D_s + \int_0^t D_s \,\mathrm{d}M_s.
\end{aligned}$$

By Lemma 5.39, $\mathrm{d}\langle M, L\rangle_s = D_s^{-1}\,\mathrm{d}\langle D, M\rangle_s$, i.e., $\tilde{M}D$ is a $\mathbb{P}$ local martingale and thus $\tilde{M}$ is a $\mathbf{P}$-local martingale. □

Corollary 5.42. *When $\mathbf{P}$ is equivalent with $\mathbb{P}$ on $\mathcal{G}_\infty$, then the families of $\mathbb{P}$-semi martingales and $\mathbf{P}$-semi martingales coincide.*

Proof. Let Y be a $\mathbb{P}$-semi martingale with decomposition $Y_t = Y_0 + M_t + A_t$. Then, by Theorem 5.40,

$$Y_t = Y_0 + \tilde{M}_t + \Big(A_t + \langle M, L \rangle_t \Big)$$

with $\mathbf{P}$-local martingale $\tilde{M}_t$. Hence, Y is a $\mathbf{P}$-semi martingale. The other case follows by symmetry of the assumptions on $\mathbb{P}$ and $\mathbf{P}$. □

Remark 5.43. To go back from $\mathbf{P}$-local martingale to $\mathbb{P}$-local martingale we observe that $\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbf{P}}\big|_{\mathcal{G}_t} = D_t^{-1}$, and $D_t^{-1} = \mathcal{E}(-L)$. So if M_t is a $\mathbf{P}$-local martingale, then by Theorem 5.40, $\tilde{M}_t = M_t + \langle M, L \rangle$ is $\mathbb{P}$-local martingale. ♦

Corollary 5.44. *Let $\mathbb{P}$ and $\mathbf{P}$ be equivalent on $\mathcal{G}_\infty$ and let X, Y be two $\mathbb{P}$-semi martingales (and thus $\mathbf{P}$-semi martingales). Then $\mathbb{P}$-a.s.*

$$\langle X, Y \rangle_t^{\mathbb{P}} = \langle X, Y \rangle_t^{\mathbf{P}}, \quad \text{for all } t \geq 0.$$

and $\langle X, Y \rangle$ is given by the approximation in Proposition 4.47.

Proof. It satisfies to use Proposition 4.47 and observe that if $\mathbb{P}$ and $\mathbf{P}$ are equivalent then the limits in $\mathbb{P}$ and $\mathbf{P}$ probability must coincide. □

Corollary 5.45. *If $\mathbb{P}$ and $\mathbf{P}$ are equivalent on $\mathcal{G}_\infty$, X is a semi martingale and H locally bounded. Then the stochastic integrals $H \cdot X$ under $\mathbb{P}$ and $\mathbf{P}$ coincide.*

Proof. The integrals coincide on elementary functions and as $L^2(\mathbb{P})$ and $L^2(\mathbf{P})$ limits coincide too, the result follows. □

If the martingale M is a Brownian motion, Girsanov's theorem theorem yields:

Theorem 5.46. *Let M be a $\mathbb{P}$-Brownian motion. Then $\tilde{M} = M - \langle M, L \rangle$ is a $\mathbf{P}$-*

Brownian motion. Moreover, if there is a progressively measurable process b such that

$$L_t = \int_0^t b_s \,\mathrm{d}M_s,$$

then under $\mathbf{P}$*, M is a "Brownian motion with drift"*

$$M_t = \tilde{M}_t + \int_0^t b_s \,\mathrm{d}s.$$

Proof. Under $\mathbf{P}$, $\tilde{M}_t$ is a continuous local martingale with $\tilde{M}_0 = 0$ and by Corollary 5.44, $\langle \tilde{M} \rangle_t = \langle M \rangle_t = t$. Hence $\tilde{M}$ is a $\mathbf{P}$-Brownian motion by Lévy's theorem (Theorem 5.20).

To prove the second claim, observe that

$$\tilde{M}_t = M_t - \langle M, L \rangle_t = M_t - \left\langle M, \int_0^{\cdot} b_s \,\mathrm{d}M_s \right\rangle_t = M_t - \int_0^t b_s \,\mathrm{d}\langle M \rangle_s$$

and M is a Brownian motion, i.e., $\mathrm{d}\langle M \rangle_t = \mathrm{d}t$. □

Remark 5.47. In applications of Girsanov's theorem, on usually starts with a measure $\mathbb{P}$ and a $\mathbb{P}$-local martingale L with $L_0 = 0$. One then defines $D_t = \mathcal{E}(L)_t$ which is a positive local martingale, i.e., super martingale. If D_t is a true uniformly integrable martingale, one then can define $\mathbf{P}$ on $\mathcal{G}_\infty$ via $\mathbf{P} = D_\infty \mathbb{P}$. To ensure that this is possible, Novikov's condition is important, c.f. Theorem 5.19. ♦

Remark 5.48. The assumption of continuity on D might appear restrictive, but it is not in many situations. E.g., in the setting of canonical Brownian motion, one has the the following theorem:

Theorem 5.49. *Let $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P}) = (C([0,+\infty)), \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, W_0)$. Then every $\mathcal{F}_t$-local martingale M is contiuous and can be written as*

$$M_t = \int_0^t H_s \,\mathrm{d}X_s$$

for a predictable process H_s (and the canonical Brownian motion X).

♦

5.7. Representation of martingales

We are going to show the representation theorem mentioned in Remark 5.48. We consider here the canonical space of Brownian motion, i.e., $\Omega = C([0,+\infty);\mathbb{R})$, $\bar{\mathcal{F}}$ and $\bar{\mathcal{F}}_t$ as in Example 4.2 and W is the Wiener measure and X the canonical process.

Theorem 5.50. *For any random variable $Z \in L^2(C, \bar{F}, W)$ there is an essentially unique process $H \in L^2(X)$ (c.f. Equation (5.6)) such that*

$$Z = \mathbb{E}[Z] + \int_0^{+\infty} H_s \, \mathrm{d}X_s.$$

The proof is based on the following lemma.

Lemma 5.51. *The vector space generated by the random variables*

$$\exp\left\{ \mathrm{i} \sum_{j=1}^{n} \lambda_j \left(X_{t_j} - X_{t_{j-1}} \right) \right\}, \qquad 0 = t_0 < t_1 < \cdots < t_n, \lambda_i \in \mathbb{R}, n \in \mathbb{N}$$

is dense in $L^2(C, \bar{F}, W; \mathbb{C})$.

Proof. It suffices to show that if $Z \in L^2(C, \bar{F}, W; \mathbb{C})$ satisfies

$$\mathbb{E}\left[Z \exp\left(\mathrm{i} \sum_{j=1}^{n} \lambda_j \left(X_{t_j} - X_{t_{j-1}} \right) \right) \right] = 0, \quad \text{for all } t_i, \lambda_i, n, \tag{5.38}$$

then $Z = 0$ W-a.s., which is equivalent ot the fact that the measure $(ZW)(\mathrm{d}w) := Z(w)W(\mathrm{d}w)$ on $(C, \bar{\mathcal{F}})$ is a zero-measure.

To this end we first show that it is a zero measure on $(C, \sigma(X_{t_0}, \dots, X_{t_n}))$ for a fixed sequence $0 = t_0 < \cdots < t_n$. By Remark 5.16 the image of ZW by the map

$$w \mapsto (X_{t_1} - X_{t_0}, \dots, X_{t_n} - X_{t_{n-1}})$$

is the zero measure, since its Fourier transform is zero. This means that this measure vanishes on

$$\sigma(X_{t_1} - X_{t_0}, \dots, X_{t_n} - X_{t_{n-1}}) = \sigma(X_{t_0}, \dots, X_{t_n}).$$

Then one shows that $W(Z\mathbf{1}_A) = 0$ for any $A \in \sigma(X_{t_0}, \dots, X_{t_n})$. A Dykin's type argument implies the claim. $\square$

Proof of Theorem 5.50. Let $\mathcal{H}$ be the subspace of $Z \in L^2(C, \bar{\mathcal{F}}, W)$ which can be written as stated. For $Z \in \mathcal{H}$ we have

$$\mathbb{E}[Z^2] = \mathbb{E}[Z]^2 + \mathbb{E}\left[\int_0^{+\infty} H_s^2 \, \mathrm{d}s\right]. \tag{5.39}$$

In particular this implies the uniqueness of the representation. On the other hand, if $Z_n \in \mathcal{H}$ is a Cauchy sequence in $L^2(C, \bar{\mathcal{F}})$, then the associated processes H_n form a Cauchy sequence in $L^2(X)$, by Equation (5.39). As $L^2(X)$ is complete, $H_n \to H$ in $L^2(X)$ and by the properties of the stochastic integral,

$$Z_n \xrightarrow[n\to+\infty]{L^2(W)} Z = \int_0^{+\infty} H_s \, \mathrm{d}X_s \in \mathcal{H},$$

i.e., $\mathcal{H}$ is complete.

Finally, taking $0 = t_0 < \cdots < t_n$, $\lambda_i \in \mathbb{R}$, $n \in \mathbb{N}$, defining

$$f(s) = \sum_j \lambda_j \mathbf{1}_{(t_{j-1}, t_j]}(s)$$

and using $\mathcal{E}_t^f$ to denote the exponential martingale $\mathcal{E}\left(\mathrm{i} \int_0^\cdot f(s) \, \mathrm{d}X_s\right)$, we obtain by Itô's formula that

$$\exp\left(\mathrm{i} \sum_j \lambda_j \left(X_{t_j} - X_{t_{j-1}}\right) + \frac{1}{2} \sum_j \lambda_j^2 \left(t_j - t_{j-1}\right)\right) = \mathcal{E}_\infty^f = 1 + \int_0^{+\infty} f(s) \mathcal{E}_s^f \, \mathrm{d}X_s.$$

Hence, the random variables of the form $\mathfrak{I}(\exp(\mathrm{i} \sum_j \lambda_j (X_{t_j} - X_{t_{j-1}})))$ are in $\mathcal{H}$. By Lemma 5.51 the space of those random variables is dense in $L^2(C, \bar{\mathcal{F}}, W)$ and since $\mathcal{H}$ is complete we have $\mathcal{H} = L^2(C, \bar{\mathcal{F}}, W)$. □

Theorem 5.52. *Every $(\bar{\mathcal{F}}_t)_{t\geq 0}$-local martingale M has a version written as*

$$M_t = C + \int_0^t H_s \, \mathrm{d}X_s \tag{5.40}$$

for a constant C and $H \in L^2_{\mathrm{loc}}(X)$ (If M is L^2 bounded, then $H \in L^2(X)$). In particular, every $(\bar{\mathcal{F}}_t)_{t\geq 0}$-local martingale has a continuous version.

Proof. If M is bounded in L^2, then $M_\infty \in L^2(C, \bar{\mathcal{F}})$, so by Theorem 5.46,

$$M_\infty = \mathbb{E}[M_\infty] + \int_0^{+\infty} H_s \, \mathrm{d}X_s, \quad \text{with } H \in L^2(X).$$

Also,

$$\begin{aligned}
M_t &= \mathbb{E}\left[M_\infty \,\middle|\, \bar{\mathcal{F}}_t\right] \\
&= \mathbb{E}[M_\infty] + \mathbb{E}\left[\int_0^{+\infty} H_s \,\mathrm{d}X_s \,\middle|\, \bar{\mathcal{F}}_t\right] \\
&= \mathbb{E}[M_\infty] + \int_0^t H_s \,\mathrm{d}X_s,
\end{aligned}$$

so the theorem holds true in this case.

If M is a local martingale, then first $M_0 = C \in \mathbb{R}$ by Theorem 3.26. Taking $T_n = \inf\{t \geq 0 \mid |M_t| \geq n\}$, using the first part of the proof, then is $H_n \in L^2(X)$ such that

$$M_t^{T_n} = C + \int_0^t H_n(s) \,\mathrm{d}X_s.$$

Using the uniqueness of the representation, for $m < n$, $H_n(s,\omega) = H_m(s,\omega)$ $\mathrm{d}s$-a.e., W-a.s. on $[0, T_n]$. We can thus construct $H \in L^2_{\mathrm{loc}}(X)$ such that $H(s,\omega) = H_m(s,\omega)$ $\mathrm{d}s$-a.e. W-a.e. on $[0, T_n]$. By the construction of the stochastic integral we then have (5.40) as required. □

Remark 5.53. As $\langle M, X\rangle_t = \int_0^t H_s \,\mathrm{d}s$, H is the Radon-Nikodym derivative of $\mathrm{d}\langle M, X\rangle_s$ with respect to the Lebesgue measure. In most current examples it is applied with H explicitly. ♦

5.8. Wiener Chaos expansion*

We give another representation theorem for $L^2(C, \bar{\mathcal{F}}, W)$. Let

$$\Delta_n = \{(s_1, \dots, s_n) \in \mathbb{R}^n \mid s_1 > s_2 > \cdots > s_n\}$$

and $L^2(\Delta_n) := L^2(\Delta_n, \mathrm{d}x)$. Define

$$E_n = \left\{ f : \Delta_n \to \mathbb{R} \,\middle|\, f = \prod_{i=1}^n f_i(s_i),\, f_i \in L^2(\mathbb{R}) \right\}.$$

Exercise 5.54. The set of linear combinations of elements of E_n is dense in $L^2(\Delta_n)$.

For $f \in E_n$, we set

$$J_n(f) = \int_0^{+\infty} f_1(s_1)\,\mathrm{d}X_{s_1} \underbrace{\int_0^{s_1} f_2(s_2)\,\mathrm{d}X_{s_2} \cdots \int_0^{s_{n-1}} f_n(s_n)\,\mathrm{d}X_{s_n}}_{=:F_{n-1}(s_1)}.$$

Observe that

$$\begin{aligned}
\|J_n(f)\|_{L^2}^2 &= \mathbb{E}\left[\int_0^{+\infty} f_1^2(s_1)F_{n-1}^2(s_1)\,\mathrm{d}s_1\right] = \int_0^{+\infty} f_1^2(s_1)\,\mathbb{E}\left[F_{n-1}^2(s_1)\right]\mathrm{d}s_1 \\
&\ \vdots \\
&= \int_0^{+\infty} f_1^2(s_1)\,\mathrm{d}s_1 \int_0^{s_1} f_2^2(s_2)\,\mathrm{d}s_2 \cdots \int_0^{s_{n-1}} f_n^2(s_n)\,\mathrm{d}s_n \\
&= \|f\|_{L^2(\Delta_n)}^2.
\end{aligned}$$

Definition 5.55. The smallest closed linear subspace K_n of $L^2(C, \bar{\mathcal{F}}, W)$ containing $J_n(E_n)$ is called the n-th *Wiener chaos.*

By the previous observation, J_n can be extended to an isometry of $L^2(\Delta_n)$ and K_n which is one-to-one. Moreover, if $f \in E_n$, $g \in E_m$ for $m < n$, then

$$\begin{aligned}
&\mathbb{E}^W\left[J_n(f)J_m(g)\right] \\
&\quad = \mathbb{E}\left[\int_0^{+\infty} f_1(s_1)g_1(s_1)F_{n-1}(s_1)G_{m-1}(s_1)\,\mathrm{d}s_1\right] \\
&\quad = \int_0^{+\infty} f_1(s_1)g_1(s_1)\,\mathrm{d}s_1 \int_0^{s_1} f_2(s_2)g_2(s_2)\,\mathrm{d}s_2 \cdots \int_0^{s_{n-1}} f_m(s_m)g_m(s_m)\,\mathrm{d}s_m \\
&\qquad \times \mathbb{E}\left[\int_0^{s_m} f_{m+1}(s_{s_{m+1}})\,\mathrm{d}X_m \cdots \int_0^{s_{n-1}} f_n(s_n)\,\mathrm{d}X_n\right],
\end{aligned}$$

so K_m and K_n are orthogonal in $L^2(C, \bar{\mathcal{F}}, W)$.

Theorem 5.56. *It holds that*

$$L^2(C, \bar{\mathcal{F}}, W) = \bigoplus_{n=0}^{+\infty} K_n,$$

where K_0 is the space of constants. Put differently: Every $Z \in L^2(C, \bar{\mathcal{F}}, W)$ can be *written as*

$$Z \overset{L^2}{=} \mathbb{E}[Z] + \sum_{i=1}^{+\infty} J_m(f_m), \quad \textit{for some } f_n \in L^2(\Delta_n).$$

Proof. we start with a technical lemma. For its statement, let h_n be the n-th *Hermite polynomial* given by

$$\sum_{n\geq 0} \frac{u_n}{n!} h_n(x) = \exp\left\{ux - \frac{u^2}{2}\right\}, \quad u, x \in \mathbb{R},$$

i.e.,

$$h_n(x) = \mathrm{e}^{\frac{x^2}{2}} (-1)^n \frac{\mathrm{d}^2}{\mathrm{d}x^2} \mathrm{e}^{-\frac{x^2}{2}}.$$

Set $H_n(x,a) = a^{n/2} H_n(x/\sqrt{a})$, $H_n(x,0) = x^n$ Then

$$\exp\left(ux - \frac{an^2}{2}\right) = \sum_{n\geq 0} \frac{n^n}{n!} H_n(x,a).$$

Lemma 5.57. *Let M be a local martingale and $M_0 = 0$. Then*

$$L_t^{(n)} = H_n\big(M_t, \langle M\rangle_t\big)$$

is a local martingale and

$$L_t^{(n)} = n! \int_0^t \mathrm{d}H_{s_1} \int_0^{s_1} \mathrm{d}M_{s_2} \cdots \int_0^{s_{n-1}} \mathrm{d}M_{s_n}.$$

Proof of Lemma 5.57. One sees easily that $\left(\frac{1}{2}\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial a}\right) H_n(x,a) = 0$ and $\frac{\partial H_n}{\partial x} = n h_{n-1}$. Hence by Ito's formula,

$$L_t^{(n)} = n \int_0^t L_s^{(n-1)}\, \mathrm{d}M_s.$$

This proves the representation of $L_t^{(n)}$ and thus the claim. □

Consider now the martingale $M_t = \sum_{j=1}^{n} \lambda_j \left(X_{t_j \wedge t} - X_{t_{j-1} \wedge t} \right)$ with $\langle M \rangle_t = \sum_{j=1}^{n} \lambda_j^2 \left(t_j \wedge t - t_{j-1} \wedge \right.$
Then

$$\exp\left(\mathrm{i} \sum_j \lambda_j \left(X_{t_j} - X_{t_{j-1}} \right) \right) + \frac{1}{2} \sum_{j=1}^{n} \lambda_j^2 \left(t_j - t_{j-1} \right) = \sum_{n \geq 0} \frac{(-\mathrm{i})^n}{n!} H_n\left(M_\infty, \langle M \rangle_\infty \right) \quad (5.41)$$

and every $H_n\left(M_\infty, \langle M \rangle_\infty \right)$ as in Lemma 5.57 (with $\lambda_{n+1} = 0$ and $t_{n+1} = +\infty$). Moreover,

$$\mathrm{d}M_s = \prod_{j=1}^{n+1} \left(1 + (\lambda_j - 1) \mathbf{1}_{(t_{j-1}, t_j]}(s) \right) \mathrm{d}X_s =: f(s) \, \mathrm{d}X_s,$$

that is

$$L_t^{(n)} = n! \int_0^t f(s_1) \, \mathrm{d}X_1 \int_0^{s_1} f(s_2) \, \mathrm{d}X_2 \cdots \int_0^{s_{n-1}} f(s_n) \, \mathrm{d}X_n,$$

proving that the RHS of (5.41) can be written as

$$1 + \sum_{j \geq 0} J_n(f_n), \quad \text{with } f_n(s_1, \ldots, s_n) = \mathrm{i}^m \prod_{i=1}^{n} f(s_i). \tag{5.42}$$

The equality in (5.42) holds pointwise, and since f is bounded with compact support also in L^2.

Hence, the statement of the theorem is true for random variables as on the RHS of (5.41). Using Lemma 5.51 the claim follows. □

6. Stochastic differential equations

6.1. Introduction

The goal of stochastic differential equations (SDE's) is to provide a mathematical model for evolution of a physical system subjected to an external noise.

Let us start with an ODE

$$x'(t) = b\big(x(t)\big) \tag{6.1}$$

which describes the development of a system with local drift b. Under suitable assumptions on b it is known that the solution of this ODE exists and is unique (locally or globally).

We now want to construct a model that locally around a point x behaves like a solution to (6.1) subjected to a Brownian noise, i.e., it looks locally like

$$x'(t) + b\big(x(t)\big) + \sigma(x)B(t). \tag{6.2}$$

We will directly consider the vector case , that is $x \in \mathbb{R}^d$, $b : \mathbb{R}^d \to \mathbb{R}^d$, $\sigma(x)$ a $d \times n$-matrix and B_t a Brownian motion in $\mathbb{R}^n$ introducing the noise.

In other words, the increments of our process behave infinitesimally like a Gaussian random variable with mean $b(x)\,\mathrm{d}t$ and covariance matrix given by

$$\begin{aligned}
\xi^\top a(x)\xi &= \mathbb{E}\left[\big(\xi^\top \cdot \sigma(x)\big(B_{t+\mathrm{d}t} - B_t\big)\big)^2\right] \\
&= \mathbb{E}\left[\Big(\big(\sigma(x)^\top\xi\big)^\top \cdot \big(B_{t+\mathrm{d}t} - B_t\big)\Big)^2\right] \\
&= |\sigma(x)^\top\xi|^2\,\mathrm{d}t \\
&= \xi^\top\sigma(x)\sigma(x)^\top\xi, \quad \text{for } \xi \in \mathbb{R}^d.
\end{aligned} \tag{6.3}$$

Hence, $a(x) = \sigma(x)\sigma(x)^\top$.

One way to construct such a model is to consider an SDE

$$\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}B_t, \quad X_0 = x_0. \tag{6.4}$$

The meaning of (6.4) is given by its integral form

$$X_t = x_0 + \int_0^t b(X_s)\,\mathrm{d}s + \int_0^t \sigma(X_s)\,\mathrm{d}B_s, \tag{6.5}$$

or coordinate wise

$$X_t^i = x_0^i + \int_0^t b_i(X_s)\,\mathrm{d}s + \sum_{j=1}^n \int_0^t \sigma_{i,j}(X_s)\,\mathrm{d}B_s^j, \quad \text{for } i = 1, \dots, d. \tag{6.6}$$

Observe that Equations (6.5) and (6.6) are well defined, given the construction of the stochastic integral, and that X is a semi martingale.

In many situations it is useful that b, σ depend on t explicitly, i.e., to consider the following generalization of (6.5):

$$X_t = x_0 + \int_0^t b(s, X_s)\,\mathrm{d}s + \int_0^t \sigma(s, X_s)\,\mathrm{d}B_s. \tag{6.7}$$

We now define the solutions of this SPDE.

Definition 6.1. Let $d, m \in \mathbb{N}$, $\sigma : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^{d\times n}$, $b : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^d$ be a measurable and locally bounded functions. A solution to the equation

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \sigma(t, X_t)\,\mathrm{d}B_t \tag{E(σ, b)}$$

is a collection:

- A filtered probability space $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions;
- An n-dimensional $\mathcal{G}_t$-Brownian motion B;
- A $\mathcal{G}_t$-adapted continuous process $X = (X^1, \dots, X^d)$ such that

$$X_t = X_0 + \int_0^t b(s, X_s)\,\mathrm{d}s + \int_0^t \sigma(s, X_s)\,\mathrm{d}B_s. \tag{6.8}$$

If $X_0 = x_0 \in \mathbb{R}^d$, we say that X solves $\mathrm{E}_{x_0}(\sigma, b)$.

There are various notions of the existence and uniqueness of the solutions to (E(σ, b)).

Definition 6.2. We say that for the SDE ($\mathrm{E}(\sigma, b)$) there is

(i) *weak existence* if for any $x \in \mathbb{R}^d$ there is a solution to ($\mathrm{E}_x(\sigma, b)$);

(ii) *weak existence and uniqueness* if for any $x \in \mathbb{R}^d$ there is a solution X to ($\mathrm{E}_x(\sigma, b)$) and all such solutions have the same law;

(iii) *pathwise uniqueness* if given $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ and B any two solutions X, X' such that $X_0 = X_0' = x_0$ are indistinguishable.

(iv) A solution to ($\mathrm{E}(\sigma, b)$) is said to be *strong* if it is adapted to the (completed) filtration of the Brownian motion B.

Example 6.3. Suppose we have a probability space $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions and β_t a Brownian motion. We consider the case $d = n = 1$.

(a) *Weak existence:* The SDE

$$\mathrm{d}X_t = 2\sqrt{X_t}\,\mathrm{d}B_t + \mathrm{d}t, \quad X_0 = 0$$

is solved by $B_t = \beta_t$ and $X_t = B_t^2$ which can be checked by Ito's formula.

(b) Let $\mathrm{d}X_t = f(X_t)\,\mathrm{d}B_t$ for $f : \mathbb{R} \to \mathbb{R}$ measurable with $|f| \equiv 1$. Then $X_t = \int_0^t f(X_s)\,\mathrm{d}B_s$ and thus X_t is a martingale with $\langle X\rangle_t = \int_0^t f(X_s)^2\,\mathrm{d}s = t$. By Lévy's theorem, X is a Brownian motion.

(c) *pathwise uniqueness:* the following equation describes the so-called geometric Brownian motion with drift:

$$\mathrm{d}X_t = \sigma X_t\,\mathrm{d}B_t + \mu X_t\,\mathrm{d}t, \quad X_0 = 1.$$

It is solved by $B = \beta$ and $X_t = \exp\left(\sigma B_t + (\mu - \frac{\sigma^2}{2}t)\right)$ as can easily be proved by Ito's formula (c.f. exponential martingales). For sake of simplicity, consider $\sigma = 1$ and $\mu = 1$. Let X, Y be two solutions on a given probability space $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ with a Brownian motion B, i.e.,

$$\mathrm{d}X_t = X_t\,\mathrm{d}B_t, \quad \mathrm{d}Y_t = y_t\,\mathrm{d}B_t, \quad X_0 = Y_0 = 1.$$

Then by Ito's formula,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{Y_t}{X_t}\right) &= \frac{\mathrm{d}Y_t}{X_t} - \frac{Y_t\,\mathrm{d}X_t}{X_t^2} + \frac{Y_t}{X_t^3}\,\mathrm{d}\langle X\rangle_t - \frac{1}{X_t^2}\,\mathrm{d}\langle X, Y\rangle_t \\
&= \frac{Y_t\,\mathrm{d}B_t}{X_t} - \frac{y_t X_t\,\mathrm{d}B_t}{X_t^2} + \frac{Y_t}{X_t^3}X_t^2\,\mathrm{d}t - \frac{1}{X_t^2}X_t Y_t\,\mathrm{d}t
\end{aligned}$$

$$= 0.$$

It follows easily that X and Y are indistinguishable.

(d) *Weak existence, no weak uniqueness:* Consider

$$\mathrm{d}X_t = 3\,\mathrm{sign}(X_t)|X_t|^{\frac{2}{3}}\,\mathrm{d}B_t + 3\,\mathrm{sign}(X_t)|X_t|^{\frac{1}{3}}\,\mathrm{d}t, \qquad X_0 = 0.$$

By Ito's formula $\beta = B$ and $X_t = B_t^3$ solve this equation, but $X \equiv 0$ is possible too.

(e) *Weak existence and uniqueness, no path wise uniqueness:* Define sign at 0 by $\mathrm{sign}(0) = 1$ and consider Tanaka's equation

$$\mathrm{d}X_t = \mathrm{sign}(X_t)\,\mathrm{d}B_t, \qquad X_0 = 0.$$

With $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ and β as usual, set

$$B_t = \int_0^t \mathrm{sign}(\beta_s)\,\mathrm{d}\beta_s.$$

By Lévy's theorem, B is a Brownian motion. Then $X_t = \pm\beta_t$ solves the equation. Indeed,

$$\mathrm{d}X_t = \mathrm{d}\beta_t = \mathrm{sign}(\beta_t)\,\mathrm{sign}(\beta_t)\,\mathrm{d}B_t = \mathrm{sign}\,\beta_t\,\mathrm{d}B_t = \mathrm{sign}\,X_t\,\mathrm{d}B_t$$

and for $X_t = -\beta_t$,

$$\begin{aligned}\mathrm{d}X_t &= -\,\mathrm{d}B_t = \left(\mathrm{sign}(\beta_t)\,\mathrm{sign}(-\beta_t) - 2\mathbf{1}_{\beta_t=0}\right)\mathrm{d}B_t \\ &\overset{(!)}{=} \mathrm{sign}(-\beta_t)\,\mathrm{d}B_t = \mathrm{sign}\,X_t\,\mathrm{d}B_t,\end{aligned}$$

where in the equality (!) we ignored the second term, since $\int_0^t \mathbf{1}_{\beta_s=0}\,\mathrm{d}s = 0$ and thus also $\int_0^t \mathbf{1}_{\beta_s=0}\,\mathrm{d}\beta_s = 0$.

We will later show that Tanaka's equation has now strong solution.

From the following theorem it follows that there are no SDE's with pathwise uniqueness and no weak uniqueness, or path wise uniqueness and no strong solution.

Theorem 6.4 (Yamada-Watanabe). *If the equation* ($\mathrm{E}_x(\sigma, b)$) *has weak existence and path wise uniqueness, then it also has weak uniqueness. Moreover, for any* $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ *and every* $\mathcal{G}_t$*-Brownian motion, there is a strong solution to* ($\mathrm{E}_x(\sigma, b)$) *for any* x.

Proof. See [4, Section IX.1]. □

Theorem 6.5 (Picard's iteration method). *Assume that* $b : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^d$ *and* $\sigma : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^{d\times n}$ *are continuous and satisfy the Lipschitz condition*

$$|b(t, y) - b(t, z)| + |\sigma(t, y) - \sigma(t, z)| \leq K \quad \text{for all } t \in \mathbb{R}_+,\ y, z \in \mathbb{R}^d \tag{6.9}$$

and the uniform growth condition

$$|b(t, y)| + |\sigma(t, y)| \leq K(1 + |y|), \quad \text{for all } t \in \mathbb{R}_+,\ y \in \mathbb{R}^d. \tag{6.10}$$

Then for any $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ *and* $(B_t)_{t\geq 0}$ *as above and any* $x_0 \in \mathbb{R}^d$ *there is an essentially unique strong solution to* ($\mathrm{E}_{x_0}(\sigma, b)$).

Remark 6.6. (a) Equations (6.9) and (6.10) are equivalent to (6.9) and uniform boundedness of $|b(t, 0)|$, $|\sigma(t, 0)|$.

(b) The continuity of b and σ in t does seem necessary for the proof below, measurability should be sufficient. The reader may check this as an exercise.

♦

Proof of Theorem 6.5. **Step 1: uniqueness:** Let X, Y be strong solutions to ($\mathrm{E}(\sigma, b)$). For $M > |x_0|$ define

$$T = \inf\{n \in \mathbb{N} \mid |X_n| \geq 0 \text{ or } |Y_n| \geq M\}. \tag{6.11}$$

Then $\mathbb{P}$-a.s. for all $t \geq 0$,

$$X_{t\wedge T} - Y_{t\wedge T} = \int_0^t b(t, X_u) - b(t, Y_u)\,\mathrm{d}u + \int_0^t \sigma(t, X_u) \cdot \sigma(t, Y_u)\,\mathrm{d}B_u. \tag{6.12}$$

Therefore, for any $t_0 \geq 0$, using the elementary inequality $(a + b)^2 \leq 2(a^2 + b^2)$ we find

$$\begin{aligned}\mathbb{E}\left[\sup_{t\le t_0}|X_{t\wedge T}-X_{t\wedge T}|^2\right] &\le 2\,\mathbb{E}\left[\sup_{t\le t_0}\left|\int_0^{T\wedge t} b(u,X_u)-b(u,Y_u)\,\mathrm{d}u\right|^2\right]\\ &\quad+2\,\mathbb{E}\left[\sup_{t\le t_0}\left|\int_0^{T\wedge t}\sigma(u,X_u)-\sigma(u,Y_u)\,\mathrm{d}B_u\right|^2\right] \\ &=:\ \mathrm{I}+\mathrm{II}.\end{aligned}\tag{6.13}$$

By Cauchy-Schwartz inequality, the first term on the RHS of (6.13) satisfies

$$\mathrm{I}\le 2t_0\,\mathbb{E}\left[\int_0^{t_0\wedge T}|b(u,X_u)-b(u,Y_u)|^2\,\mathrm{d}u\right],\tag{6.14}$$

and using the Lipschitz condition and Fubini's theorem

$$\mathrm{I}\le 2t_0K^2\int_0^{t_0}\mathbb{E}\left[|X_{u\wedge T}-Y_{u\wedge T}|^2\right]\mathrm{d}u.\tag{6.15}$$

For the second term on the RHS of (6.13), by Doob's inequality with $p=2$ component wise, we obtain the bound

$$\begin{aligned}\mathrm{II}&\le 8\,\mathbb{E}\left[\left|\int_0^{t\wedge T}\sigma(u,X_u)-\sigma(u,Y_u)\,\mathrm{d}B_u\right|^2\right]\\ &=8\,\mathbb{E}\left[\int_0^{t\wedge T}\Big(\sigma(u,X_u)-\sigma(u,Y_u)\Big)^2\,\mathrm{d}u\right]\\ &\overset{(6.9)}{\le} 8K^2\,\mathbb{E}\left[\int_0^{t\wedge T}|X_u-Y_u|^2\,\mathrm{d}u\right].\end{aligned}\tag{6.16}$$

Combining Equations (6.13) to (6.16) we find

$$\mathbb{E}\left[\sup_{t\le t_0}|X_{t\wedge T}-Y_{t\wedge T}|^2\right]\le(8K^2+2t_0K^2)\int_0^{t_0}\mathbb{E}|X_{u\wedge T}-Y_{u\wedge T}|^2\,\mathrm{d}u.\tag{6.17}$$

To control (6.17) we need the following useful lemma:

Lemma 6.7 (Grönwall). *Let f be a non-negative and integrable function on $[0,t]$ such that for some $0\le a,b<+\infty$ and all $0\le u\le t$*

$$f(u)\le a+b\int_0^u f(s)\,\mathrm{d}s.\tag{6.18}$$

Then, for all $0 \le u \le t$,

$$f(u) \le a \exp\{bu\}. \tag{6.19}$$

Proof of Lemma 6.7. Iterating the inequality (6.18) we get

$$\begin{aligned}
f(u) &\le a + b\int_0^u f(s)\,\mathrm{d}s \\
&\le a + bau + b^2\int_0^u \mathrm{d}s_1 \int_0^{s_1} \mathrm{d}s_s f(s_2) \\
&\ \vdots \\
&\le a + bau + \frac{1}{2}ab^2u^2 + \cdots + \frac{1}{n!}ab^nu^n + b^{n+1}\int_0^u \mathrm{d}s_1 \int_0^{s_1} \mathrm{d}s_2 \cdots \int_0^{s_n} \mathrm{d}s_{n+1} f(s_{n+1}) \\
&\le ae^{bu} + b^{n+1}\int_0^u \frac{(u-s)^n}{n!} f(s)\,\mathrm{d}s \\
&\le ae^{bu} + b^{n+1}\frac{u^n}{n!}\int_0^u f(s)\,\mathrm{d}s \\
&\xrightarrow[n\to+\infty]{} ae^{bu}.
\end{aligned}$$

This concludes the proof of Grönwall. □

Applying Lemma 6.7 on (6.17) with $a = 0$, $b = 8K^2 + 2t_0K^2$ and

$$f(u) = \mathbb{E}\left[\sup_{s\le u}|X_{s\wedge T} - Y_{s\wedge T}|^2\right]$$

we obtain that

$$\mathbb{E}\left[\sup_{t\le t_0}|X_{t\wedge T} - Y_{t\wedge T}|^2\right] = 0. \tag{6.20}$$

The uniqueness follows by letting $M \to +\infty$ (i.e., $T \to +\infty$) since then (6.20) can be read (recall that t_0 is arbitrary),

$$X_t = Y_t, \quad \mathbb{P}\text{-a.s. for all } t \ge 0.$$

Step 2: Existence: We iteratively define for $m \geq 0$ and $t \geq 0$,

$$\begin{aligned} X_t^0 &= x_0 \\ X_t^1 &= x_0 + \int_0^t b(s, X_s^0)\, \mathrm{d}s + \int_0^t \sigma(s, X_s^0)\, \mathrm{d}B_s \\ &\vdots \\ X_t^{n+1} &= x_0 + \int_0^t b(s, X_s^m)\, \mathrm{d}s + \int_0^t \sigma(s, X_s^m)\, \mathrm{d}B_s \end{aligned} \tag{6.21}$$

Then for $m \in \mathbb{N}$ and $t \geq 0$,

$$X_t^{m+1} - X_t^m = \int_0^t b(s, X_s^m) - b(s, X_s^{m-1})\, \mathrm{d}s + \int_0^t \Big(\sigma(s, X_s^m) - \sigma(s, X_s^{m-1})\Big)\, \mathrm{d}B_s. \tag{6.22}$$

With $M \geq |x|$ and $T_M = \inf\left\{ n \in \mathbb{N}_0 \,\middle|\, |X_n^m| \geq M \text{ or } |X_n^{m-1} \geq M| \right\}$, similarly as above we have for $t_0 \in [0, t]$,

$$\mathbb{E}\left[\sup_{s \leq t_0 \wedge T_M} |X_s^{m-1} - X_s^m|^2 \right] \leq (8 + 2t) K^2 \int_0^{t_0} \mathbb{E}\left|X_{u \wedge T_M}^m - X_{u \wedge T_M}^{m-1}\right|^2 \mathrm{d}u. \tag{6.23}$$

Obviously, $|x_0| = \sup_{s \leq t} |X_s^0| \in L^2(\mathbb{P})$. By Equations (6.9) and (6.10), $b(s, x_0)$ and $\sigma(s, x_0)$ are bounded in t and thus by (6.21) also $\sup_{s \leq t} |X_s^1| \in L^2(\mathbb{P})$. From (6.23) with $m = 1$, letting $M \to +\infty$ we see that $\sup_{s \leq t} |X_s^2| \in L^2(\mathbb{P})$ and iteratively we get

$$\sup_{s \leq t} |X_s^m| \in L^2(\mathbb{P}), \qquad \text{for all } m \geq 0,\ t \geq 0. \tag{6.24}$$

Comming back to (6.23), we can let $M \to +\infty$ and find for $t_0 \in [0, t]$,

$$\begin{aligned} \mathbb{E}\left[\sup_{s \leq t_0} |X_s^{m+1} - X_s^m|^2 \right] &\leq K^2 (8 + 2t) \int_0^{t_0} \mathbb{E}\left|X_n^m - X_n^{m-1}\right|^2 \mathrm{d}u \\ &\ \vdots \\ &\leq \Big(K^2(8 + 2t)\Big)^m \int_0^{t_0} \mathrm{d}t_1 \cdots \int_0^{t_{m-1}} \mathrm{d}t_m\, \mathbb{E}\left|X_{t_m}^1 - X_{t_m}^0\right|^2. \end{aligned} \tag{6.25}$$

With Equation (6.21) we have

$$\begin{aligned} \mathbb{E}\left|X_{t_m}^1 - X_{t_m}^0\right| &\leq 2|b(x_0)|^2 t_m^2 + 2\, \mathbb{E} \int_{t_0}^{t_m} \sigma(s, x_0)\, \mathrm{d}B_s \\ &\leq 2|b(x_0)| t_m^2 + 2 \int_0^{t_m} |\sigma(s, x_0)|^2\, \mathrm{d}s \\ &\overset{(6.9),(6.10)}{\leq} K'(x_0, t) t_m \end{aligned} \tag{6.26}$$

for $0 \leq t_m \leq t$. In (6.25) we thus obtain

$$\mathbb{E}\left[\sup_{s\leq t}\left|X_s^{m+1} - X_s^m\right|^2\right] \leq K'(x_0,t)\frac{((8+2t)K^2)^m}{(m+1)!}t^{m+1}, \quad \text{for } t \geq 0,\, m \geq 0. \tag{6.27}$$

This implies that for $t \geq 0$,

$$\sum_{m\geq 0}\sqrt{\mathbb{E}\left[\sup_{s\leq t}\left|X_s^{m+1} - X_s^m\right|\right]} < +\infty. \tag{6.28}$$

As a consequence, $\mathbb{P}$-a.s. $(X_t^n)_{n\in\mathbb{N}}$ convergence uniformly on bounded time intervals to $X_\cdot^\infty$, which can be chosen adapted and continuous. Moreover,

$$\begin{aligned}
\sqrt{\mathbb{E}\left[\sup_{s\leq t}\left|X_s^\infty - X_s^m\right|^2\right]} &\overset{\text{Fatou}}{\leq} \liminf_{p\to+\infty}\sqrt{\mathbb{E}\left[\sup_{s\leq t}\left|X_s^p - X_s^m\right|^2\right]} \\
&\leq \sum_{k=m}^{+\infty}\sqrt{\mathbb{E}\left[\sup_{s\leq t}\left|X_s^{k+1} - X_s^k\right|^2\right]} \\
&\xrightarrow[m\to+\infty]{} 0, \quad t \geq 0.
\end{aligned}$$

Therefore, for $t \geq 0$, $\mathbb{P}$-a.s.,

$$\begin{aligned}
X_t^{m+1} &\xrightarrow[m\to+\infty]{L^2(\mathbb{P})} X_t^\infty, \\
\int_0^t b(s, X_s^m)\,\mathrm{d}s &\xrightarrow[m\to+\infty]{L^2(\mathbb{P})} \int_0^t b(s, X_s^\infty)\,\mathrm{d}s, \\
\int_0^t \sigma(s, X_s^m)\,\mathrm{d}B_s &\xrightarrow[m\to+\infty]{L^2(\mathbb{P})} \int_0^t \sigma(s, X_s^\infty)\,\mathrm{d}B_s,
\end{aligned} \tag{6.29}$$

and in view continuity of X^∞, Equation (6.29) holds $\mathbb{P}$-a.s. for $t \geq 0$. Hence X^∞ is a solution to ($\mathrm{E}_{x_0}(b,\sigma)$). □

Remark 6.8. From the definition of X^m (6.21) and from the fact that $\mathbb{P}$-a.s. X^m converges to X^∞ uniformly on bounded intervals we use that for each $t \geq 0$ $\mathcal{F}_t^{X^\infty}$ is the smallest σ-algebra containing all null-sets of $\mathcal{G}$ and making X_s^∞ measurable for all $s \leq t$ and $\mathcal{F}_t^{X^\infty} \subseteq \mathcal{F}_t^B$.

This implies that X is even $(\mathcal{F}_t^B)_{t\geq 0}$ adapted (which is not obvious since $\mathcal{F}_t^B \subseteq \mathcal{G}_t$). Intuitively, X_t^∞ is a function of the noise $(B_s, s \leq t)$. ♦

6.2. Martingale Problems

Recall the heuristic discussion at the beginning of this chapter. The second approach to construct a process that looks locally like $x + b(x)t + \sigma(x)B_t$ is via so-called martingale problems: (even if the correspondence is less obvious here)

Definition 6.9. A solution to a martingale problem is a probability measure $\mathbb{P}_x$ on $(C([0,+\infty),\mathbb{R}^d,\mathcal{F})$ such that

(i) $\mathbb{P}_x(X_0 = x) = 1$;

(ii) for all $f \in C_0^2(\mathbb{R}^d)$ we have

$$M_t^f := f(X_t) - f(X_0) = \int_0^t Lf(x_s)\,\mathrm{d}s, \quad t \geq 0 \tag{6.30}$$

is an $(\mathcal{F}_t)_{t\geq 0}$ martingale under $\mathbb{P}$ with

$$Lf(y) = \frac{1}{2}\sum_{i,j=1}^d a_{i,j}(y)\partial^2_{i,j}f(y) + \sum_{i=1}^d b_i(y)\partial_i f(y), \quad y \in \mathbb{R}^d. \tag{6.31}$$

To see a first connection between the two approaches we show the following:

Proposition 6.10. *Assume that $b : \mathbb{R}^d \to \mathbb{R}^d$, $\sigma : \mathbb{R}^d \to \mathbb{R}^{d\times m}$ be measurable, locally bounded functions, $x \in \mathbb{R}^d$ and some $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ endowed with an n-dimensional $\mathcal{G}_t$-Brownian motion $(B_t)_{t\geq 0}$, a continuous adapted $\mathbb{R}^d$-valued prcess $(X_t)_{t\geq 0}$ satisfies $\mathbb{P}$-a.s., for $t \geq 0$,*

$$X_t = x + \int_0^t b(X_s)\,\mathrm{d}s + \int_0^t \sigma(X_s)\,\mathrm{d}B_s. \tag{6.32}$$

Then for any $f \in C^2(\mathbb{R}^d;\mathbb{R})$ the process Mf defined as in Equations (6.30) and (6.31) with $a(y) = \sigma(y)\sigma(y)^\top \in \mathbb{R}^{d\times d}$, $y \in \mathbb{R}^d$ is a continuous local martingale.

Proof. By Ito's formula, $\mathbb{P}$-a.s. for $t \geq 0$,

$$f(X_t) = f(X_0) + \sum_{i=1}^d \int_0^t \partial_i f(X_s)\,\mathrm{d}X_s^i + \frac{1}{2}\sum_{i,j=1}^d \int_0^t \partial^2_{i,j} f(X_s)\,\mathrm{d}\langle X^i, X^j\rangle_s. \tag{6.33}$$

From (6.32) we have

$$
\begin{aligned}
\langle X^i, X^j\rangle_t &= \left\langle \sum_{k=1}^n \int_0^t \sigma_{i,k}(X_u)\,\mathrm{d}B_u^k, \sum_{\ell=1}^n \int_0^t \sigma_{j,\ell}(X_u)\,\mathrm{d}B_u^\ell \right\rangle \\
&= \sum_{k=1}^n \int_0^t \sigma_{i,k}(X_u)\sigma_{j,k}(X_u)\,\mathrm{d}u \qquad (6.34)\\
&= \int_0^t a_{i,j}(X_u)\,\mathrm{d}u.
\end{aligned}
$$

Therefore (6.33) gives $\mathbb{P}$-a.s. for $t \geq 0$,

$$
\begin{aligned}
f(X_t) = f(X_0) &+ \sum_{i=1}^d \int_0^t \partial_i f(X_s) b_i(X_s)\,\mathrm{d}s + \sum_{i=1}^d \sum_{k=1}^m \int_0^t \partial_i f(X_s)\sigma_{i,j}(X_s)\,\mathrm{d}B_s^k \\
&+ \sum_{1\leq i,j\leq d} \int_0^t a_{i,j}(X_s)\partial_{i,j}^2 f(X_s)\,\mathrm{d}s \qquad (6.35)\\
= f(X_0) &+ \int_0^t Lf(X_s)\,\mathrm{d}s + \int_0^t \nabla^\top f(X_s)\sigma(X_s)\,\mathrm{d}B_s,
\end{aligned}
$$

which implies the claim of the proposition. □

We now establish a solid link between SDE's and martingale problems.

Theorem 6.11. *(i) If on some $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0})$ endowed with a m-dimensional $\mathcal{G}_t$-Brownian motion $(B_t)_{t\geq 0}$ a continuous adapted process $(Y_t)_{t\geq 0}$ satisfies $\mathbb{P}$-a.s. for $t \geq 0$,*

$$
Y_t = x + \int_0^t b(Y_s)\,\mathrm{d}s + \int_0^t \sigma(Y_s)\,\mathrm{d}B_s \qquad (6.36)
$$

then the law $\mathbb{P}_x$ of $(Y_t)_{t\geq 0}$ on $(C(\mathbb{R}_+;\mathbb{R}^d), \mathcal{F})$ is a solution to the martingale problem (6.9) with $a(y) = \sigma(y)\sigma(y)^\top$.

(ii) Conversely , if $\mathbb{P}_x$ is a solution to the martingale problem (6.9) then there exists some $(\Omega, \mathcal{G}, (\mathcal{G}_t)_{t\geq 0}, \mathbb{P})$ endowed with an n-dimensional Brownian motion $(B_t)_{t\geq 0}$ and an adapted process $(Z_t)_{t\geq 0}$ such that

$$
Z_t = x + \int_0^t b(Z_s)\,\mathrm{d}s + \int_0^t \sigma(Z_s)\,\mathrm{d}B_s, \quad \mathbb{P}\text{-a.s. for } t \geq 0 \qquad (6.37)
$$

and the law of $(Z_t)_{t\geq 0}$ is $\mathbb{P}_x$.

7. Relation of SDE's and PDE's

As announced in the introduction we are now going to see that SDE's can be used to provide a probabilistic expectation formulas for the solutions of certain second order partial differential equations (PDE's).

We will consider to the following Poisson-Dirichlet problem: Let $U \subseteq \mathbb{R}^d$ bounded, open and non-empty. Let $f \in C^0_b(U)$ and $g \in C(\partial U)$. We look for $u \in C^2(U) \cap C^0(U)$ such that

$$\begin{cases} Lu(x) = -f(x) & \text{for } x \in U, \\ u(x) = g(x) & \text{for } x \in \partial U. \end{cases} \tag{7.1}$$

with as before

$$\begin{gathered} Lu(x) = \frac{1}{2}\sum_{i,j=1}^{d} a_{i,j}(x)\partial^2_{i,j}u(x) + \sum_{i=1}^{d} b_i(x)\partial_i u(x) \\ a(x) = \sigma(x)\sigma(x)^\top, \quad b : \mathbb{R}^d \to \mathbb{R}^d, \quad \sigma : \mathbb{R}^d \to \mathbb{R}^{d\times m}. \end{gathered} \tag{7.2}$$

The Dirichlet problem corresponds to $f \equiv 0$ and the Poisson equation to $g \equiv 0$ in (7.1).

We will assum that b and σ are measurable, locally bounded and satisfy the ellipticity condition

$$\text{there exists } c > 0 \text{ such that } \xi^\top a(x)\xi \geq c|\xi|, \quad \text{for any } \xi \in \mathbb{R}^d, x \in U. \tag{7.3}$$

From the theory of PDE's (c.f. Gilbarg Trudinger, Elliptic PDE's od 2nd order, page 106) it is known that if b, σ satisfy (7.3) and the Lipschitz condition as in **??**, f is Hölder continuous in U, and U satisfies an exterior sphere condition: for any $z \in \partial U$ there exists an open ball B with $B \cap \ U = \{z\}$, then the problem (7.1) has a solution.

We now find a probabilistic representation to this solution.

Theorem 7.1. *Let b, σ in (7.3) be locally bounded and measurable. If u is a solution* *to* (7.1) *and* X_t *is a solution to the SDE*

$$X_t = x + \int_0^t b(X_s)\,\mathrm{d}s + \int_0^t \sigma(X_s)\cdot \mathrm{d}B_s \tag{7.4}$$

for some $x \in U$*, then the* exit time of X from U

$$T_U = \inf\{s \geq 0 \mid X_s \notin U\} \tag{7.5}$$

is $\mathbb{P}$*-integrable and*

$$u(x) = \mathbb{E}\left[g(X_{T_U}) + \int_0^{T_U} f(X_s)\,\mathrm{d}s\right]. \tag{7.6}$$

A. Notation

$\mathrm{Var}(X)$	$= \int_\Omega (X - \mathbb{E}[X])^2 \, \mathrm{d}\mathbb{P}$, the variance of X.
$\mathrm{Variation}(f)$	is the variation of a function f.
$\mathbb{E}[X]$	$= \int_\Omega X \, \mathrm{d}\mathbb{P}$ the expectation of X.
$\mathbb{P}$	mostly denotes a probability measure on a measurable space $(\Omega, \mathcal{A})$.